

15152

AN ANALYSIS OF GEOGRAPHIC VARIATION IN THE ANTIPREDATOR
ADAPTATIONS OF THE GUPPY, POECILIA RETICULATA

by

BENONI HENDRIK SEGHERS

B.Sc., University of British Columbia, 1967

A THESIS SUBMITTED IN PARTIAL FULFILMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
in the Department
of
ZOOLOGY

We accept this thesis as conforming
to the required standard

THE UNIVERSITY OF BRITISH COLUMBIA

January, 1973

In presenting this thesis in partial fulfilment of the requirements for an advanced degree at the University of British Columbia, I agree that the Library shall make it freely available for reference and study. I further agree that permission for extensive copying of this thesis for scholarly purposes may be granted by the Head of my Department or by his representatives. It is understood that copying or publication of this thesis for financial gain shall not be allowed without my written permission.

Department of ZOOLOGY

The University of British Columbia
Vancouver 8, Canada

Date MARCH 16, 1973.

ABSTRACT

The main objective of this study was to describe and explain several features of geographic variation among isolated and semi-isolated populations (Trinidad, West Indies) of the guppy, Poecilia reticulata. Three main aspects of geographic variation were considered: (i) sex ratio, (ii) body size, and (iii) antipredator behavior.

(i) Extreme deviations (favouring females) from a theoretical Mendelian sex ratio were correlated with the presence of dense populations of a small cyprinodontid predator, Rivulus hartii. Laboratory experiments revealed that this variation was not caused by genetic differences in the sex determination system. In addition, sex ratios were not correlated with sexual dimorphism in colour.

Predation experiments with Rivulus demonstrated that male guppies were not selectively attacked but were less adept at avoiding capture. Size-selective predation by Rivulus also placed males at a selective disadvantage. Whether conspicuous coloration increases the liability of males to predation has yet to be demonstrated unequivocally.

(ii) Populations of guppies sampled in 1967 and 1969 showed a stable pattern of variation in body size; differences of over 41% in body length and 200% in weight

were discovered. In compliance with Bergmann's Rule there was a significant negative correlation between body size and stream temperature. Though a substantial portion of the size variation can be explained as a direct phenotypic response to environmental differences, there is also good evidence for microevolutionary differences.

Of a multitude of potential selective factors that might be responsible for this genetic diversity, only one, size-selective predation, was investigated. Field and laboratory evidence supported the hypothesis that large guppies enjoy an advantage with respect to Rivulus predation but are more vulnerable to large characid and cichlid predators such as Hoplias malabaricus and Crenicichla alta. In the laboratory, size-selective predation appeared to be caused by differences in the handling efficiency of the predators, however, in nature the interaction of several other factors must be considered.

(iii) Field observations revealed that where characid and cichlid predators were present (and Rivulus absent) guppies, (a) were more restricted to the stream shore, (b) showed a greater tendency to school, (c) avoided a potential predator at a greater distance, and (d) had a lower alarm threshold. To assess the functional and evolutionary significance of this behavioral variation, predation experiments were conducted with samples of wild-caught and predator-naive (laboratory-bred) guppies

originating from 5 natural populations. These tests demonstrated that fish either taken, or descended, from populations exposed to characids and cichlids were relatively less vulnerable than those exposed to Rivulus.

To determine why the predator-naive guppies were selected non-randomly, a comparison was made of their habitat preferences, schooling behavior, reaction distance, and escape motor patterns. Significant interpopulation differences were found for several of the measures; generally, these were consistent with the field observations. In addition, it was hypothesized that certain populations may learn to avoid predators more rapidly.

It was concluded that much of the geographic variation in antipredator behavior is caused by genetic differences attributable to differential predation pressures. In some cases, these microevolutionary differences are apparently maintained without a major barrier to gene flow.

ACKNOWLEDGEMENTS

This thesis was supervised by Dr. N. R. Liley. I extend sincere thanks to him for introducing me to the problem considered in this study and for critically reading the manuscript. I am also grateful to my research committee, Drs. A. B. Acton, C. S. Holling, and J. D. McPhail for guidance and comments on the thesis.

For hospitality in the Zoology Section, Department of Biological Sciences, University of the West Indies, St. Augustine, Trinidad, I am indebted to Dr. B. D. Ainscough, Prof. F. G. Cope, Dr. J. S. Kenny, and Mr. R. L. Loregnard.

Senator Jeffrey Stollmeyer kindly permitted me to collect fish on his estate and Mr. Hugh Wood, Chief Fisheries Officer, Ministry of Agriculture, allowed me to maintain fish at the Freshwater Fisheries Research Station, Bamboo Grove.

Miss Shireen Imam and Mr. Robert W. DeForest gave freely of their time to help with the field work.

I profitted greatly from the advice and good humour of my fellow students, especially Mr. Peter J. Ballin, Mr. Kim D. Hyatt, and Dr. Donald L. Kramer.

Finally, I thank my parents for their patience,

and cheerful encouragement throughout this study.

This investigation was financed by the National Research Council of Canada through operating grants to Dr. N. R. Liley and postgraduate scholarships to myself.

TABLE OF CONTENTS

	Page
Abstract	ii
Acknowledgements	v
List of Tables	xii
List of Figures.	xiv
Chapter	
1. Introduction.	1
Main objectives of the study.	3
The biology of <u>Poecilia reticulata</u> :	
relevant literature	7
2. Materials and Methods	10
Introduction.	10
Collection techniques	10
Laboratory populations at Vancouver	12
Maintenance of fish	13
Observation and recording methods	14
Ecological measurements	15
Measurements on guppies	17
3. The Environment	19
Introduction.	19
Classification of streams	20
Geographic distribution of fish species	20

Chapter	Page
Predators of the guppy.	23
Distribution and natural history	
of the main predators	31
Migration and dispersal of guppies.	35
Summary of Chapter 3.	39
4. Geographic Variation in the Sex Ratio	41
Introduction.	41
Sex ratios of natural populations	42
Sex ratios of laboratory stocks	46
Relationship of predation to variation	
in the sex ratio.	49
Size hypothesis	57
Behavior hypothesis	58
Relative survival of the sexes in	
predation experiments	59
A. Experiments with wild-caught fish. . .	60
B. Experiments with laboratory stocks . .	64
Procedure for standard survival test. . .	64
Discussion of survival experiments. . . .	71
Tests of the behavior hypothesis.	75
Discussion of sex differences in	
behavior.	82
Conclusion and general discussion of	
geographic variation in the sex ratio . .	83
Summary of Chapter 4.	89

Chapter	Page
5. Geographic Variation in Body Size	92
Introduction.	92
Body size variation in natural	
populations	95
Relationship of body size and temperature .	97
Relationship of body size and predation . .	103
Evidence from the field for size	
selection	104
Experimental analysis of size selection . .	107
Mechanisms of predator selection for	
size differences in prey.	112
Conclusion and general discussion of	
geographic variation in body size	123
Summary of Chapter 5.	129
6. Geographic Variation in Behavior.	131
Introduction.	131
Field observations of behavior.	132
Distribution of guppies in the	
stream environment.	133
Schooling behavior.	137
Reaction distance to potential	
predators	141
Motor patterns used in antipredator	
behavior.	143

Chapter	Page
Population differences in escape	
motor patterns.	148
Summary of the field observations	
of behavior	150
Survival value of behavioral differences	
in wild-caught fish	151
Conclusion for survival experiments	
with wild-caught guppies.	159
Relative survival of laboratory stocks. . .	161
Discussion of results of survival	
experiments	166
Geographic variation in the antipredator	
behavior of laboratory stocks	169
(a) Microhabitat selection.	170
(b) Reaction distance to predators. . . .	182
(c) Escape motor patterns of	
individual fish	188
(d) Schooling behavior.	208
Conclusion and general discussion of	
geographic variation in behavior.	217
Summary of Chapter 6.	226
7. General Discussion and Conclusions.	231
The environment	231
Coloration and the sex ratio.	232
Body size	234

Chapter	Page
Antipredator behavior	235
Suggestions for future research	239
Literature cited	244
Appendix	261

LIST OF TABLES

Table	Page
1. Classification and physical features of streams in the Northern Range (March-June, 1969)	21
2. Natural history of the main fish predators.	32
3. The principal fish predators occurring at the main study areas.	36
4. Site attachment of guppies in the Petite Curucaye R.	38
5A. The sex ratios of natural populations (1967)	43
5B. The sex ratios of natural populations (1969)	44
6. The sex ratios of laboratory populations	48
7. The relationship of the sex ratio to the presence of fish predators	50
8. <u>Rivulus</u> abundance in two streams in 1967 and 1969.	53
9. Sex ratio of guppies at Blue Basin in 1967 and 1969	54
10. Predation by <u>Rivulus</u> on male and female guppies in a 40 liter aquarium	61
11. The origin, sex, and size of individual <u>Crenicichla</u> used in experiments at Vancouver.	67
12. Reaction distance of naive guppies to a dead <u>Crenicichla</u>	81
13. Predators and prey used in Experiment 5.1.	109

Table	Page
14. Direction of change of mean body size in 18 size selection tests	112
15. The relationship of body size and reaction distance to a predator.	115
16. Predation efficiency of <u>Crenicichla</u> on large and small guppies	117
17. Predation efficiency of <u>Astyanax</u> on large and small guppies.	119
18. Predation efficiency of <u>Rivulus</u> on large and small guppies.	122
19. The distribution of 5 populations of guppies in relation to water velocity, depth, and distance from the shore	138
20. The development of schooling behavior in 5 populations of guppies	141
21. Relative mortality of wild-caught guppies exposed to predators	156
22. Relative mortality of naive laboratory stocks of guppies exposed to <u>Crenicichla</u>	165
23. A comparison of the frequency of occurrence of 5 stocks of guppies in relation to water depth.	174
24. Escape motor patterns of 5 laboratory stocks of guppies exposed to a dead <u>Crenicichla</u>	190
25. Comparison of predatory success and prey escape behavior for 5 laboratory stocks of guppies.	196
26. Responses of naive Guayamare and Paria guppies to a simulated aerial predator	206
27. Concordance of behavioral measures taken on 5 stocks of guppies	220

LIST OF FIGURES

Figure	Page
1. Map of the northern half of the island of Trinidad, West Indies, showing the major river systems.	11
2. The distribution of fish species in the Northern Range region.	22
3. A schematic representation of the distribution of the major fish species in the northwest corner of Trinidad.	24
4. Photograph of the major fish predators of the guppy	27
5. A comparison of the sex ratio of 13 populations of guppies sampled in 1967 and resampled in 1969	45
6. The relationship of the sex ratio to the relative density of <u>Rivulus</u>	52
7. Relative survival of male and female guppies in the experimental section of the Petite Curucaye River	63
8. Relative survival of male and female Caparo stock guppies exposed to <u>Crenicichla</u>	69
9. Relative survival of male and female guppies of two stocks exposed to <u>Rivulus</u>	71
10. Predatory behavior of <u>Rivulus</u> exposed to male and female guppies of the Paramaribo stock	78
11. The ratio of predation attempts to successful captures for <u>Rivulus</u> feeding on male and female guppies	79

Figure	Page
12. The body size of adult male guppies collected at 20 sites in 1969.	96
13. A comparison of the body size of adult male guppies from 13 populations sampled in 1967 and resampled in 1969.	98
14. The relationship of mean body size of adult males to stream temperature	101
15. The relationship of <u>Rivulus</u> body size to the size of guppies taken as prey	105
16. Size selection by predators on 18 experimental populations of guppies.	111
17. The distribution of adult guppies across a section of the Paria River.	136
18. Relative survival of wild-caught female guppies of two populations exposed to either a single <u>Crenicichla</u> or two <u>Hoplias</u>	158
19. Depth profile of spontaneous swimming behavior of 5 laboratory stocks of guppies.	173
20. Habitat selection of 5 laboratory stocks of guppies placed in a depth gradient	179
21. Reaction distance of 5 laboratory stocks of guppies to a dead <u>Crenicichla</u>	186
22. Relationship of escape motor patterns to survival time in two laboratory stocks of guppies exposed to <u>Rivulus</u>	199
23. Test apparatus used for measuring the responsiveness of laboratory stocks of Guayamare and Paria guppies to a simulated aerial predator.	202
24. Mean index of cohesion for five laboratory stocks of guppies	211

Figure

Page

25. Predatory behavior of Rivulus exposed
to equal numbers of Paria and Lower
Aripo guppies (laboratory stock) 216

CHAPTER 1

INTRODUCTION

An animal faces many challenges to survival in its natural environment. Numerous biotic and abiotic factors influence the likelihood that an animal will produce viable offspring. Most animals share a common challenge to survival--predation. In this thesis I examine how natural populations of a small tropical fish are adapted to the risk of predation.

Biologists usually study predator-prey systems at either the numerical, functional, or evolutionary level. Population ecologists work primarily at the numerical level--they attempt to measure the impact of predators on the numbers of prey and how in turn prey density governs the number of predators that a given area can support (e.g. Ricker, 1954; Huffaker et al, 1963). At the functional level, factors controlling the consumption of prey are emphasized, particularly the physiology and behavior of the predator (e.g. Ivlev, 1961; Holling, 1966; Beukema, 1968).

The third level at which we can view a predator-prey system is the more general evolutionary level; an attempt is made to measure the contribution of predation to natural

selection in the prey (e.g. Cain and Sheppard, 1954; Kettlewell, 1961; Curio, 1965, 1969, 1970a; McPhail, 1969). This is the approach adopted here.

The evolutionary impact of predators on prey populations is normally difficult to assess directly. Often it is not known what predators are important or how effectively the prey populations are isolated in terms of gene exchange. Occasionally a natural situation is discovered where several populations of the same species are isolated or semi-isolated by relatively short distances and are exposed to different predation pressures. This presents an excellent opportunity to assess the impact of predation because it is one of the few ecological factors that varies among the populations.

A situation of this type was reported by C. P. Haskins and co-workers (1961) for populations of the guppy, Poecilia reticulata Peters, a tropical freshwater fish native to the rivers of north-east South America and some Caribbean islands. On the island of Trinidad, West Indies, Haskins et al. discovered an apparent correlation between the frequency and linkage of certain genes controlling colour patterns of guppies (expressed phenotypically only in the male) and the presence or absence of predaceous fish species. They suggested that two opposing selective forces were at play (cross-selection), one, possibly sexual selection, driving in the direction of more

conspicuous body markings and the other, predation, selecting for a more cryptic male phenotype.

This predator-prey interaction appeared very inviting for a comparative ethological study. At the outset of this research I intended to study selection by predators in relation to geographic variation in coloration, bearing in mind that additional morphological and behavioral variation might be discovered. It soon became obvious that a wealth of microgeographic variation in behavior does exist in both wild fish and their laboratory-reared offspring. Because I felt this discovery might give some insight into the mechanisms of the evolution of behavior, I devoted most of my time to the study of population differences in behavior (Chapter 6). Nevertheless I do include in this thesis the original work concerned with the "Haskins hypothesis" (Chapter 4) as well as an analysis of the adaptive significance of geographic variation in body size (Chapter 5). I hope to be able to demonstrate that these three aspects are interrelated and can be considered under the collective term of "antipredator adaptations".

Main objective of the study

The main objective of this study was to assess the evolutionary significance of natural variation in populations of the guppy and to determine to what extent this variation might be related to the distribution and abundance

of aquatic predators. Three aspects of this problem were considered:

1. Sex ratio

Though population differences in the sex ratios of fish are normally not genetically determined and hence do not present an evolutionary problem per se, I studied this variation in some detail because sex ratio differences could be valuable clues to the nature of differential selection by predators or other factors on colour patterns (which do have a genetic basis).

2. Body size

There is abundant evidence that animal body size and growth rates are influenced by both genetic and environmental factors. I wished to determine if predation or other environmental factors might be responsible for microgeographic variation in the size of adult fish.

3. Behavior

The third aspect concerns the antipredator behavior of guppies. I wanted to describe the variation among populations in nature and among laboratory-reared offspring of some of these natural populations. In particular I wished to test the survival value of the behavior (how is it adaptive?). My overall hypothesis was that differences in the behavior of natural populations would be correlated with the distribution and abundance of the major predators. More importantly, I predicted these differences would be

heritable.

To falsify this hypothesis it would be necessary to demonstrate that the variation observed in nature does not persist in fish raised under constant conditions in the laboratory. This assumes that laboratory stocks are genetically representative of field samples and that behavioral tests are sufficiently sensitive to detect differences, if present.

It is fair to ask at this point if such a study is indeed necessary. There is a paucity of comparative data of intraspecific variation in antipredator adaptations. This is surprising because antipredator devices occur in virtually all animal (and many plant) groups and are vitally important to the survival of the species. Therefore such a study could have implications not only for the "pure" evolutionary biologist who is attempting to assess the relative importance of selection and gene flow in the differentiation of local populations (e.g. Ehrlich and Raven, 1969), but also the applied biologist concerned with predicting the impact of predator introductions on the subsequent behavior of the prey population. What behavioral and morphological characters would be selected for under such conditions? How rapidly could prey counteradaptations evolve?

These questions are also relevant when man, in

harvesting a natural population, acts as a selective predator (Miller, 1957). More specifically, an appreciation of population variation in defensive behavior could be important in fish stocking programs. Raleigh and Chapman (1971) referring to trout fry migrations stated, "failure to match the innate behavior of the donor population to the requirements of the recipient environment has led to failure of many fish transplant efforts in the past (p. 39)." As Calaprice (1969) has warned, the continuous stocking of ill-adapted fish to a natural population can decrease the mean population fitness and possibly culminate in local extinctions.

My study is also relevant to the science of behavioral genetics, a field devoted largely to the measurement of behavioral differences among inbred strains of mice, rats, and Drosophila. Bruell (1967) has criticized past work in this field and made some useful suggestions for future research:

The wild base populations from which the ancestors of our laboratory strains were drawn are shrouded by the mist of incomplete records and thus, in most cases, we do not know anything about the natural environmental conditions under which the founders of our strains evolved.

Work will have to start with proper samples of individuals drawn from local wild populations which evolved under distinct and well-known environmental conditions.

Investigation of forms of behavior chosen on the basis of evolutionary criteria is probably the most critical requirement of future behavior-genetic analyses.

To go beyond intuition and to understand the adaptive significance of a behavior, we must know the environmental conditions under which it occurs and how it varies as environmental conditions vary.

(pp. 284-286)

In this research I shall endeavour to implement some of Bruell's recommendations.

Clearly the problem of natural variation in the morphology and behavior of animal species is very complex, encompassing a great portion of ecology, ethology, and evolutionary biology. My main aim is to identify some of the natural variation in Trinidad populations of the guppy and to assess its evolutionary significance.

The biology of *Poecilia reticulata*:
relevant literature

It is ironical that a species familiar to all amateur aquarists is virtually unknown to the ecologist. A very meagre literature exists for studies on the guppy under natural conditions. Because the guppy is small, has a short generation time (about 3 months), is always in good commercial supply, and adapts readily to aquarium culture, it has been used as a convenient bioassay organism in studies ranging from toxicology and gerontology to tissue transplantation and mutation research. In most of these studies domestic strains of the guppy were used.

The guppy has been the subject of several ethological studies. Aspects of sexual selection have been investigated by Noble (1938), Haskins and Haskins (1949, 1950), and

Haskins et al (1961). Liley (1966) has examined how the guppy is ethologically isolated from three other sympatric poeciliid fish. The conspicuous courtship behavior of the male guppy has attracted the attention of numerous workers (Breder and Coates, 1935; Clark and Aronson, 1951; Baerends et al, 1955; Rosen and Tucker, 1961).

The pioneering genetical studies of Winge (1922a, 1922b, 1927) and Winge and Ditlevsen (1948) (recently extended by Haskins et al, 1970) have established the primarily sex-linked nature of colour pattern inheritance in laboratory strains of the guppy. Winge's findings have subsequently been confirmed for natural populations by Haskins and Haskins (1951, 1954) and Haskins et al (1961).

Laboratory populations of the guppy have been used as model systems for investigating several aspects of population regulation (Breder and Coates, 1932; Shoemaker, 1944; Rose, 1959; Silliman and Gutsell, 1958; Geodakyan and Kosobutskii, 1972). However, no one has ever studied the dynamics of natural populations.

Additional work of ecological significance includes studies of feeding (Hester, 1964; Davis, 1968), growth (Svårdson, 1943, cited in Alm, 1959; Bertalanffy, 1938; Gibson and Hirst, 1955), and longevity (Comfort, 1961). Useful data on vision are given in Lang (1965, 1967), Waldman (1969), and Protasov (1970).

The systematics and zoogeography of the family

Poeciliidae have been reviewed by Rosen and Bailey (1963).

Avoidance behavior has been investigated in two laboratory studies (Werboff and Lloyd, 1963; Russell, 1967a, 1967b). Williams (1964) has briefly considered schooling behavior and Schutz (1956) touched upon the relationship between pheromones and the "fright reaction".

Finally there is the study by Ballin (1973) of geographic variation in courtship and aggressive behavior of three populations of Trinidad guppies. This is a valuable companion to my research because the study was carried out on similar genetic material but explored a different set of behavioral parameters.

CHAPTER 2

MATERIALS AND METHODS

Introduction

This research is a combination of field and laboratory study. The field observations and collections were made in the rivers of the Northern Range Mountain region of the island of Trinidad, West Indies (Fig. 1) and the bulk of the experimental work was conducted under controlled conditions in the laboratory.

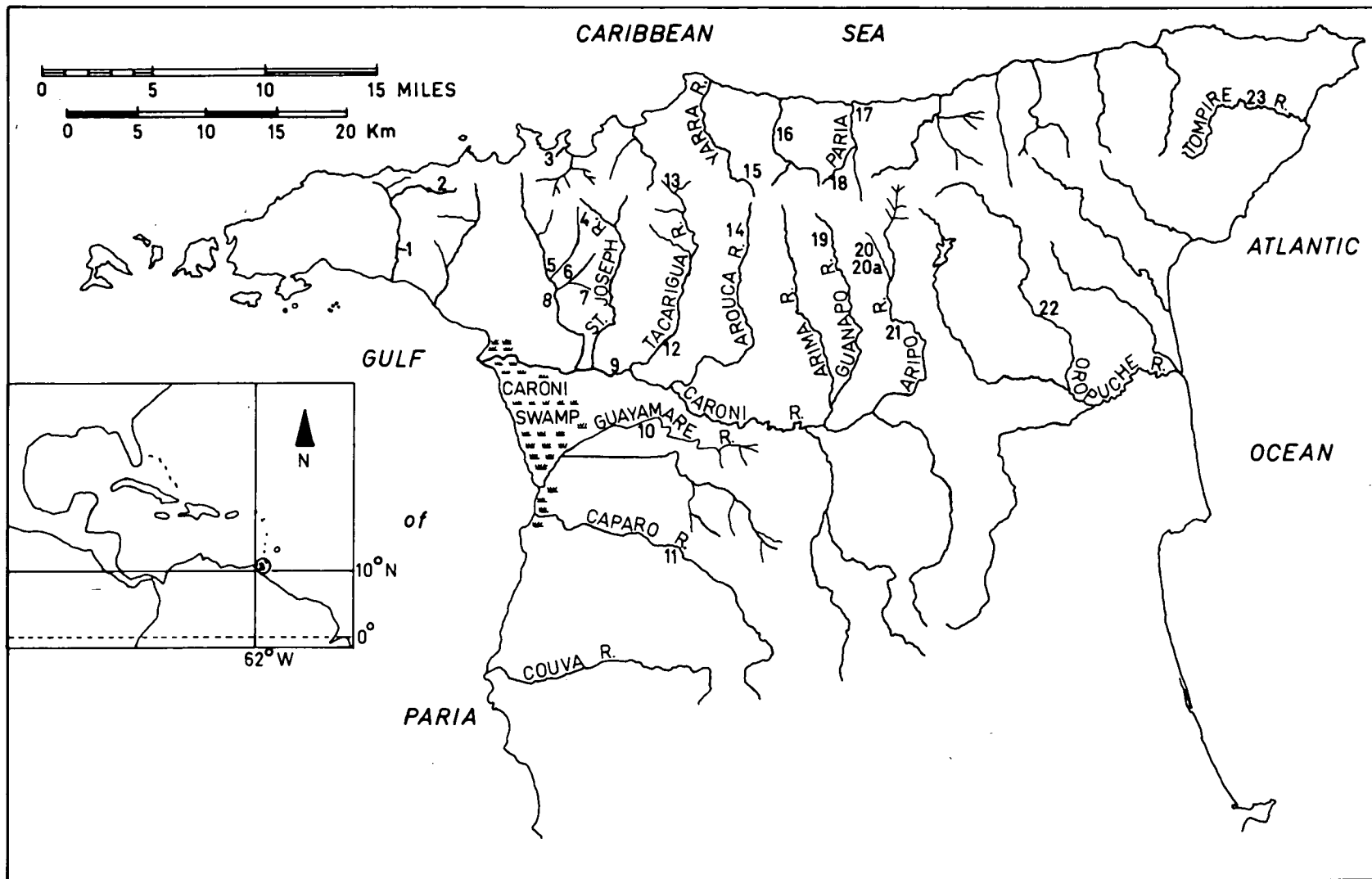
The work in Trinidad was based on two field expeditions. The first of these (June-August, 1967) provided an introduction to the problem of variation in Trinidad guppies and the second (March-June, 1969) allowed me to make more detailed collections of fish, extend my 1967 observations, and conduct some experiments with wild fish. These collection periods include parts of the wet (June-October) and dry (November-May) seasons.

Collection techniques

Guppies were collected with a fine circular dipnet or a fine "one-man seine". A variety of methods were used to collect the predators including dipnets, seines, gill-nets, castnet, and hook-and-line. The castnet proved to be

FIGURE 1. Map of the northern half of the island of Trinidad, West Indies, showing the major river systems. Collection sites are indicated by numbers corresponding to the key below. Inset map indicates the position of the island just off the north-east coast of Venezuela.

Map no.	Name of stream	Standard abbreviation
1	Sierra Leone Road	SLR
2	Blue Basin	BB
3	Maracas Village	MV
4	Upper Curumpalo	UCur
5	Lower Curumpalo	LCur
6	Grande Curucaye	GCur
7	Petite Curucaye	PCur
8	Santa Cruz	SC
9	Caroni	Car
10	Guayamare	Guay
11	Caparo	Cap
12	Lower Tacarigua	LTac
13	Upper Tacarigua	UTac
14	Upper Arouca	UArouc
15	Yarra	Yar
16	Marianne	Mar
17	Lower Paria	LPar
18	Upper Paria	Par
19	Upper Guanapo	UGuan
20	Upper Aripo (Naranjo)	UA(N)
20a	Upper Aripo (Crossing)	UA(X)
21	Lower Aripo	LA
22	Oropuche	Oro
23	Tompire Tributary	TT



the best way to capture the larger fish predators.

Collections of guppies were made for several purposes. An instant "dead" collection was made at each site of interest by placing the fish directly into a 10% formalin solution. This collection was used to measure morphometric characters and determine the sex ratio. A "live" collection was made at certain sites to obtain: (1) fish for experiments in Trinidad, (2) founders for laboratory populations at Vancouver, Canada and (3) males for the assessment of variation in colour patterns (Liley and Seghers, unpublished data).

Every effort was made to collect a random, representative sample for each stream. Because densities fluctuated from stream to stream, more time was required at some sites to collect sufficient numbers for statistical comparisons.

Laboratory populations at Vancouver

Live fish were shipped by air to Vancouver in both 1967 and 1969. These shipments included several species of predators and samples of guppies taken from representative streams. Samples from different streams were kept isolated. As many fish as practically feasible were shipped to insure that the laboratory populations (referred to hereafter as "stocks") would be representative of their respective natural counterparts.

In all cases no fewer than 50 large gravid females and 25 mature males were used to start the laboratory cultures. In 1969 the majority of the stocks started in 1967 were replenished or replaced with fresh collections. This was done to avoid inbreeding and selection in laboratory stock, i.e. to maintain as much of the natural variability as possible.

A total of 23 sites were sampled in Trinidad (Fig. 1) and fish derived from 10 of these were selected for behavioral studies at Vancouver. Several stocks were used by other workers (Henderson, unpubl.; Ballin, 1973; Morrell, unpubl.; Liley, unpubl.). I selected 5 stocks for the comparative study of antipredator behavior and used several others, including one stock from Paramaribo, Surinam (Sommeldijske Kreek), for various other experiments.

Maintenance of fish

Standard procedures for the culture of tropical freshwater fish were used. Guppies were housed mostly in 40 and 60 liter glass aquaria maintained at temperatures found in nature (24-28°C). Fish from each geographic locality were kept separate and bred in mass culture in numerous aquaria. Within each stock fish were periodically mixed to reduce inbreeding and prevent genetic drift.

The predators were housed in glass or wood-and-glass tanks of 40 to 400 liters; in Trinidad the larger

individuals were kept in concrete pools (indoor and outdoor) of 700 to 1600 liters.

All aquaria were fitted with either sub-gravel, outside, or inside filters. The tank floors were covered with approximately 4 cm of light-brown fine sand. Aquaria not used for experiments contained water plants (Ceratopteris sp., Lemna sp., Sagittaria sp.).

Illumination was provided by cool white fluorescent tubes mounted 20-30 cm above the water surface. Photoperiod was controlled by Inter-matic automatic time switches connected to the lights; the standard photoregime was 12 light--12 dark.

Guppies were fed daily with finely-ground Clark's dry fish food supplemented several times per week with chopped, live Tubifex worms. The predators were fed on whole or chopped Tubifex supplemented irregularly with live guppies. Special feeding regimes were used for experiments and will be described in the appropriate experimental sections.

Observation and recording methods

Trinidad. Under most conditions in nature, guppies are easily observed from the edge of a stream. Several of the predator species may be observed in this way also, or from a bridge over a stream. Usually upon initial approach, the predators (and guppies at certain sites) will flee and

move to deeper water or hide under boulders, submerged logs, or leaf detritus. I found that if I sat quietly for a few minutes the fish resumed their normal activity. I also made observations underwater using a snorkel and face mask; a few observations were made at night with a headlamp.

Behavioral information was directly recorded into a field notebook or dictated into a portable tape-recorder. The duration of behavior patterns was measured in the field with a stopwatch.

Vancouver. More sophisticated equipment was available at Vancouver--both Rustrak (4-channel) and Esterline-Angus (20-channel) event recorders were used to record the frequency and duration of behavior patterns. By depressing a key corresponding to a behavioral event (1 key per channel) a permanent record was made on a continuously moving chart.

Several other methods were used to study behavior. These will be incorporated into the specific experimental sections.

Ecological measurements

A detailed limnological survey was not warranted for this study but I did want quantitative measures of the most important variables that might affect the behavior of the fish.

(a) Stream dimensions. At each collection and observation site I measured the average depth and width of

the stream and recorded changes caused by rainfall.

(b) Water velocity. Velocity was measured with a surface float placed in the centre of a representative section of stream. A separate measurement was made for the portion of the stream inhabited by guppies (microhabitat). In each case the mode of 5 to 10 runs was used.

(c) Volume of flow (discharge). This was calculated using the standard formula given by Needham and Needham (1962: 104).

(d) Temperature. Temperature was measured to the nearest 0.1°C with a mercury thermometer placed in the mainstream and the microhabitats.

(e) pH. Fresh Hydrion pH paper was used to measure relative acidity or alkalinity of the water. The modal value of several determinations was used.

(f) DH. Hardness was measured with a Rila Water Hardness Test Kit and all determinations were replicated.

(g) Turbidity. In the few streams that were turbid, an improvised Secchi disc was used to measure the attenuation of light with depth.

(h) Shade (degree of overhead cover). The amount of cover was estimated on a semi-quantitative 5-point scale ranging from no shade (0) to complete shading (4).

(i) Substrate. Colour photographs were taken of the stream bottoms.

(j) Other fish species. An attempt was made to

collect all fish species that were sympatric with the populations of the guppy. Stomach contents were analysed.

(k) Population estimates. An accurate census was not made for most streams although a "catch-per-unit-effort" comparison gave a rough estimate of the relative abundance of guppies and other fish species. In one stream (Petite Curucaye) a 35 metre section was screened off and all fish were removed; in another (Paria) a visual count of guppies was made.

Measurements on guppies

(a) Sex ratio. The guppy is sexually dimorphic-- mature males are identified by the presence of a gonopodium (modified anal fin) and fully developed colour markings. Immature males may lack coloration but show a partially or fully developed gonopodium. The sex of fish smaller than immature males is uncertain and they are classed as immature. In this thesis I express the sex ratio as the number of males per one female.

(b) Length. Two measurements of length were made, standard length (tip of snout to posterior end of caudal peduncle) and total length (tip of snout to tip of tail). Some samples were measured with sliding vernier calipers to the nearest 0.1 mm. Others were measured with a ruler to the nearest 0.5 mm.

Live fish used in experiments were measured to the

nearest 0.5 mm total length. The fish were placed directly from the water into a dry petri dish. When the fish stopped moving (few sec), a ruler was passed under the dish to measure the length. This caused no mortality and proved to be the most efficient method to measure the thousands of live fish that were used during this study.

All length measurements of guppies and other fish given in this thesis are total length except where another measurement is specifically designated.

(c) Weight. A few samples were weighed to the nearest .001 g. The preserved fish were placed on a paper towel to remove surface fluid prior to weighing.

(d) Colour patterns. To record individual and population variation in the coloration of males, a scoring method was developed (Liley and Seghers, unpubl.) for black, red-orange-yellow, and iridescent (mainly blue and green) markings. Colour patterns were assessed on wild-caught fish that had been freshly killed by immersion in ice-water. In addition colour photographic slides were made of the same individuals.

CHAPTER 3

THE ENVIRONMENT

Introduction

Beebe (1952) has given a general account of the geography, climatic conditions, and plant and animal assemblages of Trinidad with special reference to the Arima Valley. This valley is located in the Northern Range Mountains near the centre of my study area. Beebe did not deal with the fish fauna nor was any attention given to the stream ecosystem.

A study of the Maracas R. (tributary of the St. Joseph R., see Fig. 1) by Thornhill et al (1966) indicated the main food webs in the stream and described the micro-habitat distribution of the fish species. A more general treatment of the ecology of streams in the Northern Range with special reference to the guppy and its predators was included in Haskins et al (1961).

Boeseman (1960, 1964) has compiled a key to the species of freshwater fish of Trinidad. This monograph reviews most of the early collections and taxonomy of Trinidad fish but lacks ecological and detailed distributional data.

In general the ecology of stream fish in the tropics has been little studied (Gery, 1969; Allen, 1969) and this applies to Trinidad as well.

Classification of streams

Using the methods outlined in Chapter 2, it was possible to erect a classification of a representative sample of the streams and rivers of the Northern Range region. This was an abiotic classification based on the width, depth, and velocity of each stream in the region surrounding the collection sites (Table 1). A more detailed breakdown of these measurements can be found in the Appendix (Table 1). Wherever possible I attempted to make measurements in dry (March, April, 1969) and wet (June, July, 1969) seasons because several of the streams are greatly affected by the change in rainfall (increase in volume of flow and turbidity).

Geographic distribution of fish species

As noted above, there is little published information regarding the distribution of fish in Trinidad. Figure 2 summarizes the results of the 1967 and 1969 collections. For specific names, consult the Appendix (Table 2).

It is clear that the size of the streams is the main factor limiting the diversity of the fish fauna though streams isolated by physical barriers may have an impoverished fish fauna and many species which might otherwise be

TABLE 1. Classification and physical features of streams in the Northern Range (March - June, 1969).

Stream type		Width (m)	Depth (m)	Velocity (m/sec)	Volume of Flow (m ³ /sec)	Temp. (°C)	Shade ^a	Turbidity ^b
Springwater	PCur GCur UCur TT	0.50-1.0	0.05-0.15	0.11- 0.29	0.0085- 0.0128	24.3- 26.2	3-4	0
Headstream	UA(N) UTac UArouc UGuan BB Yar Par	1.20-5.0	0.06-0.15	0.32- 0.67	0.028- 0.267	24.6- 27.4	2-3	0-1
Midstream	LA LTac SC Mar Oro	3.0-8.0	0.13-0.20	0.42- 1.18	0.150- 1.129	24.3- 30.0	2	1
Lowland R.	Guay Cap Car	2.0-25.0	1.5-3.0	0.33- 0.40	0.563- 22.50	26.9- 29.1	0-1	2

^a0 - no. shade

1 - small amount of shade restricted mainly to streambank

2 - medium shade (50% cover)

3 - medium to dense (75% cover with few exposed parts)

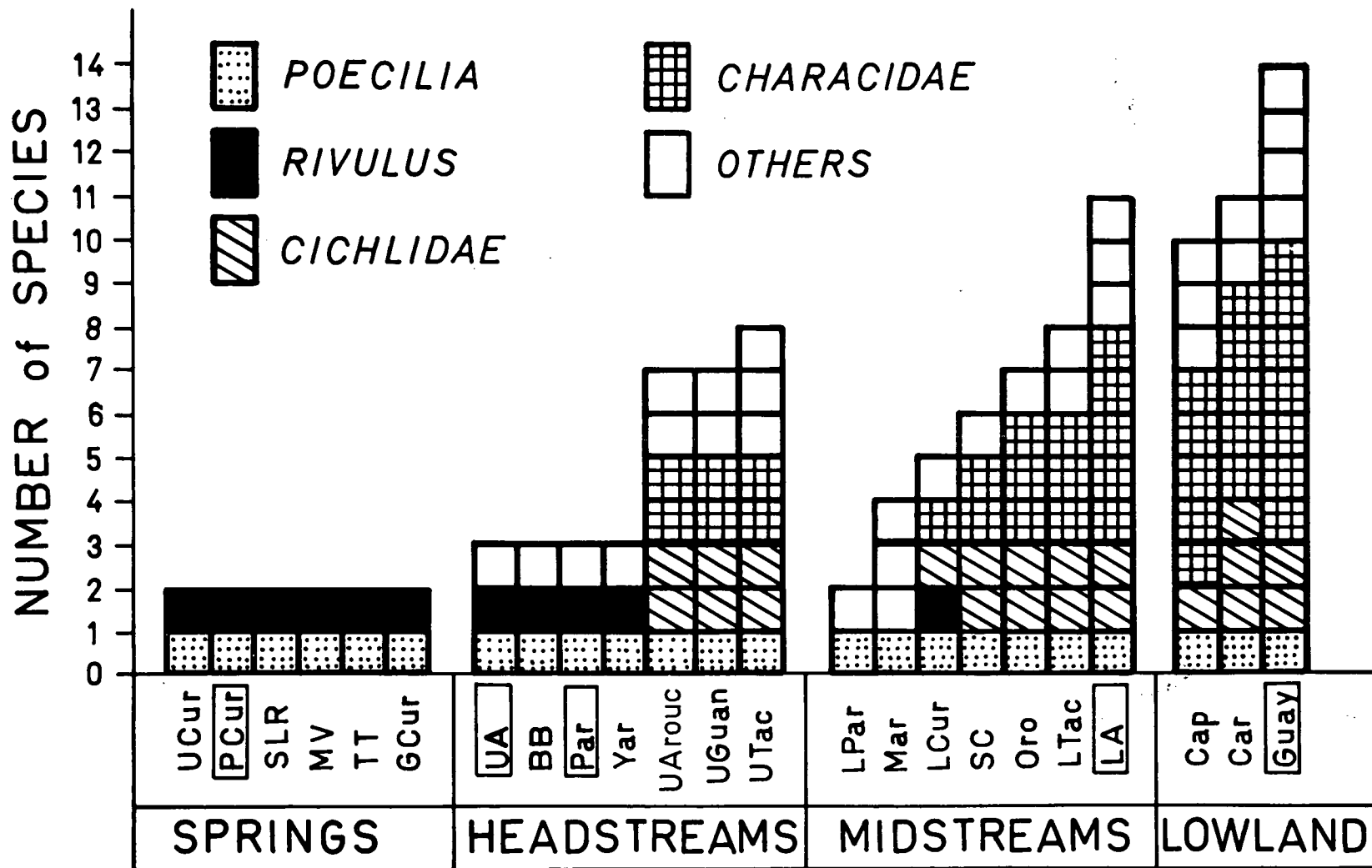
4 - very dense cover with virtually complete shading

^b0 - always clear

1 - turbid only after heavy rains

2 - turbid throughout year

FIGURE 2. The distribution of fish species in the Northern Range region. For a list of abbreviations, refer to Fig. 1. Streams outlined by a box are the principal ones that were investigated.



expected to occur are not present.

The Northern Range has served as the major barrier (no interconnecting freshwater streams) to characid and cichlid dispersal from the Caroni R. system to the north-flowing Paria, Marianne and Yarra Rivers. In streams flowing south off the Northern Range, waterfalls have prevented the access of characids and cichlids to Blue Basin and Upper Aripo Rivers.

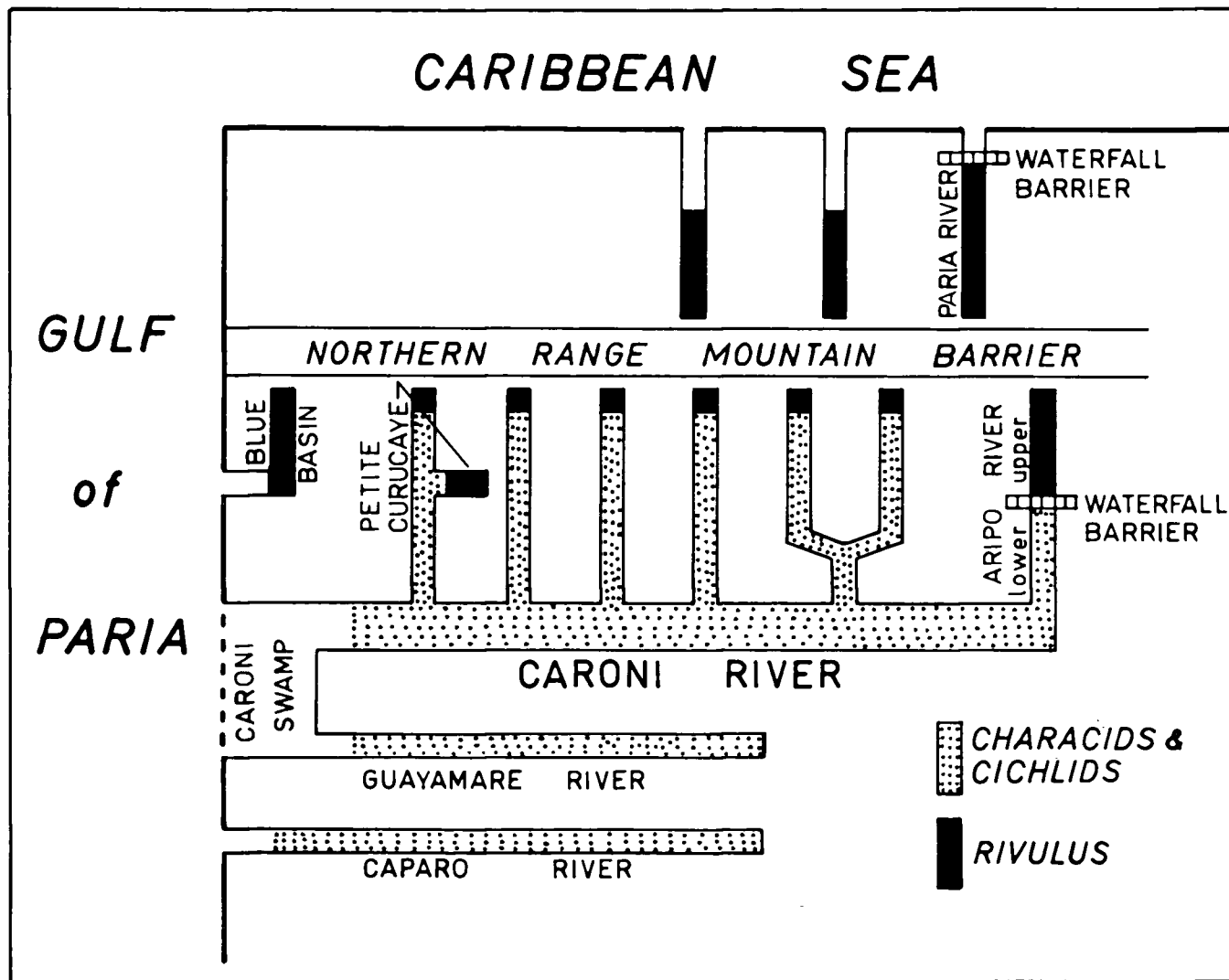
The situation at the Aripo R. is extremely interesting because a single 5 metre waterfall blocks the passage of several of the larger species to the upper regions. As I shall demonstrate below, some of these are predators of the guppy. Since guppies are distributed continuously along this river, it is possible to study populations exposed to different predation pressures over a very short distance. It is probable that the Upper Aripo R. has never contained characid or cichlid fish, hence the guppy populations have been "protected" from these species.

I have summarized the information on the distribution of the major species in schematic form (Fig. 3). Guppies are ubiquitous in this region.

Predators of the guppy

It would be expected a priori that a small abundant fish such as the guppy would be subject to considerable predation by large fish and possibly fish-eating birds,

FIGURE 3. A schematic representation of the distribution of the major fish species in the northwest corner of Trinidad. Major barriers to fish dispersal are shown. Small populations of Rivulus do occur in the vicinity of streams containing characids and cichlids but are confined to small pools or ditches.



mammals, and insects. Direct evidence of this can be obtained by observing attacks in nature or by recovering guppy remains in the digestive tract of the suspected predator.

It is often difficult to obtain extensive direct evidence of predation on a specific prey item (e.g. McPhail, 1969). Many vertebrate predators are notoriously secretive in their habits. In addition, if they are not specialist feeders, predation on a given prey species will fluctuate with the relative abundance and availability of alternate prey. Even in the so-called "stable" tropical environment, seasonal diet changes are well-known for freshwater fish (Lowe-McConnell, 1964, 1969b; Zaret and Rand, 1971).

Another problem is that piscivorous fish may go for long periods without feeding, and when they do feed, digestion may be rapid. Therefore a high proportion of the stomachs are empty or the food may be digested beyond recognition.

Finally, predators occupying a high trophic level are numerically not abundant; sample sizes are bound to be small.

I encountered all these problems in attempting to assess the relative impact of each suspected predator on the guppy populations.

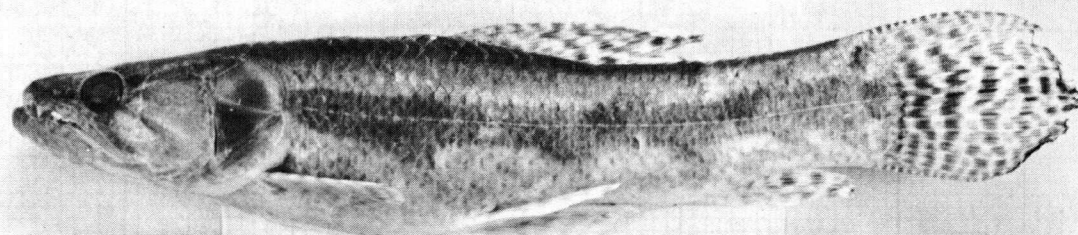
(a) Major predators. Haskins et al (1961) listed 5 species of fish predators of guppies in the Northern

Range streams (Figure 4) including 4 termed "serious": Astyanax bimaculatus, Aequidens latifrons (= A. pulcher), Crenicichla saxatilis (probably C. alta, Boeseman, 1960) and Hoplias malabaricus. The fifth predator, Rivulus hartii, was termed "less severe".

Few data were given for the basis of this classification. Presumably the body size, relative abundance, morphological adaptations, and wide distribution of the suspected predators were used as the main criteria, together with feeding tendencies in the laboratory. This would be a good first approximation though my field evidence shows that Rivulus is a more severe guppy predator than Aequidens. Aequidens was never observed to attack guppies and no guppies were recovered from the stomachs of juveniles or adults of this species (see Table 3 in Appendix for the analysis of stomach contents). In contrast 10% of all Rivulus sampled had guppy remains in their stomachs; attacks were also observed in nature.

Before the present study, little was known about the feeding habits of any of these 5 potential predators (all occur on the South American mainland; Hoplias and Astyanax have a very wide distribution). Thornhill et al (1966) drew a food web showing Hoplias and Crenicichla as predators (in the Maracas R., Trinidad) of Hemibrycon, Astyanax and the guppy (no other data given). Lowe-McConnell (1969a) and Knoppel, 1970 (cited in Roberts, 1972) found fish remains

FIGURE 4. Photograph of the major fish predators of the guppy.



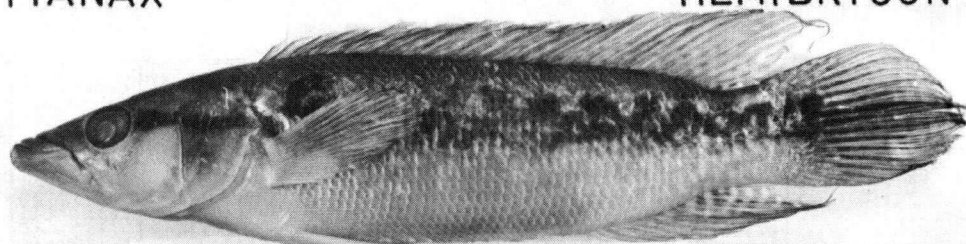
HOPLIAS



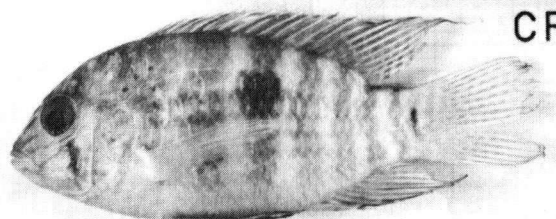
ASTYANAX



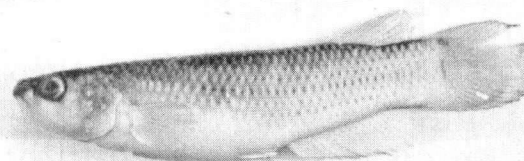
HEMIBRYCON



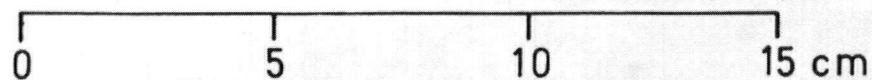
CRENICICHLA



AEQUIDENS



RIVULUS



in the stomachs of Hoplias from Guyana and Brazil respectively. Sterba (1962) stated that Crenicichla species, "are typical predatory fishes which lie in wait for prey after the manner of the European Pike, seize it with a lightning rush and swallow it head-first (p. 706)." Haskins et al (1961) claimed, "Crenicichla is a pike-like carnivore which is an extremely successful--and quite possibly a specialized Lebistes predator (p. 380)." (Lebistes reticulatus = Poecilia reticulata, Rosen and Bailey, 1963).

Although the majority of Crenicichla in my sample did have fish remains (guppies and other species) in their stomachs, I also recovered snails and insect larvae from several individuals, suggesting that this species is not a strict piscivore (as Haskins et al implied). Beebe (1925) also recovered invertebrates from Crenicichla stomachs in Guyana (shrimps from C. alta; ants and fish from C. lugubris). Though these data vitiate the thesis that Crenicichla is a specialized guppy predator, an example from one of my experiments underlines the potential of this predator under ideal conditions--63 adult male guppies were eaten by a sub-adult Crenicichla over a 48-hour period. Hoplias is also capable of a similar feat.

(b) Minor predators. There are a number of other fish species which may feed on guppies occasionally but considering either their small body size, limited distribution, or low numbers, I have assumed they have a

negligible effect on the present-day structure of the guppy populations in the Northern Range region. These include Roboides dayi, Corynopoma riisei, and Synbranchus marmoratus. I found all of these would eat guppies in small aquaria but only Synbranchus could attack and ingest adult guppies.

Terrestrial animals may also feed on guppies although Haskins et al (1961) inferred, on the basis of sighting only two herons and no kingfishers over 13 years of episodic observation, that this source of predation pressure was unimportant. Nevertheless on June 22, 1969 (1300 hrs) and June 29, 1969 (1600 hrs) I observed a pair of Little Green Kingfishers (Chloroceryle americana) sitting on an overhanging branch at the Paria R. collection site. Herklots (1961) reported this species at Paria and in the Caura Valley (Tacarigua R. area); Beebe (1952) listed it with the fauna of the Arima Valley. Belcher and Smooker (1936) reported finding scales of small fish in a nest (tunnel) of this species. They also found a nest of the Great Green Kingfisher (C. amazona) on the Madamas R. (east of Paria R.).

The Pygmy Kingfisher (C. aenae) has been seen by Junge and Mees (1961) near the mouth of a river (Burro R.?) at Maracas Bay and Herklots (1961) reported it for the Caura Valley and the Aripo road (Aripo R.) area. There have also been sightings of the Belted Kingfisher (Ceryle alcyon), a winter migrant from the north, in the region of the

Northern Range.

No details of feeding behavior or diet were given for any of these reports so the potential impact on guppy populations of kingfisher predation remains conjectural.

The fish-eating bat Noctilio leporinus occurs in Trinidad. Most observations of its natural feeding behavior have been made off the islands of the northwest coast where the bats were feeding on small marine fish. Worth (1967: 224) stated that Trinidad Noctilio use their elongated hind toes and claws to scoop up guppies as they skim over the water surface. Unfortunately no further details were given. Bloedel (1955) studied the hunting methods of N. leporinus in Panama. He demonstrated that one bat could capture 30 to 40 small fish (including the poeciliids Gambusia nicaraguensis and Mollienisia sphenops) per night from an experimental pool maintained at about four times the natural prey density. Random dipping and gaffing appeared to be the capture technique.

Suthers (1965) concluded that Noctilio could not use its sonar to locate fish below the water surface because the air-to-water acoustic impedance match is poor. However Suthers' demonstration that the bat could echolocate surface ripples suggests that in nature fish might be caught if they reveal their presence by disturbing the water surface. Certainly in smooth-flowing sections of stream and in pools, guppies (and probably Bloedel's fish) create surface

disturbances in localized areas that might aid the bat.

Undoubtedly Noctilio does eat guppies but no quantitative data are available.

There are a number of other potential predators of the guppy in the Northern Range but I have no information on their fish-feeding habits. Many of the larger reptiles and mammals are unfortunately nearing extinction on the island due to hunting pressure and habitat destruction. Thus their impact (if any) on the guppy populations is presumably steadily decreasing.

I now wish to return to the major predators and consider in more detail, some aspects of their distribution and natural history.

Distribution and natural history of the main predators

I have summarized the important features of the natural history of the main predators in Table 2. All the behavioral information is based on field observations except for the attack strategy of Aequidens and Hoplias. These observations were made on wild-caught fish in an aquarium and outdoor pool respectively. I have already stated that Aequidens is probably the least important of the 5 predators though I feel my data are not strong enough to exclude it from the predator category altogether. The nocturnal habits and low density (in clear water) of Hoplias make field observations of its feeding behavior difficult.

TABLE 2. Natural history of the main fish predators.

	CICHLIDAE		CHARACIDAE		CYPRINODONTIDAE
	Crenicichla	Aequidens	Hoplias	Astyanax	Rivulus
Maximum length (mm) ^a	200 (350)	148 (170)	392 (500)	116 (150)	105 (100)
Principal food	fish	benthic invert.	fish	aquatic insect larvae	terrestrial insects
Principal habitat	headstreams midstreams	headstreams midstreams	large turbid rivers	headstreams midstreams	springwater
Microhabitat	under logs, rocks; digs pits to hide	around logs, rocks, leaf detritus, shore vegetation	open bottom at night; under leaf detritus, in day	open midwater in pools; in riffles	ubiquitous in pools
Social organization	solitary or small groups of 2 to 5 individuals	loose aggre- gations or dispersed breeding pairs	solitary	tight schools	dispersed through- out pool
Activity period ^b	mainly diurnal	diurnal	mainly nocturnal	diurnal	nocturnal and diurnal
Attack strategy on guppies	ambush or stalk; occa- sional con- tinuous pursuit	continuous pursuit	stalk or continuous pursuit	increase in direction and velocity of normal swimming	continuous pursuit

^aLengths in brackets are the maxima given for the species by Sterba (1962); others are from the present Trinidad collections.

^bDiurnal includes periods of dim light before sunrise and after sunset.

Hoplias is primarily a fish of the larger turbid rivers; most of the individuals I caught came from the Caroni and Guayamare Rivers. Hoplias also occurs in lower numbers in all midstreams and headstreams not blocked by major waterfalls. It is absent from all streams flowing off the north face of the Northern Range.

Crenicichla shares the overall distribution of Hoplias but is less abundant than Hoplias in lowland turbid rivers. In my samples, the ratio of Crenicichla:Hoplias was approximately 10:1 for headstreams and midstreams but the reverse for lowland rivers. As a very rough estimate of the numbers of adult Crenicichla in a typical midstream, I counted 50 of these predators on a half-day excursion along 3 km of the Lower Aripo R.

Astyanax and Aequidens are distributed over the same range as the previous two species but some differences do occur. Astyanax is very prominent in lowland rivers and midstreams but tends to decrease in numbers in the headstreams. In the smaller mountain streams a morphologically similar characid of the genus Hemibrycon (probably H. taeniurus) appears to replace Astyanax. Hemibrycon is smaller than Astyanax but has similar feeding behavior (see Appendix, Table 3 for diet); one juvenile guppy was recovered from the stomach of a Hemibrycon. Generally, when I refer to Astyanax in this thesis, I am also including its smaller cousin.

Aequidens has a similar distribution to Crenicichla though I did not collect it in the Caroni or Guayamare Rivers.

I have stated before that Rivulus is found in springwaters or in larger streams devoid of characids and cichlids. It appears that its distribution is limited by the presence of the larger predators. In the laboratory the escape behavior of Rivulus is particularly ineffective in the face of the slow, stalking approaches of Hoplias or Crenicichla. Small populations of Rivulus do occur in the vicinity of the streams containing characid and cichlid predators but are confined to small pools or ditches. This may in fact be a smaller "lowland race" of R. hartii (J. S. Kenny, pers. comm.; see also Boeseman, 1960: 120).

The important point arising out of this discussion of the distribution of predators is that all populations of the guppy are not exposed to the same predators. Over most of its range in northern Trinidad, the guppy is exposed to either characid and cichlid predation or Rivulus predation. Since in the characid-cichlid situation there are predators of all sizes that can handle the smallest or largest guppy, I have termed the predation pressure in these environments as severe. Undoubtedly there are differences in predation intensities between some of these locations but I have insufficient data to warrant splitting this category.

Rivulus on the other hand is a small predator--its

major impact falls on small guppies. The severity of predation in this case must depend on the abundance and size distribution of the predator (assuming alternate prey to be in equal supply). Because Rivulus density is more easily assessed than the density of characids and cichlids, I have been able to distinguish between populations of guppies that are exposed to intense Rivulus predation and those suffering only moderate or weak predation. In some streams (e.g. Paria) Rivulus density is very low and few large individuals occur. Populations of guppies under these conditions must enjoy virtual freedom from aquatic predators.

For the detailed study of antipredator behavior (Chapter 6) I have selected 5 main populations of guppies to compare (Table 3). These populations were chosen because they encompass the range of predation pressures occurring in the Northern Range area. In addition they are geographically isolated or semi-isolated--gene exchange must be very small among them. For these reasons, they are an excellent test case for my hypothesis as outlined in Chapter 1.

Migration and dispersal of guppies

I have used the term "population" relatively loosely in this thesis--it should ideally be defined by rates of dispersal or gene migration. As Dobzhansky (1970: 240) points out, a definition of geographic isolation must.

TABLE 3. The principal fish predators occurring at the main study areas.

River	Crenicichla	Aequidens	Hoplias	Astyanax	Rivulus
Guayamare	+	- ^a	+	+	- ^b
Lower Aripo	+	+	+	+	- ^b
P. Curucaye	-	-	-	-	+
Upper Aripo	-	-	-	-	+
Paria	-	-	-	-	+

^aGuayamare does have the morphologically similar Cichlasoma bimaculatum.

^bRivulus is not absent from this geographic area but does not occur in the river itself, hence guppies are not exposed to it.

incorporate both absolute distance and the vagility of the organism.

Although I associate a "population" with a specific collection site, there is good evidence (at least for the 5 sites in Table 3) suggesting that I am dealing with quite discrete units. Haskins et al (1961) found that guppies show considerable site attachment; even during periods of sudden flash flooding the integrity of certain subpopulations remained intact. They also discovered that over a uniform stretch of the Arima R. the frequencies of certain colour patterns (known to be genetically controlled) could change abruptly. However in the same river they were able to establish permanently (at least to about 36 generations) a marker gene up to 10 km from the point of introduction. These observations suggest that subpopulations within a single river may be quite isolated but that some gene exchange must be occurring.

Site attachment was observed in an experiment (Chapter 4) conducted in the screened-off section of the Petite Curucaye R. (refer to Fig. 1, Appendix). The section was initially cleared of all guppies and then 50 male and 50 female guppies were collected from above and below the section and placed in Pool 2 (Pools 1, 3, and 4 were now "vacant"). On subsequent days I recounted all the fish, noted their distribution in the section, and returned them to Pool 2. Although I had expected considerable migration

to Pools 3 and 4, this was not observed (Table 4).

TABLE 4. Site attachment of guppies in the Petite Curucaye R.

	No. in Pool 2	No. outside Pool 2	Section total	Percent migration
June 14/69	84	2	86	2.3
16/69	80	5	85	5.9
18/69	79	2	81	2.5
20/69	76	3	79	3.8
22/69	79	0	79	0
26/69	70	5	75	6.7
July 1/69	71	2	73	2.7
				Mean 3.4

These data suggest that the fish in each pool may be a relatively isolated breeding unit (deme). Because approximately 50 pools separate the Petite Curucaye sampling site from the Santa Cruz R. (total distance of 1.6 km), gene flow is probably small between them. However, the contribution of juvenile fish to the total gene flow is unknown.

This condition is reminiscent of the unidimensional "stepping stone" model envisaged by Kimura and Weiss (1964). Even in the absence of major barriers to gene flow, isolation

by distance alone might facilitate the differentiation of local populations by genetic drift (Rohlf and Schnell, 1971) or differential selection (Ehrlich and Raven, 1969).

In addition to isolation by distance, two populations (Paria and Upper Aripo) are also isolated by major physical barriers (refer to Fig. 3). Gene exchange with the Caroni River system is probably rare though some fish undoubtedly get swept over the Aripo falls (the phenotype--size and colour--of males collected in pools immediately below the falls is intermediate between Upper Aripo and Lower Aripo; this region may be a narrow hybrid zone).

Dispersal of fish to these isolated populations must be fortuitous--terrestrial predators (bats, kingfishers) or tornadoes may accidentally drop live fish above a barrier. In recent Trinidad history at least one tornado was observed to carry small fish for a considerable distance.

Summary of Chapter 3

1. The limnology of Trinidad streams is relatively unknown.
2. In the Northern Range area the streams may be classified into four intergrading types differing in a number of abiotic and biotic parameters.
3. To a certain extent the distribution of fish species is correlated with this classification. Diversity

is greatest in the lowland rivers and least in the spring-waters.

4. The distribution of fish is also greatly influenced by the topography. The Northern Range has prevented certain species in the Caroni R. system from colonizing the rivers flowing northward into the Caribbean Sea. In some streams waterfalls have blocked the upward passage of fish.

5. It is concluded that the most important predators of the guppy are Hoplias malabaricus, Crenicichla alta, Astyanax bimaculatus, and Rivulus hartii. Guppies have been found in the stomachs of the last three species.

6. Fish-eating aerial predators occur in the regions under consideration but their impact on guppy populations is unknown.

7. The guppy is ubiquitous in its distribution in the rivers of northern Trinidad but there are important discontinuities in the distribution of predators. Some guppies are preyed upon by Rivulus; others suffer characid and cichlid predation.

8. Guppies show considerable site attachment to small portions of their home stream. Together with isolation by distance and physical barriers, this behavior probably reduces gene flow between populations and facilitates the differentiation of local populations.

CHAPTER 4

GEOGRAPHIC VARIATION IN THE SEX RATIO

Introduction

In this chapter I shall describe population differences in the sex ratio and attempt to find out why these differences occur.

The relationship between predation and the selective loss of male guppies from natural populations has received a great deal of attention from Haskins et al (1961). These workers used the sex ratio as indirect evidence for the hypothesis that male coloration is determined by the conflicting demands imposed by sexual selection and predation.

Since I originally set out to study the behavior of guppies of different colours (i.e. male colours--females of all populations are a cryptic grey-brown colour), it was important to critically assess this earlier study, firstly by extensive collecting in Trinidad and secondly, by repeating some of the experiments of Haskins et al, but examining in more detail the predator-prey behavioral interactions. I reasoned that if conspicuous coloration is indeed a liability with respect to predation, geographic differences in the sex ratio should be correlated with predation pressure (i.e. a greater loss of males relative

to females where predation is severe).

During this study of the sex ratio, additional geographic variation was discovered; this is described in the following two chapters.

Sex ratios of natural populations

Table 5 lists the sex ratios of the collections made in 1967 and 1969. The sex ratio is always expressed as the ratio of males to one female.

To determine the stability of the differences in the sex ratio, 13 of the sites sampled in 1967 were resampled in 1969. The collections were separated in time by approximately 6 generations (18-26 months). I assume that none of the adults sampled in 1969 had been born at the time of the 1967 collections (longevity in nature is probably less than 1 year, Haskins et al, 1961). The correlation between years in the sex ratio is significant (Fig. 5) suggesting that the sex ratio differences among stream sites are not simply random deviations pertaining to only one collection year, season, or sample. Note that Blue Basin was dropped from the analysis for reasons given elsewhere.

The sex ratio for all 1967 collections is 0.77; for 1969 it is 0.73. The pooled ratio for both years is 0.74 ($n = 14245$) and is significantly different from an expected 1:1 ratio ($\chi^2_{1 \text{ df}} = 305.47, p < .001$).

When the sex ratios for all streams are pooled (and

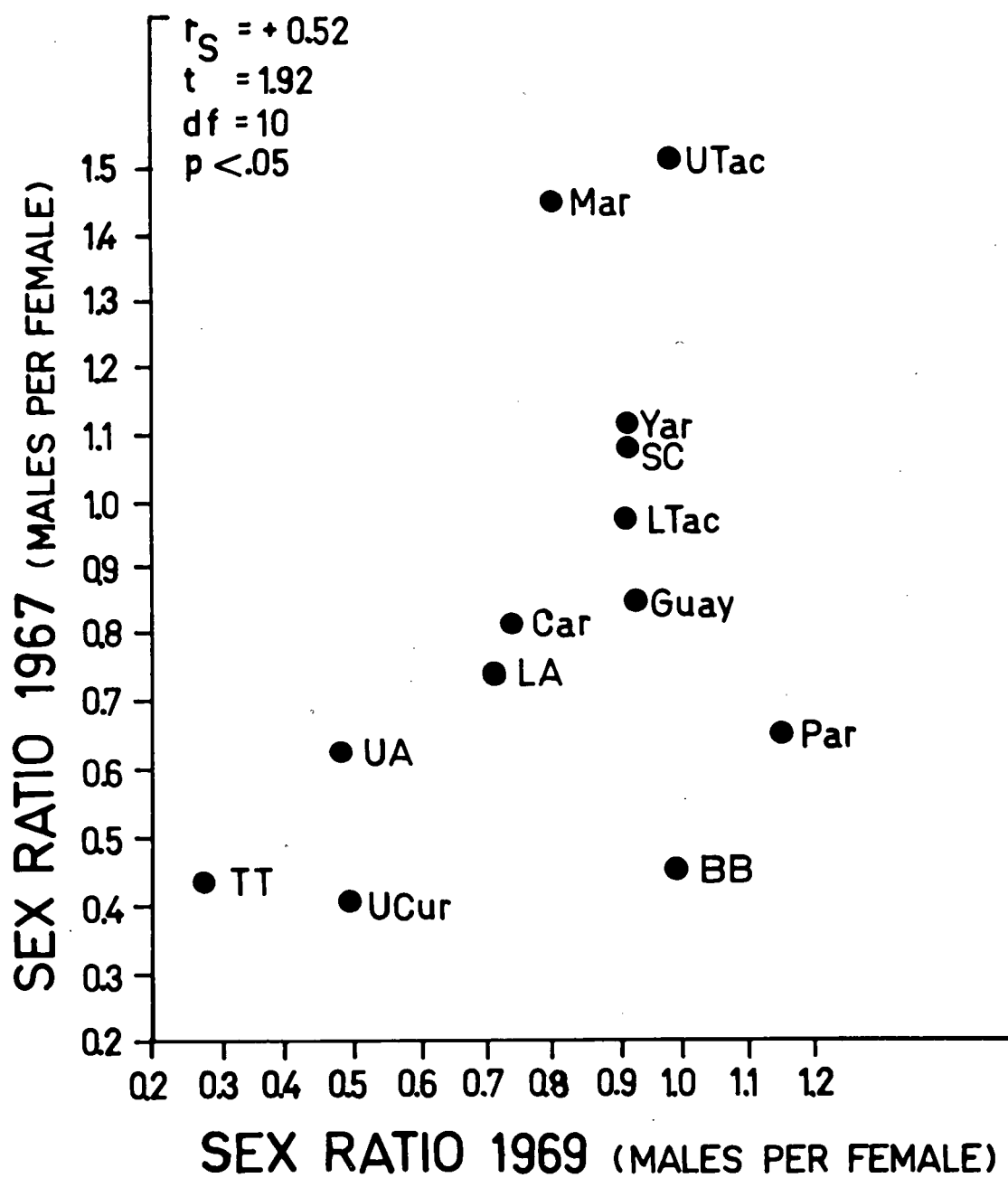
TABLE 5A. The sex ratios of natural populations (1967).

Source	Collection date	Males	Females	Immat.	N	Sex Ratio
<u>Springwater</u>						
Upper Curumpalo (1)	5/7/67	24	69	31	124	0.35
Upper Curumpalo (2)	16/8/67	53	118	16	187	0.45
total		77	187	47	311	0.41
Sierra Leone Rd. (1)	17/7/67	115	243	19	377	0.47
Sierra Leone Rd. (2)	16/8/67	210	335	446	991	0.63
total		325	578	465	1368	0.56
Maracas Village (1)	25/7/67	179	219	37	435	0.82
Maracas Village (2)	25/7/67	44	98	22	164	0.45
total		223	317	59	599	0.70
Tompire Tributary	22/7/67	50	114	10	174	0.44
Total Springs		675	1196	581	2452	0.56
Mean sex ratio (n=4): 0.53						
<u>Headstream</u>						
Upper Aripo (1)	6/7/67	70	115	79	264	0.61
Upper Aripo (2)	6/7/67	83	127	122	332	0.65
total		153	242	201	596	0.63
Upper Blue Basin	19/8/67	95	213	12	320	0.45
Upper Paria (1)	10/7/67	122	162	53	337	0.75
Upper Paria (2)	24/8/67	85	150	150	385	0.57
total		217	312	203	722	0.66
Yarra	10/7/67	65	57	3	125	1.14
Upper Tacarigua (1)	1/7/67	106	81	88	275	1.31
Upper Tacarigua (2)	17/8/67	261	161	234	656	1.62
total		367	242	322	931	1.52
Total Headstream		887	1066	741	2694	0.83
Mean sex ratio (n=5): 0.88						
<u>Midstream</u>						
Lower Paria	27/7/67	59	111	42	212	0.53
Marianne	24/7/67	97	67	6	170	1.45
Santa Cruz	5/7/67	112	102	203	417	1.10
Lower Tacarigua (1)	24/7/67	150	150	199	499	1.00
Lower Tacarigua (2)	7/8/67	268	276	849	1393	0.97
total		418	426	1048	1892	0.98
Lower Aripo (1)	12/7/67	81	118	116	315	0.69
Lower Aripo (2)	21/7/67	65	77	115	257	0.84
total		146	195	231	572	0.75
Quebrada d.H.	3/7/67	73	71	80	224	1.03
Total Midstream		905	972	1610	3487	0.93
Mean sex ratio (n=6): 0.97						
<u>Lowland Rivers</u>						
Guayamare	19/7/67	105	122	70	297	0.86
Caroni	12/7/67	92	112	351	555	0.82
Total Lowland Rivers		197	234	421	852	0.84
Mean sex ratio (n=2): 0.84						
1967 Total		2664	3468	3353	9485	0.77

TABLE 5B. The sex ratios of natural populations (1969).

Source	Collection date	Males	Females	Immat.	N	Sex ratio
<u>Springwater</u>						
Upper Curumpalo (1)	13/3/69	30	73	19	122	0.41
Upper Curumpalo (2)	13/3/69	40	70	8	118	0.57
total		70	143	27	240	0.49
Petite Curucaye (1)	13/6/69	84	251	129	464	0.34
Petite Curucaye (2)	26/6/69	43	134	85	262	0.32
total		127	385	214	726	0.33
Grande Curucaye	22/4/69	263	432	381	1076	0.61
Tompire Tributary	25/4/69	26	94	85	205	0.28
Total Springs		486	1054	707	2247	0.46
Mean sex ratio (n=4): 0.43						
<u>Headstream</u>						
Upper Aripo (1)	29/3/69	55	89	365	509	0.62
Upper Aripo (2)	13/5/69	86	164	87	337	0.52
Upper Aripo (3)	13/5/69	111	275	113	499	0.40
total		252	528	565	1345	0.48
Upper Blue Basin (1)	6/5/69	224	261	417	902	0.86
Upper Blue Basin (2)	29/5/69	194	164	305	663	1.18
total		418	425	722	1565	0.98
Upper Paria	18/4/69	292	256	212	760	1.14
Yarra	19/3/69	53	58	55	166	0.91
Upper Tacarigua	15/4/69	286	293	347	926	0.98
Upper Guanapo	2/5/69	331	449	585	1365	0.74
Upper Arouca	2/5/69	158	184	164	506	0.86
Total Headstreams		1790	2193	2650	6633	0.82
Mean sex ratio (n=7): 0.87						
<u>Midstream</u>						
Marianne	19/3/69	68	85	81	234	0.80
Santa Cruz	14/4/69	219	241	898	1358	0.91
Lower Tacarigua	5/4/69	317	350	528	1195	0.91
Lower Aripo	2/4/69	153	216	525	894	0.71
Oropuche	25/4/69	138	275	324	737	0.50
Total Midstreams		895	1167	2356	4418	0.77
Mean sex ratio (n=5): 0.77						
<u>Lowland Rivers</u>						
Guayamare	16/3/69	110	120	85	315	0.92
Caroni	7/4/69	84	114	58	256	0.74
Caparo	17/6/69	50	50	48	148	1.00
Total Lowland Rivers		244	284	191	719	0.86
Mean sex ratio (n=3): 0.89						
1969 Total		3415	4698	5904	14017	0.73
Total 1967 & 1969		6079	8166	9257	23502	0.74

FIGURE 5. A comparison of the sex ratio of 13 populations of guppies sampled in 1967 and resampled in 1969. For sample sizes refer to Table 5; for stream abbreviations refer to Fig. 1.



hence geographic differences obscured) these data corroborate the findings of Haskins et al (1961) for Trinidad guppies (sex ratio = 0.61, n = 3994) and mainland South American guppies (F. F. Bond collection, sex ratio = 0.61, n = 7825). It is clear that on the average there is a significant excess of females in natural populations. However as Table 5 reveals, all populations are not equally unbalanced and some (e.g. Lower Tacarigua) are remarkably well balanced. Therefore geographic variation may be a good clue to the mechanisms involved in the selective loss of males, a point apparently overlooked by Haskins et al (1961).

The presence of differences in the sex ratio raises the possibility that in some populations the secondary sex ratio is biased in favor of females because more female zygotes are produced (through abnormal meiosis or partial "Y" sperm inviability) or more males die before birth (guppies are ovoviviparous and the embryos spend the first few weeks of life inside the mother). I considered this possibility in my laboratory stocks.

Sex ratios of laboratory stocks

To determine if my stocks would produce an excess of females (for genetic or other reasons), I raised a large number of guppies to sexual maturity under controlled predator-free conditions. I used stocks derived from natural populations exhibiting both balanced (e.g.

Guayamare) and unbalanced (e.g. Petite Curucaye) sex ratios. The data were obtained by (a) sexing the adults in stocks that had been started with immature fish (number and length not recorded) and (b) running tests to check specifically for variation in the sex ratio. In (b) I raised only immature fish of less than 12 mm, a size well below the point where secondary sexual characters become evident.

The sex ratios obtained in these experiments (Table 6) did not differ significantly from the expected theoretical 1:1 ratio though there was an overall slight excess of females (sex ratio = 0.90, $n = 1049$). I conclude that there is no inherent tendency for any of the stocks to produce an uneven sex ratio. I assume this applies to the respective natural populations as well.

My results support the findings of Haskins et al (1961). They obtained a ratio of 0.94 when 3359 guppies from unspecified rivers in Trinidad were raised to maturity in the laboratory. However Breder and Coates (1932) working with a stock derived from a Jamaican population found the sex ratio at birth to be 0.5. Shoemaker (1944) repeated these experiments with an "Illinois" strain of domestic guppies and concluded that the ratio at birth was indeed 1:1 but that males were more susceptible to high temperatures and consequently died at a younger age; this resulted in very unbalanced sex ratios. Comfort (1961) and Haskins et al (1961) working with domestic and wild stocks

respectively, concluded that in the absence of predation there was no difference in longevity between the sexes.

Geodakyan and Kosobutskii (1972) have shown experimentally (in an unspecified laboratory stock of guppies) that, "the sex ratio of natality depends in turn on the sex ratio of the parent generation (p. 124)." It is not known how this feedback is accomplished but vision appears to play a role.

It is possible that the genetic control of sex is unstable and that racial or strain differences might account for the conflicting reports from various laboratories (Winge, 1934; Shoemaker, 1944; Kallman, 1965). For Trinidad populations the evidence for a stable XX-XY sex determination system is quite convincing.

Why then is there such striking geographic variation in the sex ratio? Selective predation on males may be part of the answer.

Relationship of predation to variation in the sex ratio

When the sex ratios of the field collections are related to the presence of fish predators (Table 7) no clear-cut correlation is discernible. In 1969 (the year for which the most complete data are available) the sex ratio ranged from 0.28 to 1.14 for streams containing Rivulus and 0.50 to 1.00 for streams with characids and cichlids.

TABLE 7. The relationship of the sex ratio to the presence of fish predators. Sex ratios are based on the 1969 collections.

RIVULUS		CHARACIDS AND CICHLIDS	
River	Sex ratio	River	Sex ratio
Tompire Trib.	0.28	Oropuche	0.50
Petite Curucaye	0.33	Lower Aripo	0.71
Upper Aripo	0.48	Upper Guanapo	0.74
Upper Curumpalo	0.49	Caroni	0.74
Grande Curucaye	0.61	Upper Arouca	0.86
Marianne	0.80	Santa Cruz	0.91
Yarra	0.91	Lower Tacarigua	0.91
Blue Basin	0.98	Guayamare	0.92
Paria	1.14	Upper Tacarigua	0.98
		Caparo	1.00

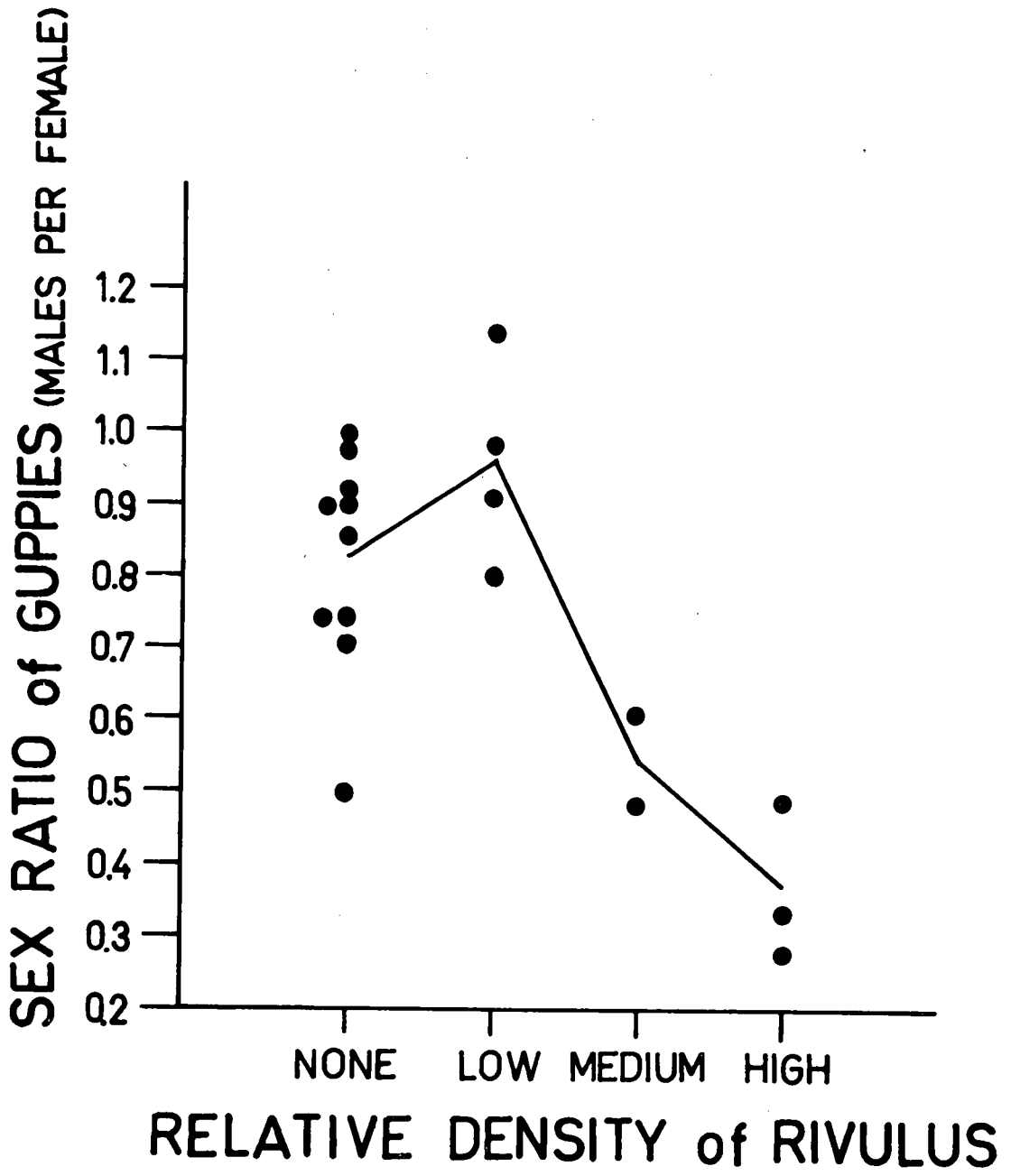
It is important to recall that in general, streams with Rivulus are smaller, cooler, and more heavily shaded, though where physical barriers prevent the immigration of characids and cichlids, Rivulus does occur under environmental conditions resembling some of the normal characid-cichlid habitats. Thus Upper Aripo, Paria, and Yarra (all with Rivulus) are physically similar streams to Upper Tacarigua and Upper Arouca (both with characids and cichlids).

The relationship between predation and the sex ratio becomes more evident when predation is assessed in terms of the abundance of predators rather than only their presence or absence (Fig. 6). This analysis could be made only for Rivulus habitats because the density estimates for characids and cichlids were inadequate. With the exception of Oropuche, the sex ratios of all populations exposed to characids and cichlids tended to be fairly balanced with only a slight preponderance of females. For these streams no relationship was found between the sex ratio and the turbidity, temperature, or size of the stream.

For streams containing Rivulus, the sex ratio was balanced when Rivulus density was low but unbalanced when high densities were observed (low density = less than 10 adults caught in 1 to 2 hours; high density = greater than 50 adults caught with same effort).

Rivulus density is also correlated to some extent

FIGURE 6. The relationship of the sex ratio to the relative density of Rivulus. Data are for the 1969 collections.



with the amount of cover and size of the stream. The most extreme sex ratios were observed in the Tompire Tributary and Petite Curucaye R.; these streams also had the smallest volume of water flow and the largest individuals, and densest populations of Rivulus.

There is some additional comparative evidence relating Rivulus abundance to sex ratios of guppies. At Blue Basin in 1969, there was a conspicuous absence of large (> 50 mm) Rivulus compared to the 1967 levels. Although twice the effort was spent searching for Rivulus in 1969 (collecting gear was identical and both collections were in the afternoon) very few Rivulus were caught or seen (Table 8). The same between-years comparison with another stream (Tompire Tributary) revealed no parallel differences suggesting the decrease was specific to Blue Basin.

TABLE 8. Rivulus abundance in two streams in 1967 and 1969. Twice the effort was spent collecting specimens at Blue Basin in 1969 compared to 1967. The effort both years at Tompire Tributary was about the same.

	BLUE BASIN			TOMPIRE TRIBUTARY		
	< 50 mm	> 50 mm	Total	< 50 mm	> 50 mm	Total
1967	21	40	61	36	18	54
1969	16	6	22	41	15	56
Total	37	46	83	77	33	110
$\chi^2_{1 \text{ df}} = 8.11, p < .005$			$\chi^2_{1 \text{ df}} = 0.29, p > .50$			

As well as a significant decrease in the abundance of large Rivulus at Blue Basin in 1969, there was also a striking change in the sex ratios of guppies (Table 9).

TABLE 9. Sex ratio of guppies at Blue Basin in 1967 and 1969.

	Males	Females	Total
1967 (<u>Rivulus</u> high)	95	213	308
1969 (<u>Rivulus</u> low)	418	425	843
Total	513	638	1151
$\chi^2_{1 \text{ df}} = 31.32, p < .001$			

The main ecological difference between years appeared to be the level of pollution through the use of this small mountain stream for the laundry of clothes. In 1967 there was little human activity in the headwater regions of Blue Basin, but in 1969 I observed several groups of women washing clothes in the stream. Pools a few hundred meters below the main laundry site had accumulations of soap foam and empty detergent bottles. A few meters below the laundry site I observed several dozen dead guppies and a dead Synbranchus marmoratus. Although I found no dead Rivulus, I put forth the hypothesis that the decrease in Rivulus numbers in 1969 was caused directly or

indirectly by pollution. Because the number of guppies was still very high (perhaps above 1967 levels), it is possible that they were not affected by the pollutants (except in the immediate vicinity of the pollution source). Therefore the reduction of the Rivulus stock could have facilitated the restoration of a balanced sex ratio in 1969.

(Of the 13 stream sites that were compared between 1967 and 1969 (Fig. 5) Blue Basin was the only one differing noticeably in its predator population. For this reason it was dropped from the calculation of the correlation coefficient.)

On the basis of the comparative field data presented above, the hypothesis that Rivulus selectively feeds on male guppies is certainly compelling; however it still relies on circumstantial evidence. Sex ratios might actually be determined by more subtle causal factors that also influence (or at least are correlated with) Rivulus densities. I attempted to get direct field evidence by examining stomach contents.

Unfortunately I found few guppies of recognizable sex in predator stomachs. Of 58 guppies recovered from Rivulus there were 10 adults or sub-adults (6 males, 1 female, 3 uncertain). All guppies recovered from Astyanax and Hemibrycon were juveniles (< 13 mm) but most of the fish remains in Crenicichla stomachs (mostly scales and vertebrae) appeared to be of larger fish, probably adult

guppies (sex unknown) and juvenile Aequidens or Astyanax. Therefore except for limited data on Rivulus (6 males versus 1 female recovered) I have no direct field evidence demonstrating males are more liable to natural predation.

For additional support of the hypothesis, I adopted an experimental approach. This seemed mandatory because even at this preliminary stage I had made several observations that were at variance with the "Haskins hypothesis":

(a) sex ratios are relatively balanced where the "more dangerous and specialized fish predators" (characids and cichlids) occur but are unbalanced (few males) where good numbers of Rivulus ("probably one of the less severe fish predators of adult Lebistes") are found.

(b) males are often greatly outnumbered by females yet are still conspicuously coloured (e.g. Upper Aripo males are characterized by large black markings that are often ocellated with iridescent pigments; however the sex ratio in 1967 = 0.63; 1969 = 0.48).

If sex ratios reflect the intensity of differential selection by predators against males on the basis of their colour, observations (a) and (b) are just the opposite of what would be expected. Furthermore, I found that (c) sex ratios are normally most unbalanced in small, clear streams (e.g. Petite Curucaye, Tompire Tributary), an environment where conspicuousness (based on either hue or contrast) is probably least important to visual predators.

These conflicting observations prompted experiments aimed at testing alternate hypotheses that might account for the observed variation in the sex ratio.

Size hypothesis

It is well known that male guppies cease growing at about the time of sexual maturity whereas females continue to grow until death. Consequently the sexes do not share the same size distribution. In the Petite Curucaye R. for example, 73% of the females exceed the size (in length) of the largest male. This is shown very clearly in the Appendix, Fig. 2). In the region of overlap in length, no sex difference in wet body weight was found.

Since many predators are size-selective in their feeding (Ivlev, 1961; Jackson, 1961; Brooks and Dodson, 1965; Mason, 1965; Galbraith, 1967; Brooks, 1968; LeBrasseur, 1969; Parker, 1971), all size classes of prey may not be equally vulnerable to a given predator. Size-selective predation might play an important role in the observed geographic trends in the sex ratio. For example I found no guppies above 23.0 mm in the stomachs of Rivulus. For the Petite Curucaye population, this upper limit coincides with the mean size of males ($22.9 \text{ mm} \pm 1.41 \text{ SD}$) but falls well below the mean size of females ($30.1 \text{ mm} \pm 5.63 \text{ SD}$). Therefore size selection alone might account for the unbalanced sex ratios observed in populations exposed to high densities of Rivulus.

Conversely, in populations liable to attack by large predators, the larger relative size of adult females might actually be a disadvantage if it is true that, "Predators prefer to devour victims of the largest possible size" (Ivlev, 1961: 85). In aquaria and large outdoor pools, half-grown Crenicichla and Hoplias easily overwhelm even the largest female guppy. However in nature, the selective impact of large piscivores on females may be balanced by (a) the greater conspicuousness of males based on colour (as suggested by Haskins et al, 1961) and (b) selection of males by small predators (Astyanax and very small Crenicichla or Hoplias).

A balanced sex ratio might therefore reflect either the absence of predation altogether, or predation stemming from a community of predators with a wide spectrum of sizes. Where "escape by growth" is possible (no large predators) the smaller sex should be least abundant. This model appears to fit the observed sex ratio variation in the Northern Range. (For experimental demonstration of size selection by predators, see Chapter 5.)

Behavior hypothesis

Casual observations of guppies in the field or laboratory quickly reveal that the sexes do not behave identically. A large proportion of the male's activity is spent in courtship while the female shows little sexual

behavior and appears to be rather wary.

It has been postulated for other vertebrates that sex differences in behavior might contribute to a greater mortality of one sex, often the male (Lack, 1954; Thompson, 1955, cited in Maher, 1970; Alm, 1959; Selander, 1965; Estes and Goddard, 1967; Holcomb and Twiest, 1970) and occasionally the female (Olson, 1965). Although direct field confirmation is generally lacking, it is suspected that in some species males are more vulnerable because they: (a) have larger home ranges, (b) are more aggressive, (c) are more active, (d) are less "timid", (e) defend territories, (f) have conspicuous courtship displays.

These male traits are presumed to aid the predator in either detecting the prey or facilitating a close approach to within attacking distance. These disadvantages are to a certain extent by-products of otherwise vital male functions. There remains the possibility that sex differences in escape behavior per se might be present.

I shall consider some of the possible sex differences in behavior later in this chapter.

Relative survival of the sexes in predation experiments

As a first step in understanding how predators might alter the sex ratio, I conducted several predation experiments with wild-caught and laboratory-reared guppies. To eliminate the "size hypothesis" as a possible explanation

for any survival differences, I controlled size as strictly as possible by matching male and female sizes.

A. Experiments with wild-caught fish

Experiment 4.1 Predation by Rivulus in the laboratory.

Methods

A large sample of guppies and Rivulus was collected from above and below the experimental section of Petite Curucaye R., brought to the laboratory and measured for size. The following day 2 female Rivulus (80 and 82 mm) were selected and placed in a 40 liter aquarium containing natural Petite Curucaye substrate (deep brown detritus and decaying citrus leaves). The tank was maintained under a natural photoperiod with natural sunrise and sunset.

Ten male and 10 female guppies (initial prey population, sexes same size) were then placed with the predators. The experimental design called for the addition of new fish (from the same collection as the initial 20 fish) to replace those eaten. To avoid large fluctuations in density and sex ratio (prey risk) these additions were made at frequent intervals during the day. The fish added were matched for size ($\bar{x}$ for males = 22.6 mm, range 20-26; $\bar{x}$ for females = 22.9, range 21-25).

Results

A total of 61 guppies were added (i.e. eaten by the predators) over a 12-day period (Table 10). This was

composed of 44 males and 17 females showing clearly that males were taken significantly more often ($\chi^2_1 \text{ df} = 11.08$, $p < .001$).

TABLE 10. Predation by Rivulus on male and female guppies in a 40 liter aquarium. Predator and prey were wild-caught.

Day	NUMBER ADDED	
	Males	Females
1	6	2
2	3	1
3	1	4
4	5	3
5	4	1
6	2	1
7	9	2
8	2	2
9	0	0
10	2	0
11	7	0
12	3	1
Total	44	17

Experiment 4.2 Predation by Rivulus in the field.

Methods

This experiment was conducted in the cleared section of the Petite Curucaye R. (refer to map and figures in the Appendix for details on this site and the absolute number of predators and prey present prior to clearing).

Following the removal of all guppies and over 90% of the Rivulus, I collected fish from above and below the

screened-off section. Eight Rivulus ($\bar{x}$ = 73.6 mm; range 63-89) were added to the section to supplement about 6 others of similar size that had not been caught in the original census. The guppies used in this experiment were matched for size ($\bar{x}$ for males = 21.5 mm, range 19-25; $\bar{x}$ for females = 21.7, range 19-25). Then 50 of each sex were placed in the section (released at 1600 hr in Pool 2).

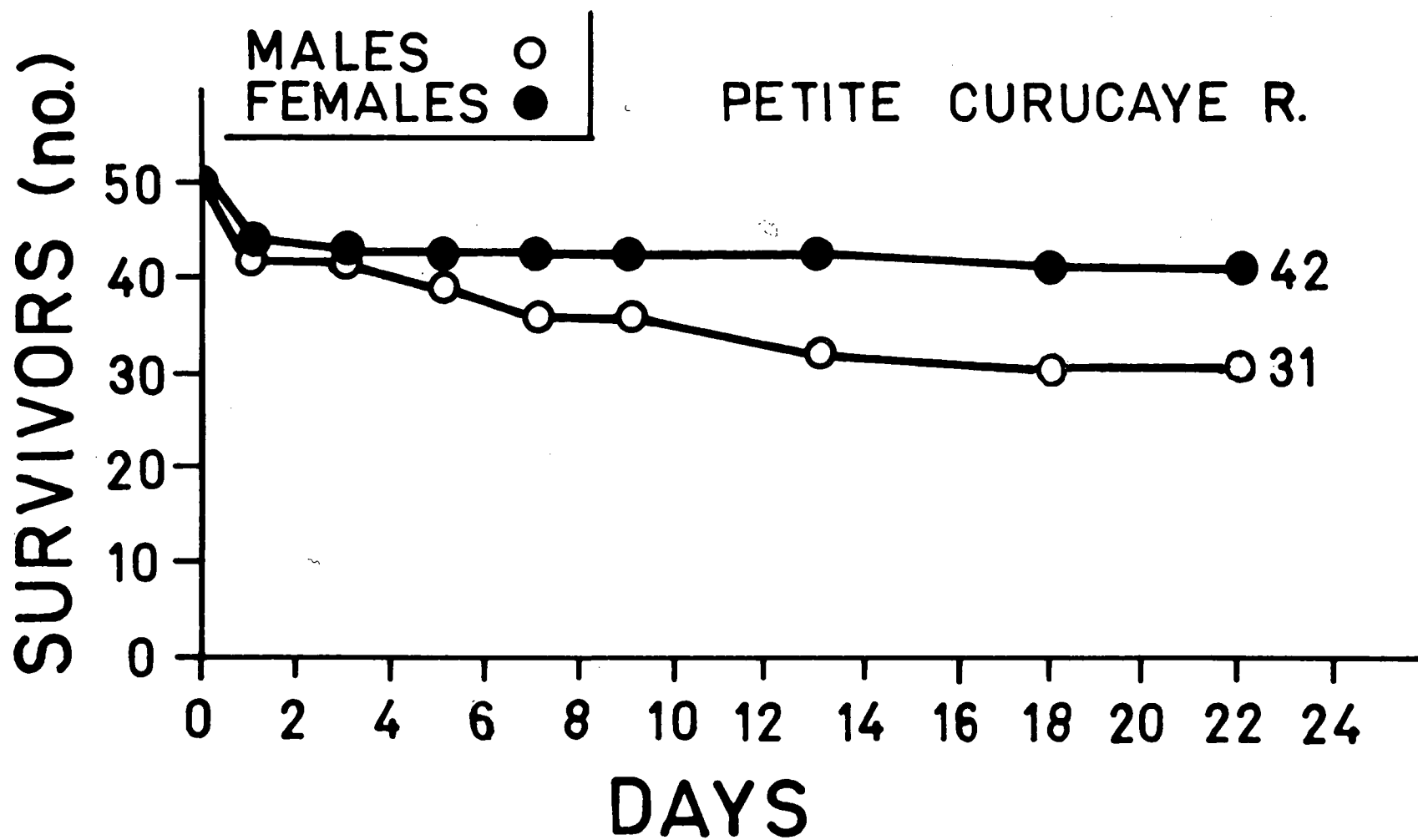
Thus a "synthetic" population of prey with a 1:1 sex ratio was created. I monitored the sex ratio at frequent intervals for 22 days. Because the natural sex ratio of this stream is 0.33, I predicted that males would disappear more quickly.

Results

By Day 22 the sex ratio of the section had stabilised at 0.74 (Fig. 7). The number of fish disappearing (8 females and 19 males, $\chi^2_{1 \text{ df}} = 3.70$) just fails to reach the .05 level of probability.

Since it was not feasible to run a control section without predators, it is not certain that Rivulus predation accounted for all the losses. However a large group of guppies collected at Petite Curucaye at about the same time as the experimental sample and placed in large, predator-free outdoor pools showed no mortality. (Handling mortality caused by capture, transport, and measurement is normally about 1%.) Also there was no evidence that the guppies could escape from the section under or through the fine

FIGURE 7. Relative survival of male and female guppies
in the experimental section of the Petite
Curucaye River.



screen. As I indicated in Chapter 3, about 95% of the guppies remained in Pool 2 during the course of the experiment.

Since guppies had been found previously in the stomachs of Rivulus from this section and numerous approaches and occasional attacks were observed during the experiment, I conclude that the greater mortality of males was caused by selective predation. This field test then confirms the laboratory results (Exp. 4.1).

B. Experiments with laboratory stocks

In addition to these 2 experiments with wild fish, I exposed laboratory stocks (i.e. predator-naive) to Crenicichla or Rivulus. The methods used in these experiments shall be outlined in considerable detail because the same procedure was used in experiments in subsequent chapters. I shall refer to it as the "standard survival test".

Procedure for standard survival test

In all tests of this type a 200 liter glass and stainless steel frame aquarium was used (dimensions: L = 92 cm, W = 48 cm, D = 46.5 cm). Four identical aquaria were set up in the same room so that several tests could be conducted simultaneously. These tanks were covered on the back and sides with black plastic; the front pane was covered with a moveable black curtain. The depth

was kept at 30 cm to allow space beneath the cover glass for guppies to "surface jump" when attacked. A large flower pot was placed in one side of the tank to serve as a refuge for the predator(s). The substrate consisted of light brown sand mixed with fine pebbles and some larger stones to 10 cm in diameter. The tank was devoid of aquatic plants since no submerged vascular plants occur in most Northern Range streams (sedges and grasses near the shore do get covered periodically in the wet season).

Normally 50 guppies of type "a" and 50 of type "b" were selected at random from stock aquaria (e.g. a = males, b = females). These were measured 24 to 48 hours before the start of the experiment. The predator was usually the resident in the test tank and the guppies were placed with it (after a 1 hour period floating in a screen basket), normally in the late afternoon. This was termed "Day 0". Survival was measured simply by removing the guppies with a large net and counting the sexes, sizes, stocks, etc. in question. They were immediately returned to the test tank. This procedure did not appear to disturb either predator or prey.

When approximately 50% of the population had been eaten, all remaining fish were measured again to test for possible size selectivity by the predator (Chapter 5, Fig. 16). When densities reached less than 15, it was possible to count fish without removing them. The test was

terminated when numbers either reached or approached zero.

When the predator was Crenicichla, only one individual was used for 3 reasons: (a) Crenicichla is a very aggressive species and dominance by one fish could lead to injury or death of subdominant fish, (b) field observations revealed that adult Crenicichla are solitary predators, and (c) Crenicichla of even juvenile size are able to consume a large number of adult guppies; to detect differences in survival with an initial density of 100 guppies per 135 liters of water, overall mortality should not exceed a mean rate of about 10 per day.

When the predator was Rivulus, usually more than one individual was used so that the predation rates would be comparable to that of a single Crenicichla. Also, social feeding appears to be part of the natural behavioral repertoire of this species; at Petite Curucaye several predators were observed to pursue a single guppy.

In all experiments predators were fed an excess of Tubifex the day before the start of the experiment; this was to prevent a predator "overkill" on Day 1 as well as to control the initial hunger level (motivational state). During the experiment prey were fed fine dried food which the predators did not eat.

Four out of the 5 Crenicichla used in experiments at Vancouver were wild-caught as juveniles and raised on a variety of foods (see Chapter 2). Rivulus were all

aquarium bred. Both species had had considerable prior experience feeding on male and female guppies. Because I was unsuccessful in breeding Crenicichla, I had to use the same individuals in many of the experiments. As a precaution against bias resulting from individual differences, I have retained their individual identity in presenting experimental results (Table 11). With Rivulus individual differences are less likely to bias results because in all standard survival tests, the results are the collective action of 6 predators. Also several "6-man teams" were used.

TABLE 11. The origin, sex, and size of individual Crenicichla used in experiments at Vancouver.

Source	Code no.	Sex	Home tank no.	Total length (mm)	
				when caught	max. reached
Lower Aripo	C-1	m	T51	75-100	205
Lower Aripo	C-2	f	T52	75-100	185
Lower Aripo	C-3	f	T53	75-100	173
Lower Aripo	C-4	m	T52	75-100	187
Vancouver supplier	C-5	m	15A	--	250

Experiment 4.3 Predation by Crenicichla on Caparo stock.

Methods

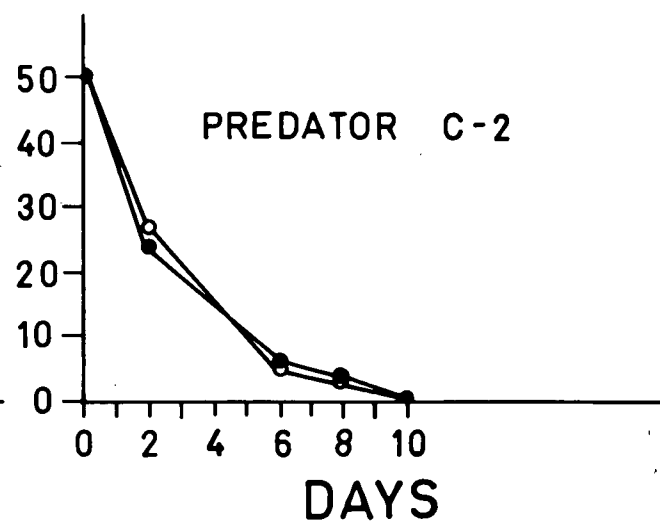
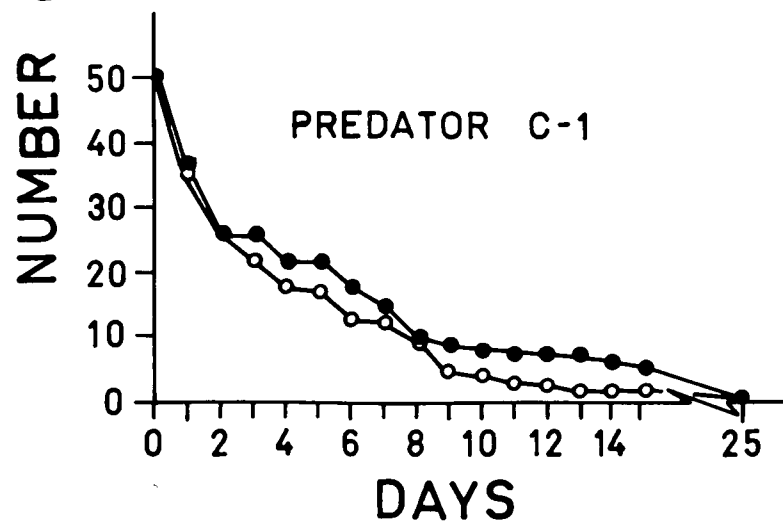
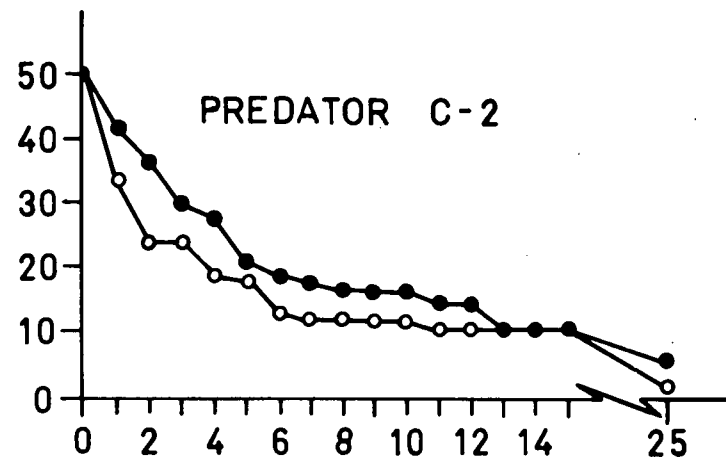
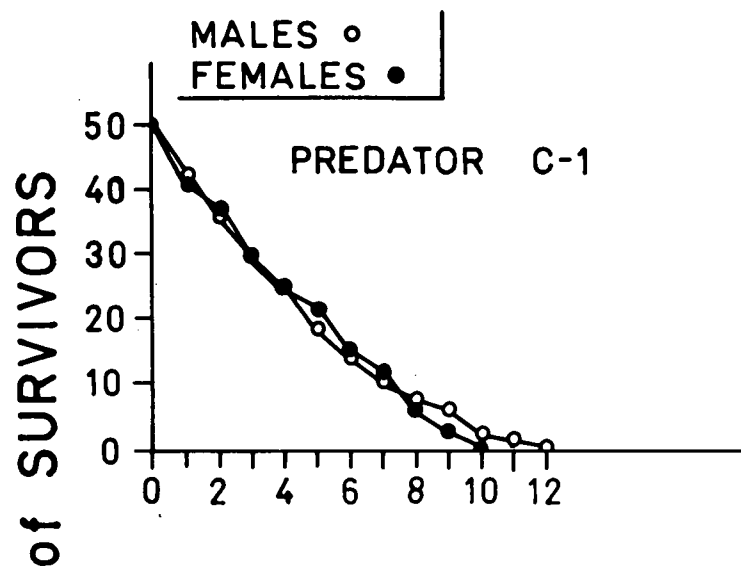
Using the standard survival test, I exposed Caparo stock males and females to Crenicichla C-1 and C-2 (2 replicates per predator). For one replicate with each predator the size of the sexes was matched as closely as possible and for the other a random sample of adults was taken (i.e. a sample with greater variance, females generally larger than males). The mean sizes and size ranges used are given below (refer also to Fig. 8).

<u>Test</u>	<u>Sex</u>	<u>Mean length (mm)</u>	<u>Range</u>
a	m	19.8	16.0 - 25.0
	f	21.8	17.0 - 32.0
b	m	20.3	16.5 - 24.0
	f	20.1	17.0 - 24.0
c	m	20.2	18.0 - 23.5
	f	20.1	16.0 - 24.5
d	m	19.5	15.5 - 24.0
	f	24.2	18.5 - 32.5

Results

It is clear (Fig. 8) that females had little if any advantage over males under my experimental conditions. This is in sharp contrast to the results obtained in Exp. 4.1 and 4.2 with wild-caught Rivulus and Petite Curucaye guppies. Even more surprising is the difference between these results and those published by Haskins et al (1961). In tests in aquaria and outdoor pools they showed a dramatic sex

FIGURE 8. Relative survival of male and female Caparo stock guppies exposed to Crenicichla.



difference in survival rate--in 2 tests with Crenicichla males were over 5 times more vulnerable than females.

Unfortunately too few details were given as to the methods used, therefore I am unable to formulate a plausible reason for the difference in our results.

Experiment 4.4 Predation by Rivulus on Caparo and Petite Curucaye stock.

Methods

The methods used were the same as in Exp. 4.3 except that body size was rigidly controlled in all 4 replicates. Two replicates were performed with Caparo stocks and 2 with Petite Curucaye stock. All 4 tests were conducted with the same 6 Rivulus (75, 75, 85, 73, 73, 85 mm) in the same aquarium. Sizes of prey are given below.

<u>Test</u>	<u>Sex</u>	<u>Mean length (mm)</u>	<u>Range</u>
a	m	19.9	17.0 - 23.0
	f	19.9	16.0 - 23.5
b	m	19.2	15.0 - 22.5
	f	19.8	16.5 - 23.0
c	m	20.4	18.5 - 23.5
	f	20.4	17.5 - 23.5
d	m	20.1	17.5 - 23.0
	f	20.5	17.5 - 23.5

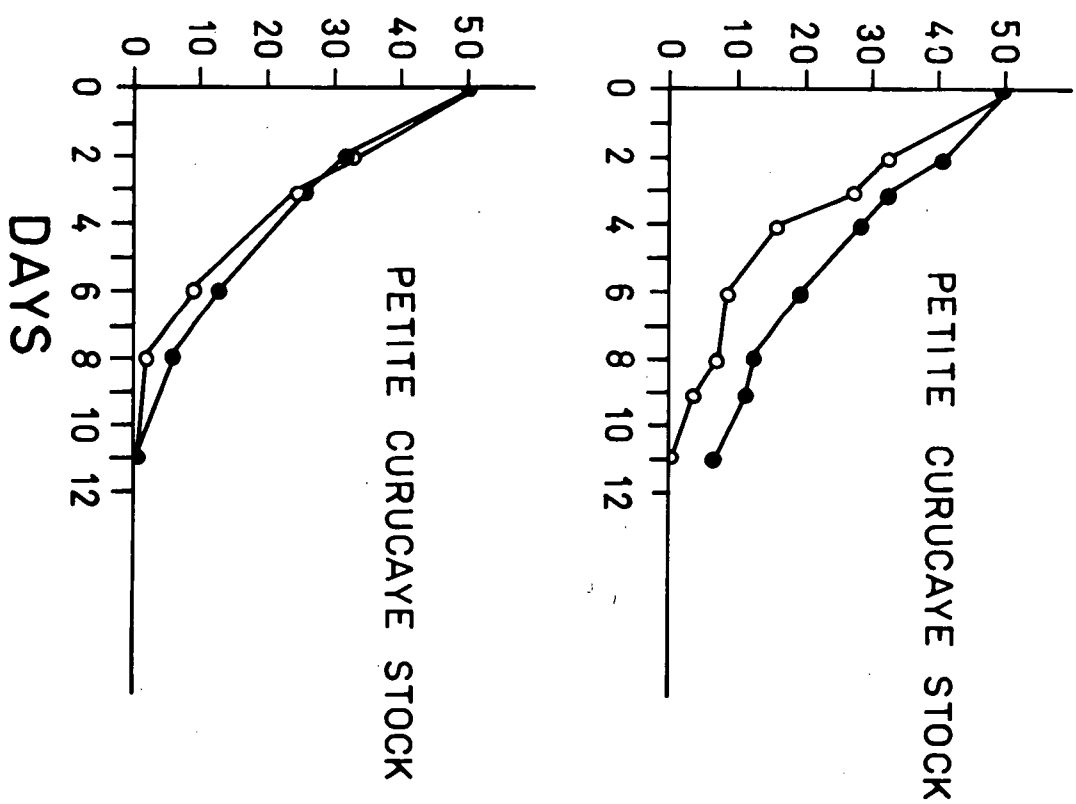
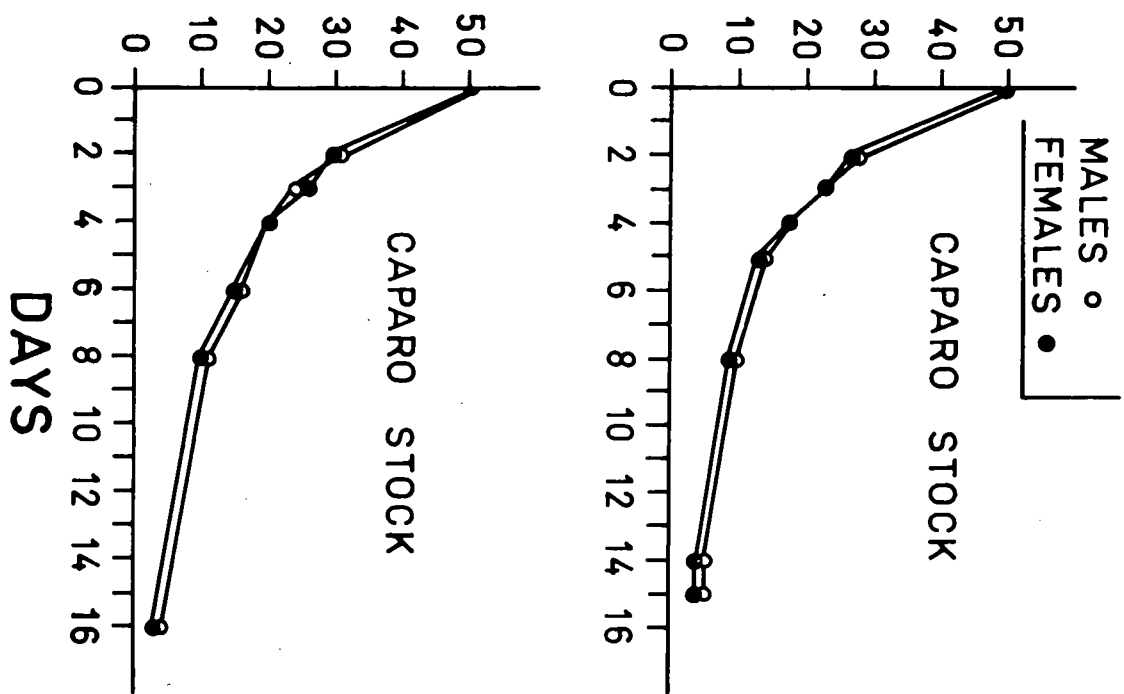
Results

Again it is obvious that males and females were taken approximately equally by the predators (Fig. 9). In

71a

FIGURE 9. Relative survival of male and female guppies
of two stocks exposed to Rivulus.

NUMBER of SURVIVORS



one of the replicates with Petite Curucaye stock, the females showed better survival but this could have been due to a chance deviation (no. eaten by Day 4 = 21 females, 34 males; expected no. by non-selective predation = 27.5, $\chi^2_{1 \text{ df}} = 2.62$, $.25 > p > .10$).

Discussion of survival experiments

Taken together, the results of the previous 4 experiments illustrate that when body size is a controlled variable, the male sex is not always more vulnerable to predators. However Exp. 4.1 and 4.2 show that under certain circumstances males are taken more often. Several reasons could be suggested for these conflicting results:

1. The difference could have been an artifact caused by a difference in experimental methods.

- (a) difference in apparatus, light regime, micro-environment, etc.

- (b) difference in measurement of relative vulnerability--Exp. 4.1 by the "replacement method" and Exp. 4.3 and 4.4 by the "standard survival test".

- (c) difference in body size of prey--fish used in Exp. 4.1 were larger.

- (d) difference in experience--fish used in Exp. 4.1 and 4.2 were wild-caught; those in Exp. 4.3 and 4.4 were laboratory-reared.

2. Possibly there is a stock difference in the

degree of sexual dimorphism (colour or behavior). Relative to their respective females, Caparo males may be less vulnerable than Petite Curucaye males.

3. Subtle sex differences might be inconsequential to a large piscivore such as Crenicichla (Exp. 4.3).

Several of these points require further elaboration.

Light is an important ecological factor in most predator-prey systems. My own observations of Rivulus in nature revealed that the greatest proportion of feeding activity occurred under dim or dark light conditions. This was also observed in Exp. 4.1: of the 61 guppies added, only 12 were added between 900 and 1800 hr (sunrise in Trinidad approximately 600 hrs; sunset, 1800 hrs). Hyatt (unpubl.) has also observed that Rivulus has remarkable scotopic vision. Since in Exp. 4.4 there was a more sudden transition from dark to light (and vice-versa) than in nature (Exp. 4.2) or Exp. 4.1, Rivulus may have been exhibiting an atypical feeding pattern. It is possible that in nature the difference in conspicuousness between the sexes is most accentuated under dim illumination; in twilight the iridescent pigments of the male may already be reflecting light while the female is still relatively invisible. Under these conditions a visual predator would probably attack and capture more males.

I have no evidence that the "replacement method"

used in Exp. 4.1 could have biased the test in favour of females. From slight differences in the size and morphology of the fish, I was able to rule out the possibility that the newly-added fish were more vulnerable than the original residents (as Beyerle and Williams (1968) found in a study of food selectivity by pike).

The problem of prey size differences may be relevant because in Exp. 4.1 the mean size of guppies was closer to the upper handling limit for Rivulus. If a sex difference in escape behavior does exist, it is possible that it only becomes really important as the predator encounters greater difficulty in subduing the prey. As I suggested above, this condition may not be applicable to large Crenicichla.

Finally there is the question of prior experience. It is possible that the wild-caught Rivulus used in Exp. 4.1 and 4.2 had been conditioned to attack male guppies. This conditioning might occur if, (a) males are easier to capture than females of a similar size, or (b) Rivulus has difficulty distinguishing female guppies from juveniles of its own species (immature Rivulus and female guppies are roughly the same size, shape, and colour, but differ considerably in their locomotory behavior; inexperienced aquarists often mistake one species for the other).

Alternative (b) implies that female guppies enjoy an advantage through a fortuitous form of Batesian mimicry. Certainly there is no evidence of cannibalism in natural

populations of Rivulus--I examined over 300 stomachs of this species and found no trace of predation on conspecifics. Attacks on conspecifics have been observed in the laboratory but often the fish were egested unharmed, suggesting predation may be inhibited by a taste factor (if species recognition fails at the visual level).

I made a preliminary attempt to test this mimicry hypothesis. In 2 pilot experiments concerned primarily with the avoidance conditioning of guppies to adult Rivulus, I purposely added 15 juvenile Rivulus of 16 to 37 mm (size of female guppies) to act as potential negative reinforcement against attacks on female guppies (initial prey population 50 males and 50 females). However in both tests the survival rate of males and females was virtually identical suggesting the presence of "models" did not decrease the relative rate of predation on females. Possibly I used too few "models" in these experiments (cf. Brower, 1960).

Tests of the behavior hypothesis

The results of the previous experiments have shown that under certain circumstances male guppies are more vulnerable to Rivulus predation. This suggests that (a) Rivulus hunts non-randomly or (b) Rivulus hunts randomly but males are less able to avoid capture. In the absence of direct observations of predatory behavior, it is impossible to decide which mechanism is operative. I shall

now test the assumption (implicit in the study of Haskins et al, 1961) that there are no sexual differences in escape behavior.

Experiment 4.5 Predator behavior of Rivulus and escape behavior of male and female guppies.

Methods

Six Rivulus (60 to 70 mm) with prior experience in feeding on male and female guppies were placed in a 40 liter aquarium. After starving the predators for 24 hours (approximately the time required to completely digest a meal of 3 guppies), 10 male and 10 female guppies were added to this aquarium. They were first placed in a screen basket for 20 min and then released. These guppies were all first and second generation fish of the Paramaribo stock. They had never been exposed to predators before and were all matched closely for size (range--21 to 24 mm).

As soon as the fish were released, the predatory behavior of Rivulus was recorded on a Rustrak event recorder for 60 min. After this the survivors were removed and discarded. This procedure was replicated 10 times with the same 6 Rivulus. The following behavior patterns were recorded:

(a) Approach--the movement of a predator towards a guppy. This may vary in speed from a slow movement to a rapid dart. No attempt was made to separate these. The

pursuit of a guppy by Rivulus can be separated into phases of acceleration and deceleration--each renewed acceleration received a single "approach" score.

(b) Attack--a successful approach. The predator makes contact with the prey.

(c) Capture--a successful attack. The prey is normally swallowed immediately (i.e. capture = kill).

Results

The results (Fig. 10) reveal that females and not males were actually approached and attacked more often (Wilcoxon test; approach: $T = 4$, $n = 10$, $p < .02$; attack: $T = 4$, $n = 8$, $p < .05$). The probability levels are for a two-tailed test (no prediction made).

The most interesting result however, was that more males than females were killed indicating it took less effort to capture males successfully (kills for matched pairs: $T = 0$, $n = 9$, $p < .01$; total kill: $\chi^2_{1 \text{ df}} = 7.51$, $.01 > p > .005$).

When the predatory attempts are expressed as a ratio per single kill (predatory "effort"), there is no overlap between the sexes for any of the 10 replicates (Fig. 11). Because the prey were all of similar length and weight, it is clear that females were more adept at avoiding capture. Even when contacted by the predator they were able to dart away to safety more often. Males however, were usually captured in 3 or less attacks under

FIGURE 10. Predatory behavior of Rivulus exposed to male and female guppies of the Paramaribo stock.

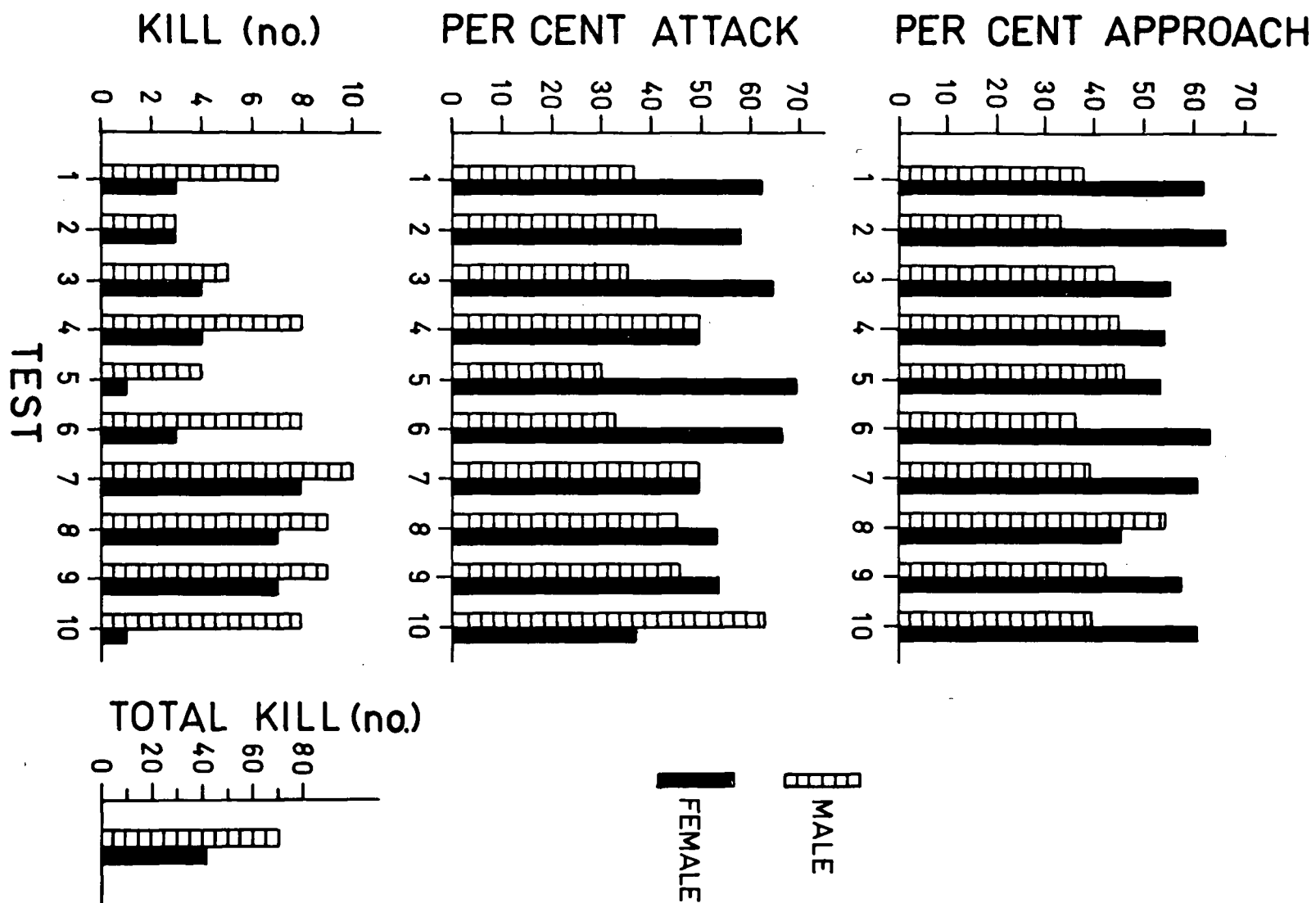
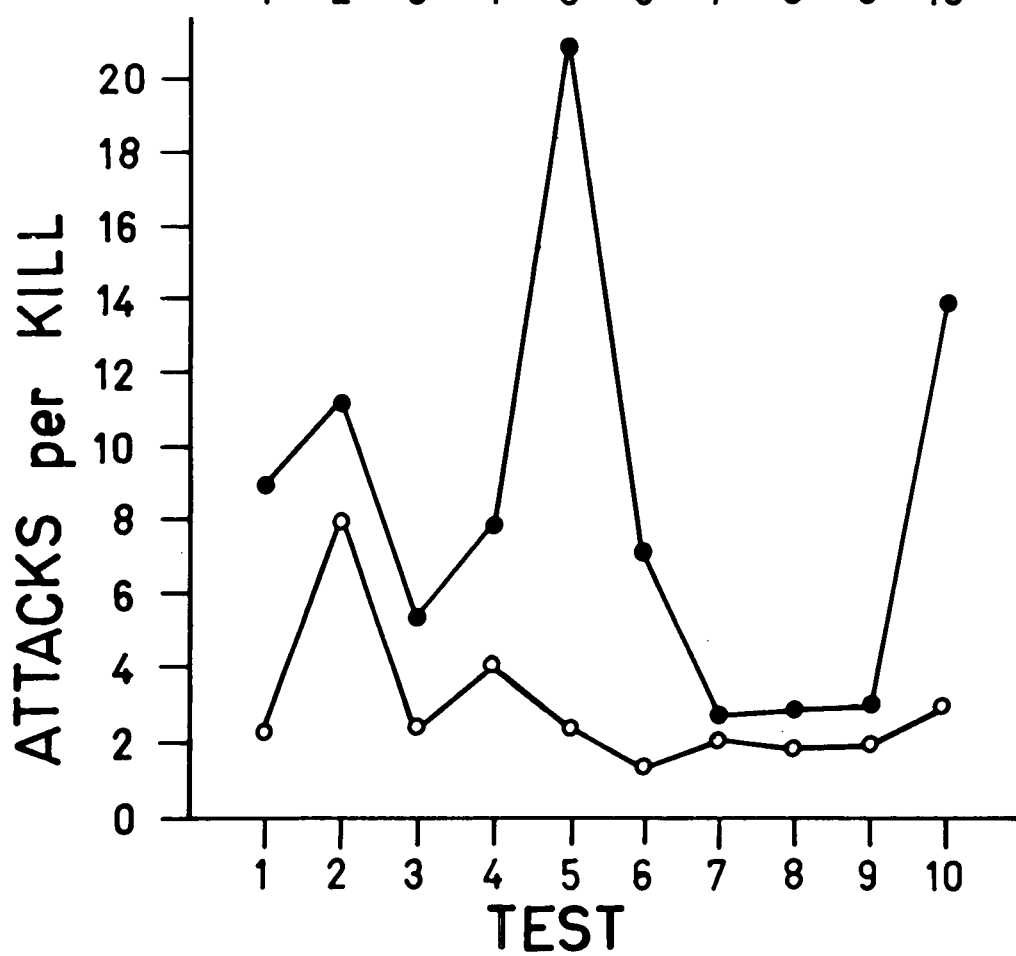
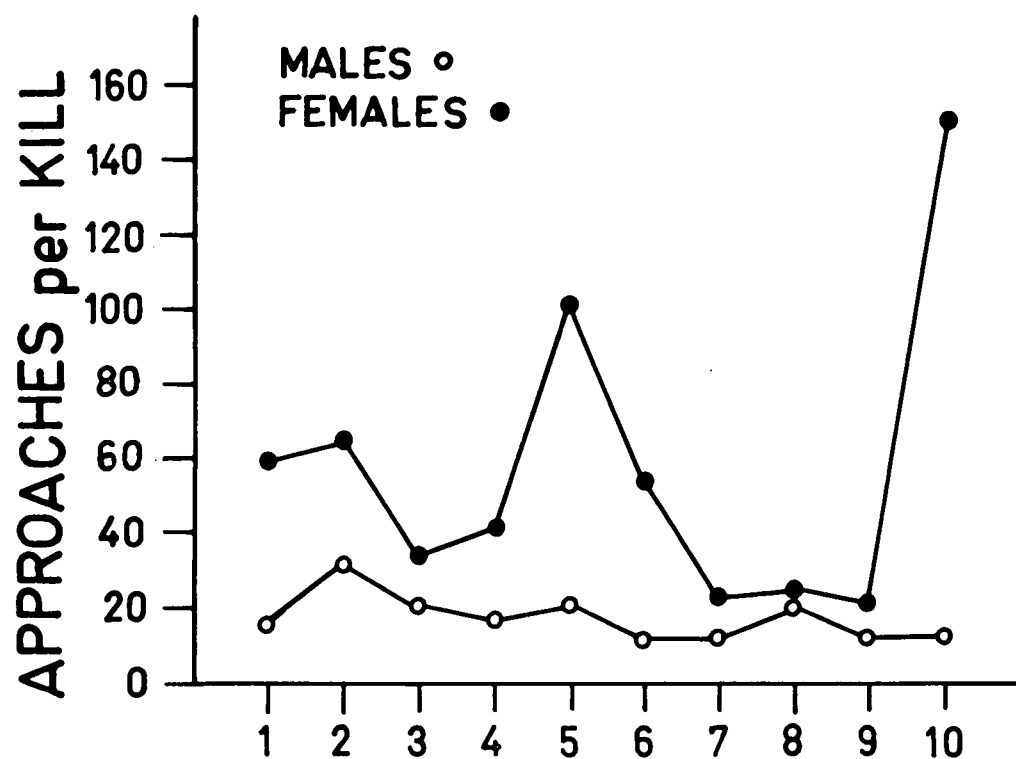


FIGURE 11. The ratio of predation attempts to successful captures for Rivulus feeding on male and female guppies.



these experimental conditions.

The greater number of approaches directed towards females should not be taken to mean that the predators preferred females. To some extent this greater score results from my inability to distinguish between an initial approach starting a bout of chasing (i.e. a choice of prey) and approaches within a bout.

In conclusion this experiment demonstrates that there are sex differences in escape behavior, at least for Paramaribo stock guppies.

Evidence from other experiments for sex differences in antipredator behavior

Although Exp. 4.5 demonstrates clearly that females are more adept at avoiding Rivulus predation, the nature of this sex difference was observed on only a qualitative level. Because of the flurry of activity during the experiment, I concentrated on the behavior of the predators; it was not feasible to simultaneously measure the escape behavior of the guppies.

However, in experiments designed to measure behavioral differences among stocks (Chapter 6), I often used males and females and quantified their behavior separately. Without going into detail here, I found no sex difference in (a) the form and frequency of escape motor patterns, (b) the tendency to swim at a particular depth or remain in a shore refuge, and (c) schooling

behavior. I did find a difference in reaction distance to a potential predator.

The details of the methods used in measuring reaction distances are given in Chapter 6. Briefly, what I measured was the distance at which individual guppies showed avoidance behavior to a potential predator, in this case a dead Crenicichla. For all 5 stocks tested (Table 12) the mean reaction distance of the females was greater than the males although these differences were very slight for some of the stocks. Though the overall difference between the sexes failed to reach statistical significance ($p > .05$), a trend does seem to be apparent.

TABLE 12. Reaction distance of naive guppies to a dead Crenicichla.

Stock	Mean reaction distance (cm) ± standard error		n
	Males	Females	
Lower Aripo	14.0 ± 0.64	15.6 ± 0.98	25 per sex
Guayamare	11.7 ± 1.29	11.9 ± 1.11	25
Upper Aripo	12.3 ± 0.72	13.0 ± 1.12	25
Petite Curucaye	6.8 ± 1.09	10.2 ± 1.09	25
Paria	7.8 ± 0.94	10.4 ± 1.12	25
Total	10.53 ± 0.49	12.20 ± 0.51	125 per sex

Discussion of sex differences in behavior

It is noteworthy that the greatest sex difference in reaction distance is in Petite Curucaye stock, a stock derived from a natural population with a very unbalanced sex ratio in favour of females. Unfortunately I do not have comparable data on the reaction distances of Caparo guppies but there is no reason to suspect that these would be very different from Guayamare stock (Table 12), i.e. no sex difference. Earlier I drew attention to the possibility of stock differences in the degree of sexual dimorphism and hinted that this might have played a role in the survival experiments (Petite Curucaye and Caparo stocks were used).

These data suggest that the balanced sex ratios observed in populations exposed to severe characid-cichlid predation may in part be a result of the greater similarity in the antipredator behavior of the sexes. However, in populations exposed to Rivulus predation, the sexes are apparently more dimorphic and consequently a greater percentage of males are lost each generation.

For individual males to be able to "afford" this increased risk, viz. predation, some survival benefits must accrue from the retention of sexual dimorphism. Strong sexual selection is usually inferred to be the mechanism that places a premium on certain male traits that would otherwise be a liability (Selander, 1965). I suggest escape behavior may be one "concession" made to

sexual selection in populations where characid and cichlid predation is absent. More work needs to be done to confirm this. A series of experiments similar to Exp. 4.5 but with Paria and Upper Aripo males and females might be profitable.

This suggestion does not vitiate the "size hypothesis"--both aspects of differential mortality may be operative. For example, in the Petite Curucaye R. a male guppy may become a target of Rivulus because it is small, or it allows the predator to approach too closely. The pattern of geographic variation in the sex ratio is consistent with both these ideas.

Although the discovery of a behavioral difference in the escape ability of the sexes (Exp. 4.5) may not be applicable to all populations of guppies, it certainly does underline the need for a cautious interpretation of the results of simple survival experiments (cf. Haskins et al, 1961). Colour is but one of many attributes of a prey that determines its relative vulnerability.

Conclusion and general discussion of geographic variation in the sex ratio

In this chapter I have described the pattern of geographic variation in the sex ratio and have used a comparative and experimental approach to discover why these differences occur. The main conclusion that can be drawn from this work is that population differences in the sex ratio are related to differences in predation. However,

this relationship is not simply a function of sexual dimorphism in colour. I have demonstrated that several other sex differences are relevant to differential predation, especially in environments infested with Rivulus.

Few comparable studies appear to have been conducted on the relationship between the sex ratio and predation. Nevertheless there has been considerable debate about the theoretical aspects of population variation in the sex ratio. One view, which might be termed "anti-selectionist", is that, "Changes in the quantity and the quality of the food supply are the main causes of alterations in the sex ratio (Nikolskii, 1969: 130)." From a review of fish population studies, Nikolskii (1969) asserts that females predominate when density is low and food is abundant while the opposite conditions are conducive to an excess production of males. These "adaptive" alterations in sex ratio are not seen as hereditary changes but occur via unspecified metabolic and hormonal routes.

The opposing school adheres more closely to Fisher's theory (Fisher, 1958: 162): "the action of Natural Selection will tend to equalize the parental expenditure devoted to the production of the two sexes" Williams (1966: 152) concluded that, "there is no evidence from data on sex ratios to support the concept of biotic adaptation." In a broad review (mostly of insects), Anderson (1961) found no good evidence for a feedback from

the sex ratio to population density. In two more recent studies of insects however, extreme crowding and food shortage did result in altered sex ratios; in one case it was caused by an increased mortality of females (Dingle, 1966) and in the other (Feinberg and Pimentel, 1966), a genetic change in the primary sex ratio "evolved". In both these examples the competition was probably far more severe than Nikolskii (1969) had implied was necessary for sex ratio changes in fish populations.

Although I have very little quantitative data on the food supply and relative densities of populations of the guppy, there appears to be no obvious correlation of these factors with the sex ratio. In 1969 very dense populations occurred at Paria and Petite Curucaye Rivers, yet Paria had a balanced sex ratio (1.14) and Petite Curucaye did not (0.33). By the same token very low densities were observed at Caparo, Guayamare, and Marianne Rivers, and very high densities at Santa Cruz and Lower Tacarigua Rivers; however all populations showed a balanced sex ratio (1.00, 0.92, 0.80, and 0.91, 0.91 respectively). In summary, my results do not appear to support Nikolskii's (1969) hypothesis that population differences in the sex ratio are governed by food supply or density.

I concur with Haskins et al (1961) that an excess of females in any given population is probably caused by a

greater vulnerability of males to predators, but I differ in my interpretation as to how predators exert this selective effect. In a search for alternate mechanisms to colour I discovered that size selection was important; also males were found to be less adept at escaping Rivulus predation, though this may not be true for all populations. Whether courtship activity increases the vulnerability of males is unknown. It seems intuitively obvious that when a male is in full courtship display, he is attending to stimuli from the female. Not only may his display jeopardize his camouflage but his concentration on the female may allow a predator to approach more closely. It should be possible to measure the reaction distance to predators of displaying and non-displaying males in nature.

Although I was unable to test the "conspicuousness hypothesis" directly, the circumstantial evidence suggests that the colour of the males may be only a minor factor in the strict sense of camouflage. There remains, however, the possibility that predators can become conditioned to selectively attack the males because this sex is less proficient at escaping. Where large characid and cichlid predators abound, this disadvantage may be counterbalanced by the greater energy return to be obtained from the larger females.

It would be of interest to see if my findings apply to other species of sexually dimorphic fish. Unfortunately

I could find few studies where both extensive data on sex ratios and predators were available. Krumholz's (1963) observation that sex ratios in the poeciliid Gambusia manni tended to be more balanced in the absence of predation support my findings although I observed no senile "thin and gaunt" or "emaciated" males and females under such conditions. Where predation was thought to be severe, Krumholz always found more females than males but it is not known why males had higher death rates. (Sexual dimorphism in size and colour is considerably less in G. manni than in the guppy.) In G. patruelis, George (1960) found that females were more responsive to predator models than males (cf. my Table 12). Unfortunately, when he presented the Gambusia to a real predator (Esox americanus), the sex of the survivors was not recorded.

In the Diamond Killifish, Hastings and Yerger (1971) noted that on the average males outnumbered females about 2:1 ($n = 2127$). Although they did not speculate why this unusual sex ratio occurred, it is noteworthy that this species inhabits weedy areas in very shallow water, the male is slightly larger than the female, and colour dimorphism is seasonal. Possibly a small fish predator that can penetrate shallow water selectively feeds on small fish (cf. Rivulus). Females would be more abundant in this fraction of the population, opposite to the situation in the guppy.

Liley (1966) found that at several sites in Georgetown, Guyana, the sex ratio of 3 species of poeciliids (including the guppy) appeared to be related to the clarity of the water. Collections from green and cloudy water showed a roughly balanced ratio while those from clear water contained fewer males. Liley suggested that in clear water the males are more vulnerable to visual predators. However it is interesting to note that the most extreme ratio was found for P. picta, even though P. picta males are less coloured than P. reticulata males (females of both species are nearly identical). Perhaps sex differences other than colour play a more important role in determining vulnerability.

With respect to water clarity and Trinidad populations of P. reticulata, I have already pointed out that where the guppy is exposed to the major predators, it tends to have a balanced sex ratio in either turbid (Caparo, Guayamare) or clear (Upper Tacarigua, Upper Arouca) water. The outstanding exception to this generalization is the Oropuche R.; it has characid and cichlid predators and is clear for most of the year, yet the sex ratio is 0.50 ($n = 413$). Although this river was sampled only once, I do not think the ratio resulted from sampling error. Thus I am unable to explain this ratio from my experience with other similar environments. Perhaps this is one case where the "Haskins hypothesis" is valid, i.e. severe predation by

Crenicichla, Hoplias, etc. resulting in a very unbalanced sex ratio in favour of females.

Summary of Chapter 4

1. Collections of guppies from Trinidad in 1967 and 1969 yielded a total of 23502 fish; of these 6079 are males, 8166 females, and 9257 immature. The sex ratio is 0.74 and is significantly different ($p < .001$) from an expected Mendelian ratio of 1.00.

2. There are large interpopulation differences in the sex ratio. The ratios found in 1967 are positively correlated ($p < .05$) with those observed in 1969.

3. Sex ratios approaching unity were found in predator-free laboratory stocks of guppies started from immature fish. This is evidence against the hypothesis that geographic variation in the sex ratio is caused by genetic differences in the sex determination system.

4. Sex ratios are relatively balanced in populations exposed to either characid and cichlid predators, or low densities of Rivulus hartii. Very unbalanced sex ratios in favour of females occur where Rivulus density is high.

5. In one stream a change in Rivulus density between 1967 (high) and 1969 (low) was correlated with a significant ($p < .001$) restoration of balance in the sex ratio.

6. Sex ratios do not appear to be correlated with geographic variation in male coloration.

7. A model incorporating the size distribution of the sexes and the size-selective feeding behavior of the predators appears to fit the observed sex ratio variation in the Northern Range streams.

8. Experiments with wild-caught guppies and Rivulus conducted in the field and laboratory demonstrated that males were more vulnerable to predators even when body size was a controlled variable. However in experiments using laboratory stocks of guppies as the prey and either a single Crenicichla or six Rivulus as the predator(s), no sex difference in survival was apparent. I sought an explanation for these conflicting results in terms of differences in experimental design and prior experience of the predators.

9. In a separate experiment of Rivulus predation on a laboratory stock of guppies, males were not selectively attacked but were less adept than females at avoiding capture. Consequently more males were killed than females ($p < .005$).

10. The reaction distance of female guppies to a potential predator is larger than that of males, however there appears to be geographic variation in the degree of sexual dissimilarity in this trait.

11. In conclusion, the geographic variation in the

sex ratio of guppies in the Northern Range region of Trinidad appears to be caused by differential predation on males. Other biotic and abiotic factors seem to be involved only to the extent that they govern the distribution and abundance of predators. The relative vulnerability of the male sex is dependent upon the interaction of the sizes and hunting tactics of the predators, and the sex differences in the size and behavior of the prey. Although it seems reasonable that conspicuous coloration is a liability, this has yet to be demonstrated unequivocally.

CHAPTER 5

GEOGRAPHIC VARIATION IN BODY SIZE

Introduction

Population differences in body size are a conspicuous aspect of geographic variation in Trinidad guppies. In this chapter I explore some of the environmental factors and selection mechanisms that might be responsible for the observed size trends. In particular I wish to evaluate the hypothesis that differences in body size are to some extent adaptations to size-selective predation.

In spite of the voluminous literature devoted to problems of fish growth and body size (see Alm, 1959 for review) there is very little known about the genetic component of size variation in natural populations. In general, fisheries models view body size in terms of a phenotypic response to a given temperature and food supply. This is reasonable because unlike most other animals, fish generally have extremely flexible growth and maturity rates that are very sensitive to environmental conditions (Weatherley, 1966).

Although it is well-known that differences in body size are heritable (Calaprice, 1969) and that fishing mortality may be size-selective (Nikolskii, 1969; Ricker,

1969), the resultant growth and size responses to selective fishing are usually interpreted as a change in age structure or competition for food. Miller (1957) reviewed the early literature and concluded, "There is no clear-cut evidence of exploitation having caused any heritable changes in fishes (p. 803)."

With this paucity of data on selective mortality in managed fish populations, it is not surprising that considerably less is known about natural selective mortality, in particular the response to size-selective predation. The best examples for this phenomenon are studies on invertebrate prey organisms (Mason, 1965; Brooks, 1968, 1971).

Two studies of size-selective predation on fish by piscivores have suggested that heritable changes in growth rates and body size may have occurred in the evolutionary history of the species. Jackson (1961, 1965) argued that the impact of the tiger-fish, Hydrocynus vittatus on small fish (less than 20 cm) has selected for fish species that are large as adults (but see Fryer, 1965 for critique). More recently Parker (1971) has speculated that in the Bella Coola R., chum salmon fry have "evolved a strategy" to outgrow a small, size-selective predator (coho salmon parr). For evolution in the opposite direction, Roberts (1972: 134) has commented that minute fish in the Amazon and Congo R. systems may be less vulnerable to predaceous

fish because they are, "below the size threshold for attack. . . ."

While predation may be an important selection mechanism in some cases, body size must inevitably be a compromise between numerous complementary and conflicting selective forces. Hamilton (1961) has tabulated 7 selective forces which might interact to cause intraspecific size trends in birds; most of these are peculiar to homeotherms or concern problems of flight.

For numerous animals natural selection may favour an optimal body size for: (1) food-getting (Brooks and Dodson, 1965; Estes and Goddard, 1967); (2) resisting abiotic stresses such as high water velocity (Hubbs, 1940; Hartman, 1969), or wave action (Struhsaker, 1968; Berry and Crothers, 1968); (3) avoiding non-predatory interspecific interactions (Hamilton, 1961; Soulé, 1966); (4) securing some form of mating advantage (Hanson and Smith, 1967; Hartman, 1969).

With respect to this last selective force, it is important to note that Henderson (unpubl.) found that aggressive male guppies were able to "orient" (i.e. court) more to females than docile males, yet Ballin (1973) could detect no clear-cut correlation of aggressive behavior and body size. This implies that mating success may not be related to size per se. Presumably male size, colour, courtship intensity, and aggressive behavior all interact

in sexual selection.

With this brief introduction to the problem of the adaptive significance of intraspecific size differences, I shall now document the size variation of Trinidad guppies and search for ecological correlates of this variation.

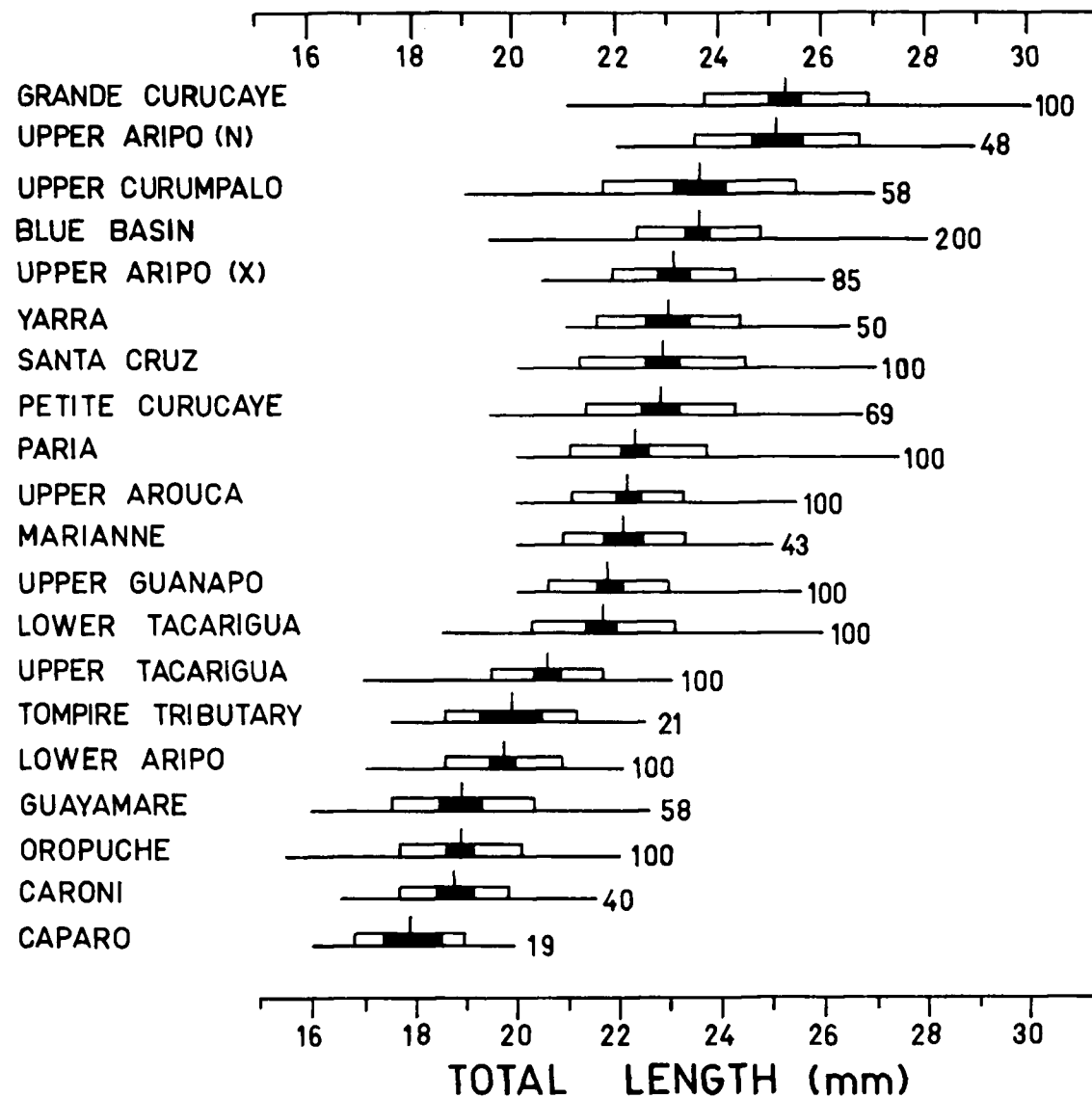
Body size variation in natural populations

Figure 12 gives the mean size and the variation about the mean for the 20 populations sampled in 1969 (rank order). The sizes of mature males are used in this comparative analysis because once the male has attained full coloration and complete gonopodium development, size is not a function of age. Hence the sample variance is smaller and the overall population trends are easier to discern. Nevertheless, albeit that female body size is indeterminate, there is a significant positive correlation between male and female total length (9 populations were compared: Spearman rank correlation, $r_s = + 0.72$, $t = 2.74$, $df = 7$, $p < .05$).

Figure 12 reveals that there is considerable variation among populations ranging in mean size from 25.3 to 17.9 mm. This may not appear to be a large difference in absolute terms, but for a small fish such as the guppy it represents a three-fold difference in wet body weight (150 mg versus 50 mg).

By arranging the sizes in rank order, the size

FIGURE 12. The body size of adult male guppies collected at 20 sites in 1969. Vertical line indicates the mean, horizontal line the total range, black rectangle ± 2 SE, open rectangle ± 1 SD. The sample size is shown at the upper end of the total range.



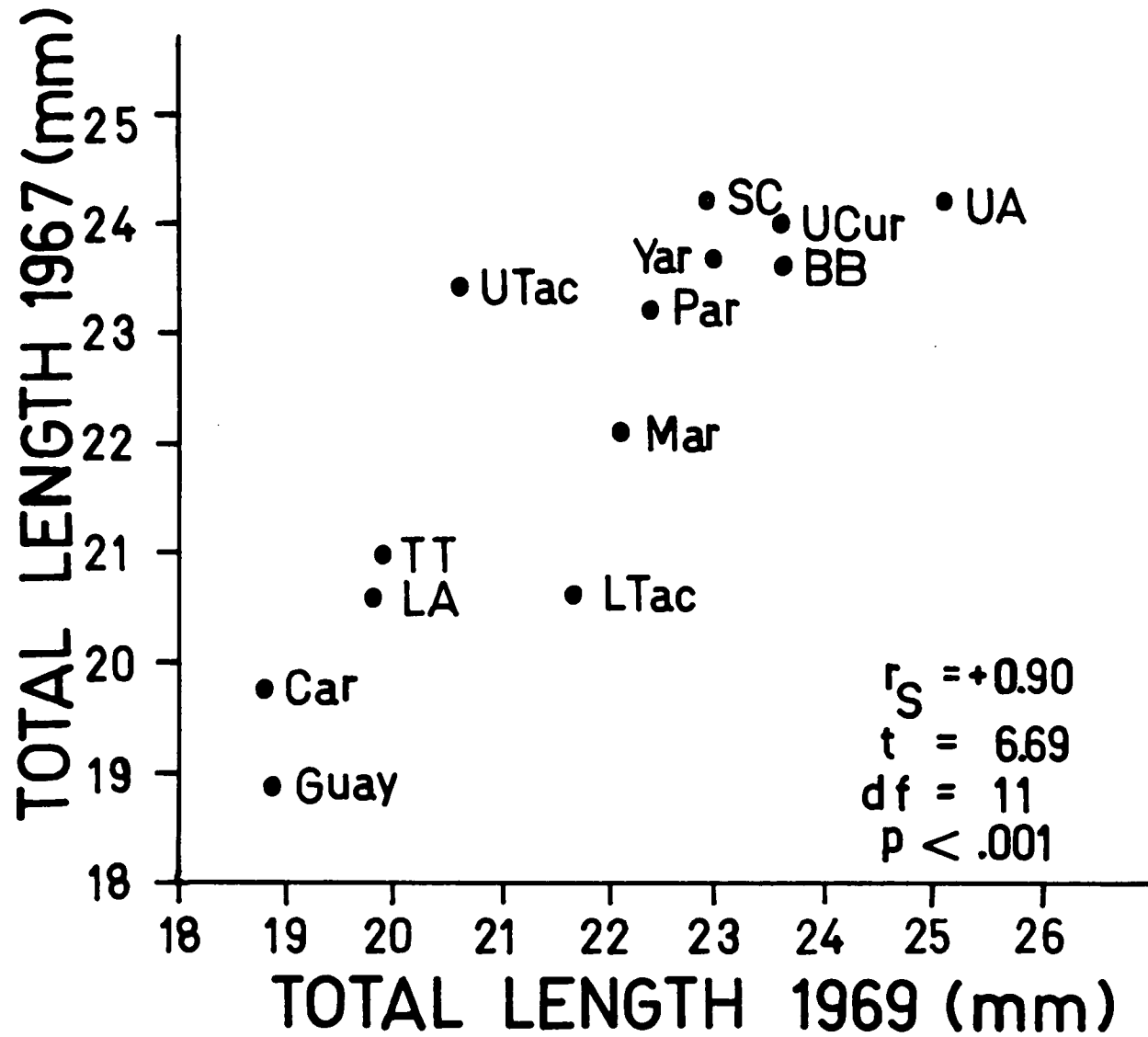
trends appear to follow a smooth, gradual cline perhaps indicating a phenotypic response to an environmental gradient rather than a genetic response to a geographically variable selective factor. But when these size trends are viewed in terms of actual geographic distances, the clines are often very steep suggesting selective factors may be important. For example, the change in mean body length from Upper Aripo (X) to Lower Aripo, a distance of only 5 km, reveals a size difference of nearly half the maximum mean difference for all the Trinidad populations.

To determine if these size differences are a stable feature of the populations, I compared the 1969 measurements with those taken approximately 6 generations before in 1967 (Figure 13). Though only 13 populations are comparable between years, the correlation is highly significant ($r_s = + 0.90$, $t = 6.69$, $p < .001$). This consistency is indeed remarkable, especially for the lowland rivers and midstreams, which are unstable environments with respect to seasonal changes in temperature, volume of flow, and perhaps food supply for fish.

Relationship of body size and temperature

Temperature is known to have a considerable effect on the growth, adult body size, and meristics of fish populations (reviews: Brown, 1957; Barlow, 1961; Shontz, 1962; Paloheimo and Dickie, 1966). The relationship between

FIGURE 13. A comparison of the body size of adult male guppies from 13 populations sampled in 1967 and resampled in 1969. For stream abbreviations refer to Figure 1.



temperature and body size is not a simple one, especially if food supply is also a variable (Brett et al, 1969). In the majority of fish studied, warm temperatures hasten growth and sexual maturity; since growth decreases or stops at maturity, populations in warm water usually consist of smaller individuals (Gunter, 1950). This has been demonstrated experimentally for the tropical freshwater cyprinodontid, Cynolebias adloffi (Liu and Walford, 1966): males raised at 16 C were 19% longer and 76.3% heavier than those kept at 22 C; in addition the life-span was doubled at the cooler temperature.

In a study of pre-adult growth of domestic guppies, Gibson and Hirst (1955) noted that faster growth and maturity occurred at 23 and 25 C than at higher or lower temperatures. Also, larger females were found at the cooler temperatures (20, 23, 25 C).

Even under constant temperature (24 C) and food regime, Bertalanffy (1938) noted considerable differences in the growth rate and asymptotic weight of 3 "breeds" of domestic guppies. He concluded these differences were genetic.

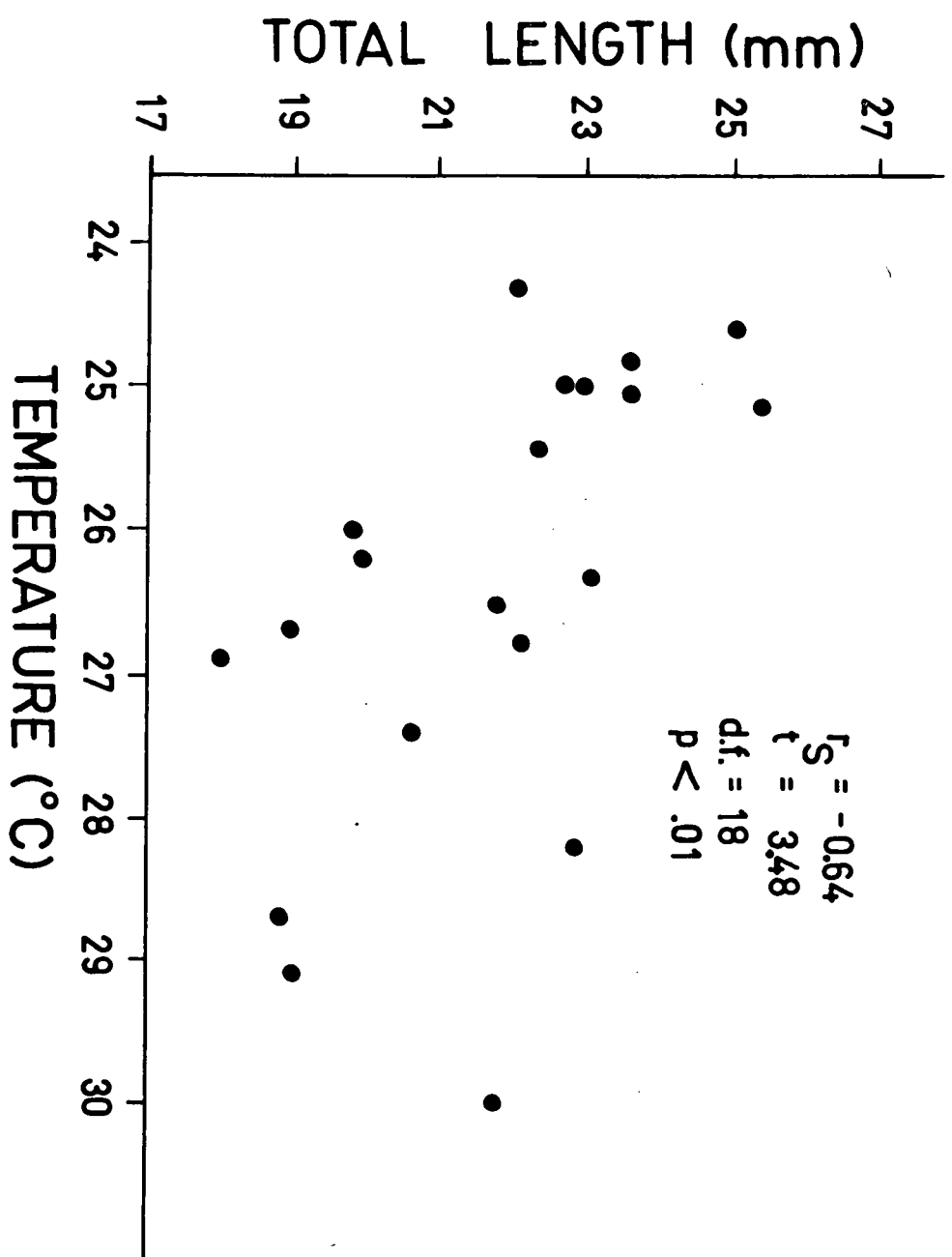
Liley (unpubl. data) also provided experimental evidence that genetic differences in body size were present in several stocks of Trinidad guppies. His results showed that temperature had a significant effect (fish at 23 C grew to a larger size than those at 28 C; cf. Liu and

Walford, 1966) but more importantly, at either temperature, the fish descended from populations which had typically large-sized individuals grew larger than those derived from small-sized wild founders.

My own data (Figure 14) reveal that temperature is at least indirectly involved with the observed size trends illustrated in Figure 12 (refer to Table 1, Appendix for temperature variation, daily changes, etc.). Samples of adult males taken from populations living in cool water were significantly larger than those taken from warmer water ($r_s = (-) 0.64$, $t = 3.48$, $p < .01$), in compliance with Bergmann's Rule (Ray, 1960). This relationship need not be a causal one because other environmental factors (e.g. food supply, predator pressure) may be correlated with temperature and could possibly affect body size.

There is some convincing field evidence however, that both temperature and predation are important factors. In the Upper Aripo R. I sampled two subpopulations of guppies and discovered that the males of each differed by 2.0 mm in mean length (UA(N) = 25.1 mm, $n = 48$; UA(X) = 23.1 mm, $n = 85$). Since these subpopulations are close to each other and the ecological conditions (in 1969) were virtually identical (roughly equal densities of guppies and Rivulus were found), this size difference seemed puzzling at first. However I soon discovered that UA(X) had a much more variable (and generally warmer) temperature

FIGURE 14. The relationship of mean body size of adult males to stream temperature.



than UA(N). The reason for this difference was that the water of the stream below UA(N) was diverted through a series of ditches used to irrigate water cress beds. These cress beds were very exposed (no cover) hence on a sunny day the temperature of the water entering the cress beds area was up to 4.0 C cooler than the water leaving it. This warmer water subsequently flowed to the UA(X) site and was probably instrumental in producing phenotypically smaller fish (cf. Liley's laboratory experiments).

The important point, however, is that in this same stream, the mean length dropped another 3.3 mm in samples taken about 5 km downstream. Since the temperature at this lower site was approximately the same as at UA(X), it is unlikely that the size decrease could be a direct effect of temperature. The most conspicuous difference between these two sites is predation. The upper location is above a waterfall serving as a barrier to characid and cichlid dispersal from below (details in Chapter 3). Because the distribution of predators is such that populations of guppies in cool water are ipso facto exposed to Rivulus predation, it is not possible to find other good "natural experiments" where guppies have been exposed to large predators but cool water (i.e. < 26 C).

It is of interest to note that where Rivulus density is low and the water is cool (Paria, Yarra, Marianne), the guppies are not as large as at sites having similar temperatures

but good numbers of Rivulus (Grande Curucaye, Upper Aripo (N)). But another confounding variable may be involved: according to Hynes (1970: 340) there are some data to show that trout grow faster in harder water. My data on this are limited (see Table 1, Appendix), but in addition to having a lower density of Rivulus, Paria, Yarra, and Marianne Rivers also have softer water.

Relationship of body size and predation

If it is assumed that the phenotypic variation in body size of fish (same age and sex) maintained under a constant temperature and food regime represents genetic variation, then natural selection for body size could be directional, stabilizing, or disruptive. In an unstable environment there could be a shift from one mode of selection to another.

When these simple models of quantitative genetics are viewed with regard to the selective impact of a number of predators of different sizes, the situation is immediately very complex. At any given point in time and space, a single prey may be exposed to potential predators of different (1) species, (2) sizes, (3) age classes, and (4) motivational states (e.g. hunger). All of these factors are known to influence the probability that a prey of a given size will be attacked and captured, not to mention the host of size-specific antipredator adaptations the prey

might have (swimming speed, proximity to cover, visual acuity, etc.). For a diverse predator community (e.g. Lower Aripo and Guayamare Rivers) the number of combinations of these factors would be enormous. To study only a few of these size-specific interactions would be a separate project in itself and I did not attempt this here.

In the analysis that follows, I wish to determine if size differences of the magnitude observed in natural populations (see Figure 12) are of any consequence to the detection and capture efficiency of average-sized Rivulus (a small predator) and characids and cichlids (small to very large predators).

Evidence from the field for size selection

Except for Rivulus and Astyanax I have little direct evidence that predators are in fact size-selective in their predation on guppies. The results of the stomach analyses for Rivulus reveal a positive curvilinear relationship of predator size to maximum prey size (Figure 15). In my samples the cut-off point beyond which even very large Rivulus did not capture guppies is 23.0 mm. The significance of this for sex ratio variation was outlined in Chapter 4.

For Astyanax the largest of 18 guppies recovered was a 13.0 mm juvenile. I should stress that this is not a good estimate of the size capabilities of the Astyanax population

FIGURE 15. The relationship of Rivulus body size to the size of guppies taken as prey. The data are taken from several natural populations.

as a whole--my sampling was very biased towards small individuals because the methods were inadequate to capture many of the very swift larger fish. The samples included only 3 individuals larger than 100 mm (fork length) though many individuals greater than 100 mm were seen in large, roving schools of approximately 20 to 100 fish each. Presumably these fish can normally handle adult guppies of all sizes but sampling by electrofishing or poisoning would be required to confirm this.

The field data for the other predators are even less satisfactory because either no guppies were found in the stomachs or the remains were too digested to get size measurements. In one Crenicichla the remains were of fish probably larger than 25 mm. In addition, at the Oropuche R. collection site (at 1500 hr) I observed an unsuccessful attack on a large (> 30 mm) female guppy by a Crenicichla of about 200 mm.

On the basis of size alone, Hoplias would be expected to select out even larger prey than Crenicichla. It is possible that adult Hoplias prey mainly on species larger than the guppy. At Lower Aripo, one juvenile Crenicichla of 65 mm was recovered from the stomach of a 205 mm Hoplias supporting this idea. The characteristic tooth puncture marks of Hoplias were also found on a moribund Astyanax (> 90 mm fork length) at the Oropuche R. and on several occasions at Caroni and Guayamare Rivers I observed small

characids of unknown species jump out of the water in response to pursuit from a very large predator (probably Hoplias).

In summary, the field observations indicate that Rivulus and small to medium-sized Astyanax prey selectively on small guppies (i.e. fish on the lower tail of the size distribution); scanty field data on adult Crenicichla and Hoplias suggest these predators probably prey more heavily on large guppies.

Experimental analysis of size selection

Two basic questions were asked in this section: can size selection be demonstrated experimentally, and if so, what are the mechanisms? My prediction was that in a sample of guppies differing in body size, the large predators would select out large fish first while the small predators would do the reverse.

It should be stressed that the prey offered in subsequent experiments differ in size but also in age; by coincidence some of the fish will be of the same age and differ in size due to heredity but this is impossible to assess. I shall argue that if size selection can be demonstrated in a synthetic experimental population, it might also have an evolutionary impact on natural populations, assuming body size to be a heritable trait and that "average-sized" predators exert the greatest selective force.

To answer the first question, I exposed large groups of guppies differing in body size to predators. As predation proceeded I noted if a shift occurred in the mean size of surviving prey. Because this did not require close observation of behavior, I used the largest possible arena for these tests so that predator and prey were not unduly crowded.

For the detailed observations of predatory behavior I had to sacrifice reality to obtain precision in measurement. Thus smaller aquaria were used and fewer prey, and the predators were starved sufficiently to insure that they would attack during the observation sessions.

Experiment 5.1 Predation by several species on guppies of different sizes.

This experiment is actually a summary of 18 separate experiments. Some of these were designed especially for testing size selectivity; others were mainly for tests dealing with antipredator behavior (Chapter 6). All the experiments had one common feature: there was variability in the sizes of guppies presented to the predators.

Methods

Table 13 lists the stocks, predators, etc. used in these tests. The guppies were measured the day before the start of each test and re-measured at about the 50% mortality point. The predators were not fed any other food

TABLE 13. Predators and prey used in Exp. 5.1.

Test	Duration (days)	Stock(s) ^a	Sex ^a	Predators	Mean size of prey (mm) ^f		
					before	after	change
a	4	PCur	m+f	<u>Rivulus</u> ^b	20.43	20.92	+0.49
b	3	PCur	m+f	<u>Rivulus</u> ^b	20.30	20.81	+0.51
c	2	Cap	m+f	<u>Rivulus</u> ^b	19.89	20.50	+0.61
d	3	Cap	m+f	<u>Rivulus</u> ^b	19.52	20.05	+0.53
e	4	Cap	m+f	<u>Rivulus</u> ^c	18.42	19.90	+1.48
f	3	SC	f	<u>Hoplias</u> ^d	27.8	26.4	-1.4
g	4	LA+PCur	f	C-1 ^e	21.81	22.60	+0.79
h	2	LA+Par	f	C-1	21.88	21.86	-0.02
i	4	LA+UA	f	C-1	22.31	22.24	-0.07
j	3	LA+Guay	m	C-1	20.75	20.65	-0.10
k	3	LA+Par	m	C-1	20.70	20.62	-0.08
l	2	LA+UA	m	C-2	21.35	21.17	-0.18
m	3	LA+Par	f	C-4	21.44	20.89	-0.55
n	2	LA+Guay	m	C-4	18.93	18.99	+0.06
o	3	Cap	m+f	C-1	20.18	20.13	-0.05
p	3	Cap	m+f	C-2	20.18	20.30	+0.12
q	4	Cap	m+f	C-1	20.81	20.68	-0.13
r	2	Cap	m+f	C-2	21.83	21.78	-0.05

^aInitial number for each stock or sex equal (e.g. 50 LA + 50 PCur).

^bThree males and 3 females (75, 75, 73, 73, 85, 85 mm).

^cThree males and 3 females (75, 75, 82, 85, 88, 90 mm).

^dTwo wild-caught specimens (175 and 205 mm).

^eC = Crenicichla (refer to Table 11).

^fFor sample sizes, refer to Fig. 16.

during the tests. All tests except Exp. 5.1 (f) were conducted in 200 liter tanks set up as described before (standard survival test). Exp. 5.1 (f) was conducted in a 1600 liter outdoor concrete pool with a depth of 50 cm; the Hoplias and guppies used in this test were wild-caught and had been in captivity for only a few weeks.

Results

I have presented the results as frequency histograms of the initial sizes and the sizes at about 50% mortality (Figure 16). The mean "before" and "after" sizes are given in Table 13. To rule out size changes caused by growth, only the results of experiments completed in less than 5 days are given.

The size changes are analysed statistically as simply the direction of the change in the mean body size following predation (Table 14). Although the size changes for most of the tests with Crenicichla were small, 9 out of 12 were in the predicted direction; overall the results cannot be accounted for by chance. This indicates some selection must have been occurring since it was known that the predators could handle all the sizes of prey that were offered.

FIGURE 16. Size selection by predators on 18 experimental populations of guppies. Upper histogram before selection, lower histogram after. Sample sizes are indicated and the direction of mean body size change following predation. Refer also to Table 13.

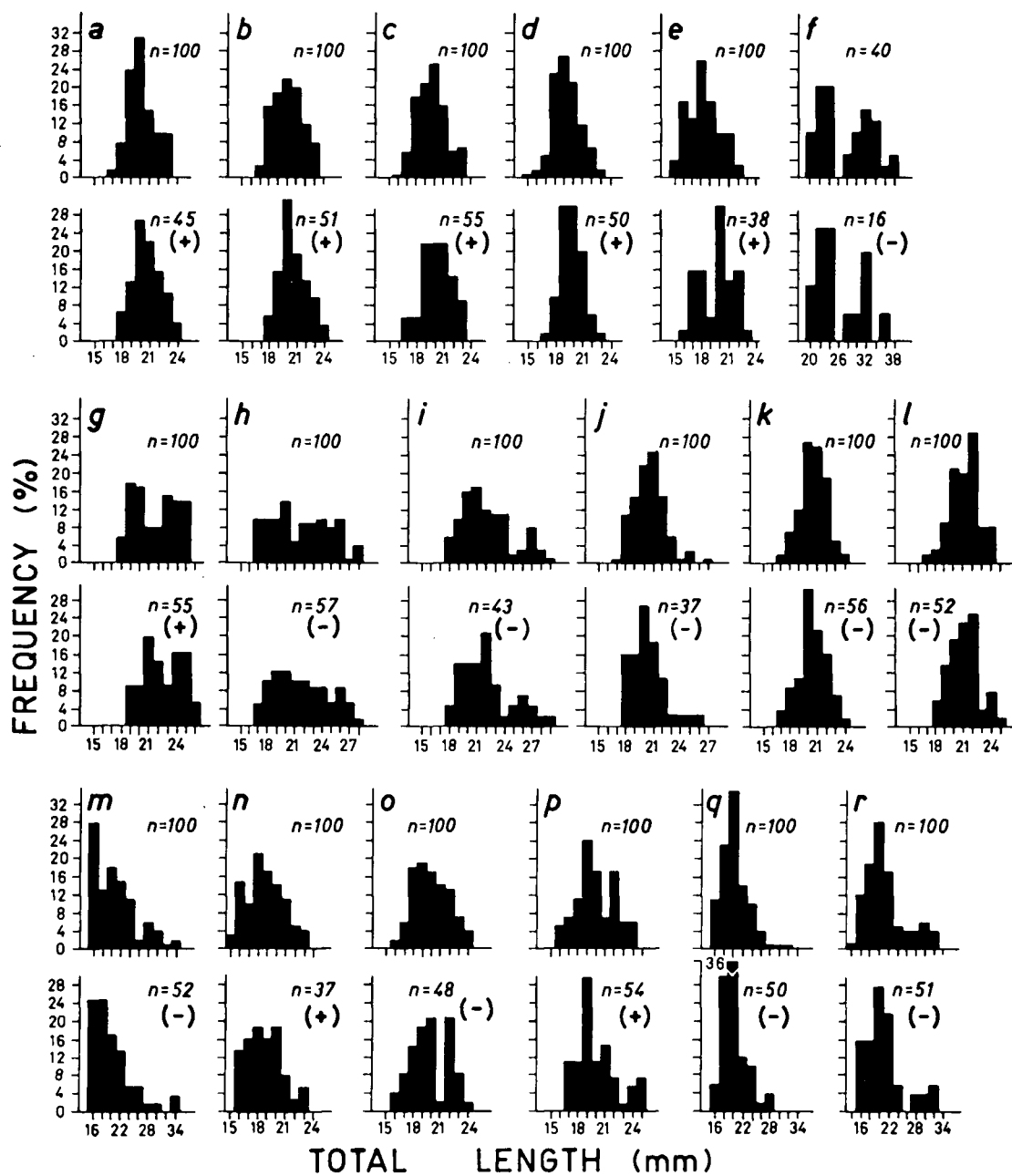


TABLE 14. Direction of change of mean body size in
18 size selection tests

Predator	Direction of change in mean body size		
	+	-	total
<u>Crenicichla</u> & <u>Hoplias</u>	3	10	13
<u>Rivulus</u>	5	0	5
Total	8	10	18
p = .01 (Fisher Exact Probability Test, one-tailed)			

Mechanisms of predator selection
for size differences in prey

The previous experiments have shown that large and small predators exert directional selection (in opposite directions) when preying upon laboratory populations of guppies which exhibit a range of body sizes comparable to those of natural populations. This selection was much more evident in tests with Rivulus and Hoplias than tests with Crenicichla.

Several mechanisms might have been operative in these experiments:

(a) Relative conspicuousness of prey. Visual predators usually can detect large prey at greater distances than small prey; at a given distance, the former subtend a larger visual angle on the predator's retina. In the few fish species that have been studied (cod, Brawn, 1969;

mackerel and mullet, Protasov, 1970; trout, Ware, 1971), the reaction distance to moving prey (or other objects) increases as a linear or curvilinear function of prey size.

(b) Accessibility of prey. Prey of different sizes are often found in different microhabitats, e.g. large guppies are normally found further from shore and over deeper water; consequently they are more accessible to large predators.

(c) Escape behavior of prey. Because swimming speed is positively correlated with body length (Bainbridge, 1960), a large fish is less likely to be overtaken by a predator than a small fish of the same species. Also a large fish may detect a predator at a greater distance because it has better visual acuity. If I apply Protasov's (1970: 81-82) data on guppies to my stocks, it is evident that the largest guppies used in Exp. 5.1 had almost twice the visual acuity of the smallest (measured as the minimal angle of resolution).

(d) Handling efficiency of predator. The prey-capturing apparatus of most predators functions optimally over a limited size range of prey. Prey above or below this range are handled (i.e. grasped and swallowed) with greater difficulty and prey escape is more probable.

(e) Conditioning of predator. Although anatomical constraints will place limits on what size of prey can be handled, a predator may actually select prey over a much

narrower range, presumably determined by the most favorable schedule of reinforcement attainable from a given size spectrum of prey. For example, if juvenile Aequidens (40-50 mm) are added to an aquarium containing guppies and a predator (Crenicichla or Hoplias), the predator quickly shifts his attack to Aequidens. However, these larger prey can only be swallowed with great difficulty (if at all). After several encounters with Aequidens, the predator learns to ignore them and resumes feeding on guppies.

(f) Hunger level of predator. As the hunger level of a predator rises, the size range of acceptable prey increases towards the limits of handling ability (and occasionally beyond). Starved Crenicichla and Hoplias will attack newborn guppies (5 mm); starved (and naive) Rivulus will attack female guppies (> 30 mm) too large to swallow.

Though all of these factors probably interact in nature to determine the relative vulnerability of a guppy of a certain size, mechanisms (a), (c), (d), and (e) seem to be the most likely factors that affected selection in Exp. 5.1; (b) and (f) can be ruled out because no refuge for small fish was provided and the predators could feed ad libitum.

I also have some evidence that size-related differences in vision (mechanism (c)) may have been unimportant. In Chapter 6 I measured the reaction distance of 5 stocks

of guppies to a dead 190 mm Crenicichla; in one test the "predator" was motionless and in the other it was "animated". Though significant stock differences were found (details in Chapter 6), a non-parametric test of the association or independence of body size and reaction distance (Table 15) revealed these 2 variables were independent in all 10 tests (5 stocks x 2 treatments). Thus a large guppy is just as likely to show avoidance at a large distance as a small guppy, and vice-versa.

TABLE 15. The relationship of body size and reaction distance to a predator. For each stock and treatment, the sample was dichotomized at the median body length and reaction distance; scores falling above (+) or below (-) the medians were pooled in a 2 x 2 contingency table. Independence was tested by chi square. (n = 125 females; scores falling on a median were omitted)

	<u>Predator not moving</u>		<u>Predator moving</u>	
	<u>Reaction distance</u>	<u>Reaction distance</u>	<u>Reaction distance</u>	<u>Reaction distance</u>
	(-)	(+)	(-)	(+)
Body length	(-) 28	25	30	25
	(+) 29	29	25	31
$x^2 = 0.089, .80 > p > .70$		$x^2 = 1.09, .30 > p > .20$		

However, I do not know if experience with, or visual cues emanating from, a live Crenicichla might affect

large and small guppies differently. Perhaps the greater visual acuity of large guppies only becomes significant when the fish have been conditioned to avoid a predator from a greater distance (i.e. > 40 cm). With experience, the reaction distance to live Rivulus increases almost 50%; for Crenicichla this is over 100% (casual observations during standard survival tests).

To examine mechanism (d)--handling efficiency--more closely, I conducted 3 short pilot experiments where I recorded directly the behavior of predators exposed to large and small guppies.

Experiment 5.2 Handling efficiency of Crenicichla.

Methods

Five large (24-27 mm, $\bar{x} = 26.1$) and 5 small (18-20.5 mm, $\bar{x} = 19.4$) female guppies of the same stock were placed in a 400 liter aquarium with Crenicichla C-5. The predator had not been fed for 24 hr (i.e. it was moderately hungry). Approach, attack, and capture scores were recorded on a Rustrak event recorder for 60 min after the introduction of the prey.

Though I already knew that this predator could handle guppies larger and smaller than those offered, the object of this experiment was to see if a mean size difference of 6.7 mm (typical of adult fish in nature) might affect the handling efficiency of a large predator.

Results

Although this test was not replicated, it is clear that it took many more approaches (pursuits without prey contact) to capture a small guppy than it did to capture a large one (Table 16). Even when the small fish were attacked, they could sometimes escape the predator's attempted grasp. Attacks on large fish were 100% successful under these experimental conditions (no prey refuge, very clear water, bright illumination).

TABLE 16. Predation efficiency of Crenicichla on large and small guppies.

Frequency/hr	Small guppies	Large guppies
Approach	28	13
Attack	5	4
Capture	2	4
Approach:Capture	14:1	3.25:1
Attack:Capture	2.5:1	1:1

Although this experiment was conducted in a large aquarium, the size differences did not appear to affect the detection of the prey from a distance (the predator reacted to even the smallest guppy from the maximum available distance--1.2 m), but it is possible that the smaller

guppies were less easily tracked during the crucial milliseconds prior to the opening of the jaw. This might have resulted in an error in the direction of attack. With high-speed cinematography it might be possible to ascertain the precise mechanism that gives small guppies an edge over large guppies in dealing with a Crenicichla. In largemouth bass most prey-capture failures occur when the mouth is opened too soon (Nyberg, 1971).

Experiment 5.3 Handling efficiency of Astyanax.

Methods

The same information as in the previous experiment was desired, however a smaller predator and smaller aquarium were used. Two Astyanax of 65 mm (fork length), one per 40 liter aquarium, were placed with female guppies of 2 size classes (5 per class) differing by a mean length of 7.3 mm ($\bar{x}$ large = 24.8; $\bar{x}$ small = 17.5). Before each test the predator was starved for 24 hr. The test was repeated once for each predator.

Results

Overall the results reveal that large guppies were approached and attacked more often than small ones, but the total capture success was about the same (Table 17). This means that the predators were more efficient in handling small fish but spent more time pursuing the large prey. It is interesting that there were individual differences in the

TABLE 17. Predation efficiency of Astyanax on large and small guppies.

Frequency/hr	Small guppies			Large Guppies		
	Test 1	Test 2	Total	Test 1	Test 2	Total
<u>Astyanax 1</u>						
Approach	10	9	19	29	22	51
Attack	3	1	4	16	10	26
Capture	1	0	1	2	2	4
<u>Astyanax 2</u>						
Approach	13	27	40	37	63	100
Attack	2	6	8	4	9	13
Capture	1	4	5	0	0	0
<u>Pooled</u>						
Approach			59			151
Attack			12			39
Capture			6			4

ability to handle large guppies. Astyanax 1 was successful on 4 of 26 attacks while Astyanax 2 was unsuccessful on all 13 attacks. The reason(s) for this difference is unknown; the predators were the same size, had had an identical feeding history in the laboratory, and were probably siblings.

The results of this experiment support the hypothesis that a small predator selects out small prey; handling efficiency may be one mechanism favoring the escape of large guppies. The field evidence (stomach samples) indicates that in nature, the selection of Astyanax is even more biased towards small guppies. Presumably accessibility and escape behavior (swimming speed) play a more important role under natural conditions. Wild Astyanax might also be conditioned to attack only small guppies because, as I have demonstrated, they are relatively easier to handle. Finally, it is possible that the Astyanax in the experiment were hungrier than their wild counterparts and hence attacked more large guppies; Astyanax caught in nature (in the day-time) normally have over one-half of their stomachs full of food.

Experiment 5.4 Handling efficiency of Rivulus.

The field (Figure 15) and laboratory (Figure 16) evidence has already indicated that Rivulus preys mostly on small guppies; the upper limit for the field is about

23.0 mm and for the laboratory it is about 26.0 mm. Thus Rivulus could not take the majority of guppies in the "large" category in Exp. 5.2 and 5.3. However it is still of interest to quantify the predatory behavior of Rivulus attacking guppies near the upper limit because this is the size range of the majority of male guppies. It is important to know how much effect a few millimetres length difference in mature males has on the capture efficiency. I predicted that even slightly larger fish would be more difficult to capture.

Methods

For this experiment it was important to obtain 2 size classes of males differing by only a few mm, each with a small size variance, and of the same stock. By good fortune I was able to use mature males of the Guayamare stock which had been raised at either a low or high temperature by N. R. Liley in his experiments on the effects of temperature on body size. The fish from the "cool" treatment were all large ($\bar{x}$ = 22.8 mm, range 22.0-24.0) and the ones from the "warm" treatment were all small ($\bar{x}$ = 18.1, range 16.5-19.5). My tests were carried out at a constant intermediate temperature.

Four Rivulus (2 of each sex) of greater than 70 mm were used as the predators. The experiment was performed in a 40 liter aquarium. For each test, 2 male guppies of the 2 size groups were placed in the aquarium with the

predator (after the guppies had had a 15 min "calming down" period behind an opaque partition in one corner of the tank).

The same behavioral measures were taken as for the previous tests with Crenicichla and Astyanax. Since there were fewer prey in this experiment, all prey were consumed before a test was terminated. A test was conducted on 10 consecutive days. In most tests each of the predators consumed 1 prey; thus they were all at comparable hunger levels at the start of each new test.

Results

Small guppies were caught with greater ease than large ones (Table 18). Both the approach and attack scores are significantly different between the size groups (Wilcoxon Signed Ranks Test, approach: $T = 5$, $n = 10$, $p < .01$; attack: $T = 4$, $n = 10$, $p < .01$; both tests one-tailed). There was no evidence that the predators were attracted to either size class (the order of captures was random), but once an attack sequence had been initiated, it took more effort (approaches and attacks) to capture the large prey. It would be expected that with increasing experience the predators would preferentially attack the small guppies but this discrimination was not observed over the 10-trial period of this experiment. There was an overall slight improvement in the ability to handle guppies (compare approaches and attacks, tests 1-5 vs. 6-10), but

TABLE 18. Predation efficiency of Rivulus on large and small guppies.

Test	Total length of prey (mm)				Survival time (sec)	Approach		Attack		Capture sequence
	small		large			small	large	small	large	
1	18.0	19.0	23.5	22.0	235	27	22	11	10	LLSS
2	19.0	17.0	23.0	22.5	155	14	32	5	17	SLSL
3	16.5	18.5	23.0	23.5	115	13	27	7	13	SLSL
4	18.0	17.0	22.0	22.0	65	16	24	4	9	SLLS
5	18.5	19.0	22.0	24.0	65	15	19	6	5	SLLL
6	18.5	18.5	22.0	24.0	55	17	16	6	7	LLSS
7	19.0	19.5	24.0	22.0	90	15	26	4	8	SLLL
8	17.0	18.5	22.5	23.0	100	16	34	4	12	LSSL
9	16.5	18.0	22.0	22.0	60	9	18	2	10	SLLL
10	17.0	19.0	23.0	23.0	35	10	14	2	4	SLSL
Mean	18.1		22.8		97.5	15.2	23.2	5.1	9.5	

this was not size-dependent. The survival time (time to consume all 4 prey) also decreased over the course of the experiment, due to (a) the increased proficiency in handling guppies and (b) a decrease in the latent period following the release of the prey from behind the partition.

Conclusion and general discussion of geographic variation in body size

In general, the pattern of geographic variation in body size (Figure 14) conforms to Bergmann's Rule as applied to poikilotherms by Ray (1960), viz. fish taken from populations living in cool water are larger than those residing in warmer water. The physiological mechanisms responsible for the apparent size adaptations to temperature are unknown. However, this is not the primary concern here.

Superimposed upon the overall temperature-size trend is geographic variation which cannot readily be explained by genetic or phenotypic responses to temperature, i.e. the fish appear to be too large or small for a particular temperature regime (e.g. the populations falling near the upper and lower extremes in Figure 12).

Though it is possible that differences in food availability might result in retarded or accelerated growth rates, thereby determining the size of males at maturity (cf. Svärdson, 1943, cited in Alm, 1959: 97), I feel this is important in only 2 streams, Lower Tacarigua and Santa Cruz. Both of these have been enriched through

human activity. The resultant increase in primary productivity appears to have favored the growth of guppies to a larger size than in comparable "natural" streams.

Nevertheless, Liley's experiments with Guayamare and Upper Aripo stocks (nutrition controlled) have provided evidence that genetic differences in body size are present. With additional experimental work, it is probable that genetic differences will be confirmed among other populations.

The question now arises as to why natural selection has favored large-bodied animals in some environments and small-bodied ones in others. Temperature (via the Baldwin effect?) is but one of a multitude of selection factors that might conceivably affect body size. I have mentioned some of the other factors in the introduction; Ray (1960) lists over 16 biotic and abiotic factors that might be involved but many of these are not applicable to the guppy. For some of the more plausible mechanisms, information is either unavailable (feeding specialization, interspecific competition) or inconclusive (sexual selection, see Ballin, 1973). Clearly the problem warrants an intensive study.

In this chapter I have made a preliminary attempt to test one of the many alternate explanations for the size trends--the hypothesis that variation in body size may reflect an adaptation to size-selective predation. This idea arose out of the observation that guppies in the Lower Aripo R. are much smaller than in the Upper Aripo R., though

apart from predation, the ecological conditions (including temperature) are very similar.

For large size to be an advantage, a fish should be able to "outgrow" a predator, and conversely, for small size to be adaptive it should reduce the vulnerability of a fish to a predator that requires large prey in its diet. Certainly in the Aripo R., the requisite predators are present and I have assembled field and laboratory evidence to show that large guppies do enjoy an advantage with respect to Rivulus predation, but are more vulnerable to Crenicichla or Hoplias.

Predation might also act on body size indirectly. Where the large predators are present, guppies are found in very shallow water at the stream edge (see Chapter 6). This restricted environment presumably exposes the fish population to new selection pressures favoring efficient feeding, reproduction, etc. in very confined surroundings. Thus an initial behavioral response to escape characid and cichlid predation may have indirectly resulted in the evolution of small fish. Parents producing large offspring will be less fit because their progeny are (a) more attractive to large predators and (b) less able to cope with life in shallow water. Both disadvantages may reduce the reproductive potential of the offspring; hence genes favoring large body size should be gradually eliminated from the population.

If predation is partly responsible for the size trends, there should be other examples of a predator-body size correlation. Overall, of the 10 populations above the median mean size of males, 8 are exposed to Rivulus and 2 to characids and cichlids; for the 10 populations below the median, only 2 are exposed to Rivulus and 8 to characids and cichlids. Thus there is a good association ($p = .05$, Fisher Exact Probability Test, 2-tailed) of small body size with characids and cichlids, and large body size with Rivulus. However, I have already cautioned that temperature and predator distribution are also associated ($p < .01$, Fisher Exact Probability Test, 2-tailed), so it is impossible, except in isolated cases as the Aripo R., to determine which mechanism is operative. Perhaps careful searching in the Northern Range might uncover other "natural experiments" where predation and temperature are not confounded.

Though the association of large body size with size-selective predation by Rivulus is an attractive hypothesis, there are several anomalies in the field collections (Figure 12) that militate against this simple idea as a satisfactory explanation for the size trends. If Rivulus does exert directional selection on body size, the largest guppies should be in populations exposed to the densest populations of Rivulus. This clearly is not always the case since the mean body size of males at Tompire Tributary

and Petite Curucaye (both with dense Rivulus populations) is well below the maximum of 25.3 mm (Grande Curucaye).

For Tompire Tributary, the small size may be partly attributable to warm temperature (26.2 C), sampling error ($n = 21$), and a different genetic history (it is isolated from the Caroni system). But this does not account for the small males at Petite Curucaye. Petite Curucaye is a tributary of Grande Curucaye and is identical to it in temperature, pH, hardness, substrate, etc.; both fish populations in these streams are undoubtedly closely related historically and are probably connected at present by some gene flow (male colour patterns are very similar). However, there are 2 conspicuous differences between them: P. Curucaye R. is smaller and has a greater abundance of Rivulus.

Thus it would appear that the size (and hence the velocity) of the stream might be an important selective factor. There are at least 2 ways this could operate. The first is the mechanism I alluded to earlier for populations of guppies "forced" to live in a very small environment--small size may have definite advantages, partly counteracting the greater vulnerability to Rivulus.

The second factor is water velocity. Though guppies generally select microenvironments where water velocity is well below the stream maximum, large body size (with its concomitant effect on swimming speed) might allow a fish

to maneuver more easily up and down or across a fast-flowing stream. This may have advantages in intraspecific behavior and escape from Rivulus and terrestrial predators. This might explain the large body size of Grande Curucaye males.

I suggest that in the Petite Curucaye R. there is literally no place for the fish to go. This presents unique survival problems which cannot be solved by large body size and swimming speed alone. I explore the anti-predator behavioral adaptations of Petite Curucaye guppies in the following chapter.

Possibly size-selective predation per se has no bearing whatsoever on geographic variation in body size. If this is true, the discovery that the body size of the majority of adult males exposed to Rivulus falls near or above this predator's upper handling limit must be coincidental.

As with characid and cichlid predation, body size must be interpreted not only in its direct effects on the handling efficiency and preference of the predator but also on the other selective factors that predator avoidance might impose upon the prey.

Finally, it should be recalled that many of the populations in the Caroni system are isolated only by distance. Geographic trends may simply reflect the degree to which local populations are able to preserve adaptive features in the face of gene flow from other populations.

Summary of Chapter 5

1. Populations of guppies sampled in 1967 and 1969 show consistent differences in adult body size. Differences of over 41% in body length and 200% in weight (mean measurements on adult males) have been found.

2. In compliance with Bergmann's Rule there is a significant negative correlation ($p < .01$) between stream temperature and the body size of guppies. Field and laboratory observations indicate that a strictly phenotypic response to environmental temperature is inadequate to account for the size trends, i.e. much of the geographic variation appears to have a genetic basis.

3. A relationship exists between the distribution of fish predators and the body size of guppies. Small guppies occur where there are primarily large predators and large guppies predominate in streams infested with a small predator. In most streams the potential selective impact of temperature and predation is difficult to assess because these 2 environmental variables are confounded.

4. Laboratory experiments demonstrate that large and small predators exert directional selection (in opposite directions) when preying upon laboratory populations of guppies which exhibit a range of body sizes comparable to those of natural populations.

5. One aspect of size-selective predation, the handling efficiency of the predator, is shown to be an

important factor determining the relative vulnerability of guppies of different body sizes.

6. In conclusion, size-selective predation is a plausible mechanism for the evolution of body size differences in guppies. However, it does not adequately explain all the geographic variation--other selective factors must be involved, including water velocity, the size of stream, and the indirect effects of antipredator behavior.

The adaptive significance of geographic variation in body size can only be understood in terms of the action and counteraction of numerous selective factors, coupled with an appreciation of the magnitude of gene flow.

CHAPTER 6

GEOGRAPHIC VARIATION IN BEHAVIOR

Introduction

In this chapter I shall describe geographic variation in the behavior of guppies and endeavor to correlate this variation with features of the habitat. In particular, I am interested in ascertaining the significance of the differences. Are these differences heritable? Do they make functional sense in terms of the major challenges to survival?

The emphasis is on population differences in anti-predator behavior--the behavioral components enabling an animal to share an environment with its predators. I have already attempted to justify this research on several grounds (see Chapter 1); most important is the potential insight to be gained into the mechanisms affecting the evolution of behavior.

The working hypothesis was erected following the discovery of behavioral differences in several situations in nature. To examine these differences more closely and to control for the effects of environmental differences, I studied behavior in the laboratory, using mostly the

offspring of samples taken from 5 representative populations (Lower Aripo, Guayamare, Upper Aripo, Petite Curucaye, and Paria). These river sites were chosen because they represent the range of ecological conditions where guppies are commonly found (at least for Northern Trinidad, see Chapter 3). They also encompass the range of predation pressure occurring in this region of Trinidad (Chapter 3, Table 3). Therefore, if behavioral variation does represent (to some degree) a microevolutionary phenomenon, it should become apparent in a comparative study of these 5 populations, especially the populations exposed to either extreme of predation pressure.

Field observations of behavior

My observations are based on the study of guppies in most of the streams sampled for body size and sex ratio variation (for methods, refer to Chapter 2). In addition, casual observations were made on populations in a stream at Paramaribo, Surinam, and in pools and ditches in Georgetown, Guyana (both are mainland South American populations).

I did not make a detailed quantitative comparison of the behavioral differences in nature. This section is then mostly descriptive.

Distribution of guppies in the stream environment

The distribution of fish across and along a stream is seldom random. In the streams I examined, differences in

depth occurrence and proximity to the shoreline were noticeable. In some populations fish were tightly bunched near each side of the stream while in others they were spread quite uniformly across the stream.

Several reasons could be suggested for these differences. Obviously abiotic factors can influence distribution. Since guppies are small fish and not particularly strong swimmers, they do not occupy water flowing faster than 0.3 m/sec. In sections of streams with a high water velocity, guppies are found in back-eddys or at the extreme edge of the stream "clinging" to the shoreline; the water velocity usually approaches zero m/sec in these regions.

Even along fast-flowing streams however, there are regions with pools and wider, smooth-flowing sections of lower velocity. If water velocity is the only factor restricting the movement of guppies, the fish should spread out in these sections. This is not observed in all populations, suggesting that in some cases other factors are playing a role, possibly the distribution of food or the threat from large predators.

(a) The effect of predators

There is good field evidence that predators may have an effect on the distribution of guppies. As mentioned before, the Aripo R. is divided into an "upper"

and "lower" region by a series of falls (limestone ledges). One of these waterfalls is 5 m high, completely blocking the upward passage of characids and cichlids. The velocity and depth of water above and below the falls are comparable but the distribution of guppies differs markedly.

To collect samples of fish from below the falls, it was necessary to sample the very edge of the stream and the side-pools; only a small number of fish were seen near the centre of the stream (mostly large females). Above the falls, the situation was strikingly different--both males and females were distributed across the stream. This is indirect evidence that the shoreline offers some protection in avoiding characid and cichlid predators.

To test this idea, I took some guppies from a shore refuge at Lower Aripo and placed them 1-2 m from shore over deeper water. Almost immediately I observed several Astyanax pursuing the guppies as they darted back to shore. The Astyanax discontinued their pursuit once the guppies reached the shallower water.

The depth at which guppies occurred was also variable. Where characid and cichlid predators were present, the few fish that did venture from shore swam very close to the surface of the water. Where these predators were absent, the guppies were distributed throughout the water column. In a few streams (Paria, Yarra, Upper Aripo) I had the impression that most of the fish were in the bottom

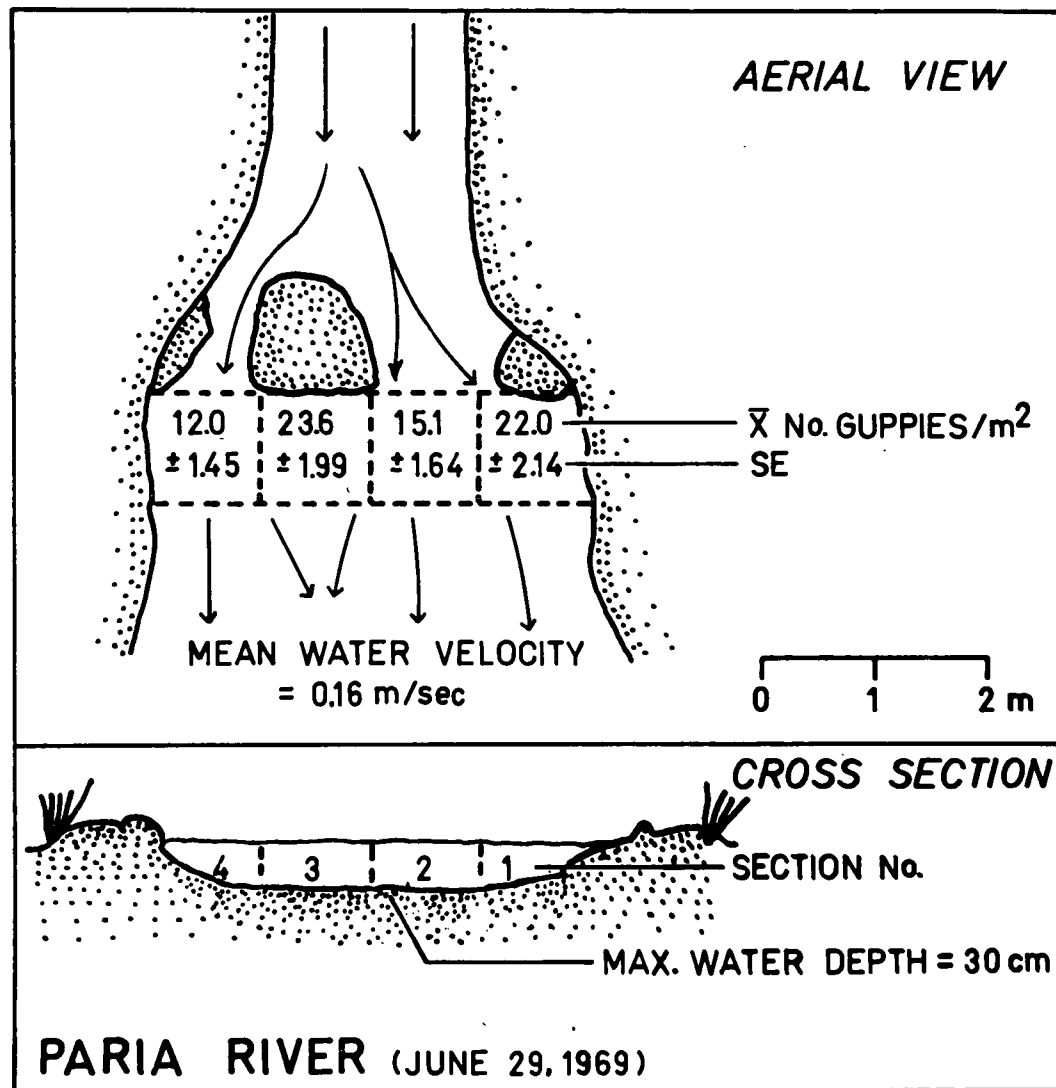
one-third of the water column.

(b) The effect of water velocity

To determine if water velocity has an effect on the distribution of guppies in a stream virtually devoid of aquatic predators, I made a census of fish along a 4-meter transect across the Paria R. (Figure 17). This transect was divided into four 1.0 m^2 sections with sections 1 and 4 representing the regions on either side of the stream and 2 and 3 as the two inner sections. The important difference among these sections was that 2 of them were protected from the full stream velocity by large boulders immediately upstream. This meant that most of the water passed through the other 2 sections with an increased velocity. One of the protected sections was at the shore, the other near the center of the stream. Thus it was possible to separate the effect of velocity on the distribution of guppies from the affinity for the shoreline.

To estimate the number of fish in each section, I proceeded from section 1 to 4 in sequence and counted very rapidly all the fish in each section. This was replicated 16 times with a 4-min interval between census periods. The fish moved rapidly in and out of these sections so there was considerable variability in the numbers seen at any one time. However the data in Figure 17 demonstrate quite clearly that fewer fish occurred in the sections with a high

FIGURE 17. The distribution of adult guppies across a section of the Paria River. Arrows indicate the direction of water flow. Refer to text for census methods.



water velocity, regardless of the proximity to the shore. This suggests that in the absence of predation by large fish, guppies distribute themselves throughout the available environment; however, water velocity can limit the time fish spend in any particular area.

There is some additional evidence that velocity is not the only factor limiting the distribution where predation is severe. At certain times of the year a small "lake" is formed behind the Guayamare irrigation dam (located 0.5 km below the collection site on this stream). Although the water velocity in this lake was essentially zero, the guppies were nevertheless distributed around the periphery. The majority were within 1 m of the shore. Since large Hoplías were caught in this lake, the survival value of this behavior is probably linked to predator avoidance.

Table 19 summarizes the observations on distribution for the 5 principal populations under study. It should be stressed that this information is only for a typical section of stream having both a depth and velocity gradient. Obviously there will be sections of even the Paria R. that are too shallow for guppies to be far from the surface, or too fast to allow the fish to be spread across the stream.

Schooling behavior

Besides differences in the gross distribution of fish across a stream, I found differences in schooling

TABLE 19. The distribution of 5 populations of guppies in relation to water velocity, depth, and distance from the shore.

Stream	Water velocity (m/sec)		Distribution of guppies	
	Mainstream	Microhabitat of guppies	Position in water column	Position across stream
L. Aripo	0.61	0 - 0.20	near surface	near shore
Guayamare	0.40	approx. 0	near surface	near shore
P. Curucayé	0.29	0 - 0.20	ubiquitous	no preference
Upper Aripo	0.55	0 - 0.20	near bottom	no preference
Paria	0.16	0 - 0.16	near bottom	no preference

behavior. I use Keenleyside's (1955) definition: "any fish aggregation can be considered a school, provided the fish are together because they are reacting to each other, and not because of similar reactions by each individual to a common external stimulus (p. 183)."

Schools of guppies may have several configurations. In fast-moving water the fish are already confined to the stream edge so it is difficult to separate the aggregating effect of the external environment from the social force of schooling. In such regions, however, it is common to see groups of 5 to 50 fish, including both males and females, moving together up or down the stream, parallel and close to the shore. These are "linear" schools because the depth and width of the school is very small--the fish appear to follow each other in "single file" fashion. This type of school was commonly seen in some portions of the Lower Aripo R.

If water velocity is less restrictive to movement, larger schools may be observed. Several large elliptical or nearly spherical schools were observed in smooth-flowing sections of the Lower Tacarigua R. Counts for these large schools ranged from 75 to over 200 adult fish. Elliptical schools are oriented with the largest diameter parallel to the direction of flow.

When large schools are disturbed by a predator (fish or human), they may break up into sub-schools and reform

into a single unit after a few seconds. This splitting varies with the intensity of the disturbance: a slow, gradual approach will elicit only a cleaving of the school around the threatening object (e.g. large Aequidens) followed by immediate reformation.

Sudden approaches (especially by a potential predator hiding under rocks or in the shore vegetation) will elicit a rapid fan-like scatter of the fish, each individual moving in his own particular direction away from the stimulus source. At Lower Tacarigua R. I was able to follow the majority of such dispersed individuals and they all appeared to re-unite.

Schooling is noticeably poorly developed in some streams. Fish in the Paria R. behave very much as individuals; the most common social groups involve fish engaged in courtship activity (e.g. 3 males following a female). These groups do not persist for more than a few seconds as the males move off to court other females.

A number of streams have fish populations exhibiting an intermediate development of the schooling response. The cohesion may be quite loose and distances between fish large, i.e. the fish do move together but in a less organized manner. If these fish are threatened, they often escape in groups but quickly dissociate once the emergency has passed. This is followed by a rapid resumption of feeding and courtship behavior.

When the tendency to school is related to the predators in the surrounding environment, the trend is that where characid and cichlid predators occur, schooling behavior is more prevalent (Table 20). Also, in streams with Rivulus, schooling is better developed where this predator is very abundant (e.g. P. Curucaye).

TABLE 20. The development of schooling behavior in 5 populations of guppies

Stream	Predator(s)	Schooling behavior
Guayamare	characids and cichlids	well-developed
L. Aripo	characids and cichlids	well-developed
P. Curucaye	<u>Rivulus</u>	intermediate
U. Aripo	<u>Rivulus</u>	poorly-developed
Paria	<u>Rivulus</u>	absent

Reaction distance to potential predators

Reaction (= reactive) distance is defined here as the distance from an external stimulus at which a fish responds with an overt change in behavior. In the context of antipredator behavior, this usually entails a change in the direction, velocity, or form of swimming behavior. The term is thus roughly equivalent to "flight distance" (Walther, 1969, for Thomson's gazelle), "approach distance"

(Heatwole, 1968, and Johnson, 1970, for lizards), and "Feindabstand" (Curio, 1969, for Darwin's finches). Sometimes a distinction is made between the distance at which a prey will approach a predator (avoiding distance) and the distance at which a prey flees from the approach of a predator (flight distance).

It is regrettable that I did not have the foresight to accurately measure this important behavioral parameter in situ. However, I did note differences in reaction distance on a more gross qualitative scale. When I approached streams containing the larger predators, the guppies were obviously very reactive. They avoided the net at a considerable distance; although I could see many fish at any given time, these were very difficult to capture. They usually spread out at rapid speed, moved to deeper water, or further from shore. This was especially true at Caroni, Guayamare, Lower Aripo, Caparo, and Oropuche Rivers.

In some streams it was much easier to approach the fish with a dipnet. The extreme situation was at Paria where the fish could be caught by hand. These fish would also pick at my legs even when these large "objects" were moved. This short reaction distance to a potential predator is probably related to the virtual absence of all aquatic predators from the Paria R.

Motor patterns used in antipredator behavior

A number of distinct motor patterns (= fixed action patterns) were observed during encounters between guppies and their predators in nature. These will be introduced here but studied in greater detail in a later section where they have been quantified for the laboratory stocks.

I have been able to distinguish 6 main escape motor patterns that are used by individual guppies. These are distinct from group responses such as schooling but may also be elicited in a number of animals simultaneously, should a group suddenly encounter a predator. Here I shall consider the case for a 1 predator--1 prey system.

1. Weak avoidance drift

This behavior pattern is seen most commonly when a potential fish predator or strange object has appeared in the guppy's visual range at a considerable distance. On perceiving the predator, the guppy may either stop swimming momentarily, or actually move closer to the predator. At a certain critical distance, the guppy turns its body lateral to the predator and moves some distance one way, and then turns around and repeats it in the opposite direction. At all times the eyes are kept on the predator (with binocular fixation, head held slightly downwards), as if the fish was surveying it from different angles. The fish remains roughly equidistant from the predator during this "inspection".

There appears to be a conflict of approach-with-drawal motivation involved in this behavior because the fish may alternately move closer to, and then away from the predator. Should the predator move, especially in the direction of the guppy, the behavior reverts to one of the other motor patterns described below.

In some respects, the body configuration in weak avoidance drift and avoidance drift (below) is reminiscent of the weak sigmoid display of the male guppy during courtship. I use the term "drift" because the fish appears to be sculling mainly with its pectoral fins and the body is held rather rigidly.

2. Avoidance drift

This pattern is a more stereotyped form of the previous one. The body is held more rigidly and a definite sigmoid position is assumed. The dorsal and caudal fins are usually spread and held rigidly. There is less movement towards or away from the predator than in the previous pattern; turns in direction are smooth and rapid.

3. Turn around

This pattern is simply a sudden cessation of swimming activity towards a potential predator coupled with a rapid "about turn", followed by swimming in the opposite direction.

4. Rapid dart

This motor pattern occurs most often when a predator has actually moved towards a guppy, or has appeared suddenly at a close range (e.g. ambush). To describe all the details of this pattern would require slow-motion cinematographic analysis because the pattern lasts only a fraction of a second. During this time the fish may have moved as much as 20-30 cm. It appears that the body is held in a rigid position and thrust is applied by the pectoral and caudal fins in a short burst, propelling the fish usually upwards in the water column and occasionally out of the water. It is possible for the guppy to maintain the same position it had before the dart was initiated, i.e. it seems to swim backwards. In this way, the binocular fixation on the predator is uninterrupted but the guppy is at a safer distance.

A rapid dart may intergrade with a turn around and be followed by a bout of rapid jerky swimming movements (zigzags, etc.) that are very irregular and difficult to describe. These latter movements appear to fall into a class of behavior termed "protean", i.e., "behavior which is sufficiently unsystematic to prevent a reactor predicting in detail the position or actions of the actor (Humphries and Driver, 1970, p. 286)."

5. Surface skim

A surface skim is elicited when an attack sequence is in progress or has just ended unsuccessfully. The fish swims rapidly away from the predator, just below the surface water film often making a ripple on the surface. This is commonly seen in nature when a school is approached suddenly by a predator and is often accompanied by one or more surface jumps.

6. Surface jump

In this behavior pattern, the fish actually leaves the water once or maybe a dozen times in succession. It occurs normally only when a predator has struck or is about to strike. However, it can also be elicited by tapping on an aquarium containing wary guppies that have recently been attacked by a predator. Surface jumps are a conspicuous feature of schools of guppies, numerous fish leaving the water simultaneously.

It is also difficult to see the details of the sub-components of this behavior because of its short duration. It appears that in certain cases (single jumps) the guppy reverses its position in mid-air and returns to the water with its head facing towards the point of exit from the water. In multiple surface jumps the orientation is forward and the fish literally skips along the water surface. The last form was seen when a large female guppy was pursued by

a Crenicichla (Oropuche R.); about a dozen separate jumps were counted before the guppy finally reached safety. A very similar response ("surface breaking") has been described for the poeciliid Gambusia patruelis during encounters with pickerel (George, 1960).

The 6 patterns described above, though viewed here as separate entities, may be part of a single attack-escape sequence. As a predator moves in the vicinity of a guppy, the guppy may show weak avoidance drift at first but shift to a rapid dart if the predator approaches too closely. If an attack occurs, surface skims and even surface jumps may be elicited. Thus the motor patterns appear to be arranged on a scale of reactivity ranging from "precautionary" behavior (avoidance drift, turn around) to "emergency" behavior (rapid dart, surface skim and jump).

Presumably the sensory-motor coordination centres operate on a system of rising thresholds with levels appropriate for each motor pattern. These levels do not appear to be fixed. This can be readily demonstrated by moving a model of a predator at a constant velocity towards a group of predator-naive and a group of experienced guppies. The first group will show avoidance drift behavior while the second will respond with rapid dart or possibly surface jump, i.e. the threshold for the release of emergency patterns is lowered with experience. This threshold change is not only specific to visual stimuli since the

responsiveness to mechanical disturbances (water vibrations) is also influenced by experience with predators.

Though the acoustico-lateralis and olfactory systems may function in predator avoidance, the primary sensory modality appears to be vision. All the escape motor patterns can be elicited when a predator "attacks" a group of guppies isolated in a clear glass jar. I did not study chemical communication between guppies, or between guppies and predators, but observations during actual predation revealed no behavioral changes of the type attributed to "Schreckstoff" (Schutz, 1956; Pfeiffer, 1962). Similarly, I observed no overt responses to samples of water taken from aquaria containing predators (cf. Göz, 1941; George, 1960; Reed, 1969; Rüppell and Gösswein, 1972).

Disturbances do not always arise from within the aquatic medium; terrestrial predators also evoke responses from guppies. Objects moved over a stream elicit a "down-dart" response. The fish moves very quickly to a deeper portion of the water column; in shallow streams it may even come to rest on the substrate.

Population differences in escape motor patterns

Although all the patterns described above can be elicited in all the populations of guppies it was evident that quantitative differences were present under natural conditions. I saw virtually all the surface skim and

surface jump behavior in streams containing characid and cichlid predators, even when the "predator" was the end of the dipnet pole. Using the same stimulus, it was difficult to elicit these responses from guppies who were not exposed to characids and cichlids. It appears, therefore, that there may be geographic variation in the degree of stimulation necessary to elicit certain escape motor patterns.

The down-dart response was also geographically variable. This quantitative difference could be demonstrated when an object was passed with a uniform velocity over a group of newly-caught guppies from Paria and Guayamare Rivers that had been placed in adjacent opaque water pails. The Guayamare fish scattered rapidly downwards in a "panic" reaction but quickly re-surfaced. The Paria guppies on the other hand moved slowly and directly downwards (no violent scatter) to the bottom of the pail. Often they remained motionless on the bottom for over 10 sec; by this time all the Guayamare fish had fully recovered and were swimming normally near the surface again. Thus in addition to a population difference in the intensity of the response and the rate of recovery, there also seems to be a qualitative difference in the form of the behavior. I could never elicit a "Paria response" from Guayamare guppies, in fact the only other population showing this response was Yarra.

Summary of the field observations of behavior

The previous section has been mainly a qualitative review of some of the differences in the microdistribution and antipredator behavior of guppies in streams with different predators.

The distribution of fish in terms of depth in the water column and proximity to the shore is affected in part by abiotic factors including the streambed morphology and water velocity. However, I put forth the hypothesis that the microdistribution of fish is also a sensitive indicator of differences in the quality of predation. Where large predators occur, the littoral restriction of guppies may in itself be an antipredator mechanism.

Large differences were observed in the development of schooling behavior. Schooling is a conspicuous feature of guppy behavior in streams with characid and cichlid predators but tends to be less prominent in streams with only Rivulus.

Gross estimates of reaction distances were made in the field and these were also correlated with the distribution and abundance of predators. In the absence of significant aquatic predation (Paria) the reaction distance to a threatening stimulus was very small, whereas it was large in streams with characid and cichlid predators.

The guppy may use at least six fairly distinct motor

patterns in response to potential predators. These patterns appear to be organized in a ranked scale with the lowest pattern appearing under conditions of a mild change in the animal's immediate environment and the highest under direct attack conditions. No qualitative differences in these patterns were seen among populations exposed to different predators, but quantitative differences were found. It appears that the higher levels of escape patterns are elicited by a less severe threat in streams with characid and cichlid predators than streams with only Rivulus. Therefore, fish exposed to different predators have different alarm thresholds.

The response to an object passed over the water surface is a stereotyped downward "freeze" at Paria while it is a more sudden random scatter at Guayamare. This might be related to the relative importance of aerial and aquatic predators in each stream.

Survival value of behavioral differences in wild-caught fish

If my hypothesis is correct that much of the variation in behavior observed in nature is related to differences in the impact of predators, it should be possible to test fish taken from "low predation" and "high predation" regions with the natural predators. It would be predicted that guppies taken from populations which showed (1) long reaction distances to potential predators, (2) well-developed

schooling behavior, (3) easily-elicited escape motor patterns and (4) well-developed habitat selection (close to surface and shore) would have better survival when placed experimentally with the natural predators.

This prediction makes no assumptions regarding the causation of the phenotypic differences in behavior--they could be the result of experience with, or natural selection by predators, or an interaction of both effects. Here I am only concerned with demonstrating whether or not the differences have survival value. The separation of experiential and genetic factors will be dealt with later.

My original plan had been to test the different populations in "field experiments" where fish from a low predator stream (showing concomitant behavioral differences) would be tested for survival in a predator-infested stream. Samples from the latter stream would be handled identically and serve as a control. Also I wanted to see if perhaps a well-developed anti-characid and cichlid response might in fact be disadvantageous in a stream infested with Rivulus. Such transplant experiments are usually viewed by ethologists as the best way to test the survival value of behavior.

For numerous reasons I had to abandon my field tests and perform the experiments in aquaria and outdoor pools. The main problems with the field tests centered on maintaining "guppy-proof" screen enclosures (I did not want to "pollute" a stream by the accidental release of foreign

genotypes). During the rainy season the highly variable water level and the abundance of suspended macrophytic material and detritus in the water made this work very difficult. A secondary problem resulted from human interference with materials left at field sites.

Experiment 6.1 Relative survival of wild-caught guppies.

Methods

An "indoor" and "outdoor" series of experiments were designed to test for survival differences among samples of guppies taken from 4 populations. The first series was done using either indoor aquaria of 80 liter capacity or an indoor concrete pool of 700 liters (depth 50 cm). These were maintained at ambient temperature and natural photoperiod.

The second series was done in 2 adjacent outdoor concrete pools of 1600 liters (depth 50 cm). The water in the indoor experiments was completely clear while the water in the outdoor experiments was very turbid with phytoplankton (Secchi disc disappeared at 35 cm).

In the indoor tests, prey were added at frequent intervals during the day to replace those eaten (replacement method). The relative number of 2 or more prey types added over a period of 5 or more days of predation was used as the index of vulnerability.

In the outdoor tests, relative survival was simply

measured as the number of fish remaining from an initial 50 individuals of each population; periodical censuses were made with a "one-man seine".

In all experiments the predators and prey had been in captivity for only a short period of time. Prey sizes were matched between populations as a control for potential size selectivity by the predators.

Because my knowledge of the discrimination learning of the predators was minimal, I usually used female guppies for these tests. Females are all morphologically similar, regardless of their population of origin; this was a convenient control for the possibility that predators might learn to associate the population-specific colour patterns of the males with their ease of capture. In addition it would rule out selection based on conspicuousness per se (though as I have pointed out in Chapter 4, good evidence for this is still lacking).

To distinguish among the female types, those of one population were marked on either the lower or upper caudal peduncle with a #30 hypodermic needle that had been dipped in a concentrated Trypan Blue (vital stain) solution. Fish were not placed with predators for at least 24 hr after marking. The mortality during this recovery period was less than 1%. The mark faded noticeably after the first day but remained visible on close inspection for at least a week. Where prey survived longer than this, they were re-marked

as soon as the mark became almost imperceptible.

In a control experiment, I placed 50 marked and 50 unmarked females of the same population with a juvenile Crenicichla. The survival rate of each group was identical suggesting the marking procedure did not affect the vulnerability of the fish to predators.

Results

Table 21 shows the results of the indoor experiments (replacement method) giving the relative number of fish added each day over the course of the experiment. χ^2 values were calculated to test the null hypothesis that the difference between the total observed and expected number of fish eaten could be explained simply by sampling fluctuations (for 1 df Yates's correction for continuity was applied).

It is clear that the null hypothesis could be rejected in most experiments. In tests with a juvenile Crenicichla (Table 21 A, B, and C) significantly more fish of the "low predation" populations (Paria, Upper Aripo, Petite Curucaye) were taken than the "high predation" population (Lower Aripo).

The results for a replicate of one of these tests with a juvenile Hoplias (Table 21 D) were less significant but in the same direction.

The results with an adult Crenicichla preying on males from 3 populations indicated that Paria was most

TABLE 21. Relative mortality of wild-caught guppies exposed to predators. Tests A, B,C--95 mm Crenicichla, 80 liter aquarium; Test D--95 mm Hoplias, 80 liter aquarium; Test E--200 mm Crenicichla, 700 liter indoor pool.

Day	TEST A		TEST B		TEST C		TEST D		TEST E		
	Lower Aripo	Upper Paria	Lower Aripo	Upper Aripo	Lower Aripo	Upper Petite Curucaye	Lower Aripo	Upper Aripo	Lower Aripo	Upper Aripo	Paria
1	0	0	2	2	0	0	0	1	0	0	0
2	0	0	3	10	5	8	9	11	-	-	-
3	1	7	5	15	7	11	4	10	3	6	9
4	2	9	6	7	0	3	5	12	-	-	-
5	3	9	5	4	2	9	2	2	-	-	-
6	1	6	2	8			7	4	12	12	20
7							13	11			
8							2	7			
9							6	15			
10							3	4			
11							7	8			
12							7	9			
13							10	7			
14							4	5			
Total	7	31	23	46	14	31	79	106	15	18	29
χ^2 (1 df)	13.92		7.01		5.69		3.65		χ^2 (2 df)	= 5.25	
p	< .001		< .01		< .02		.10 > p > .05		.10 > p > .05		

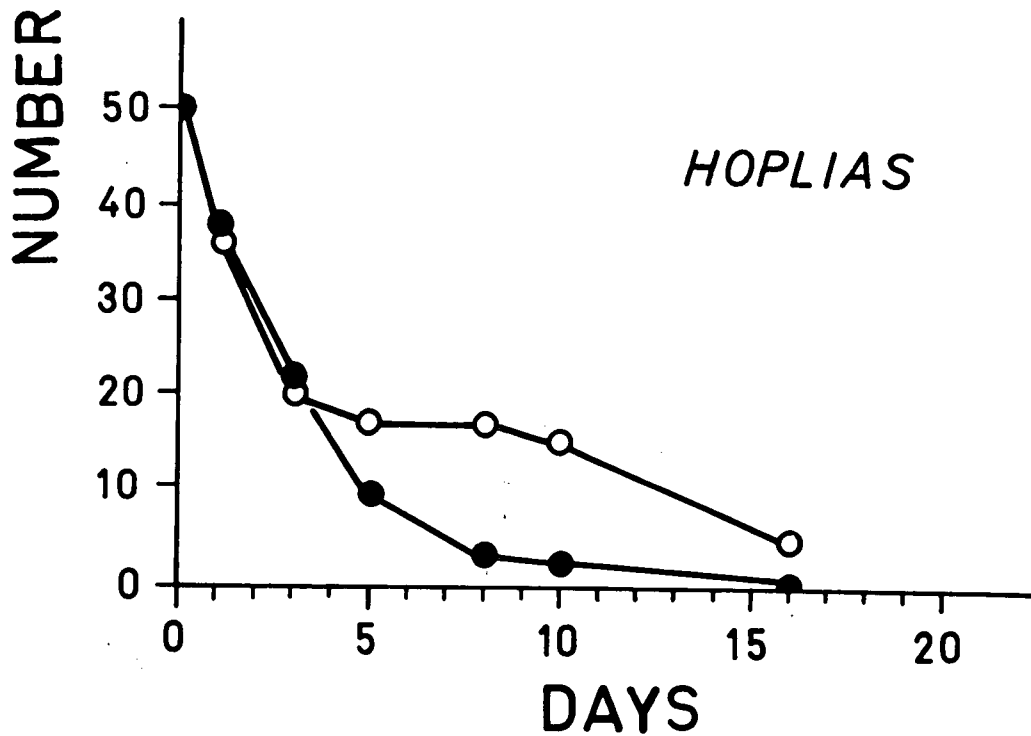
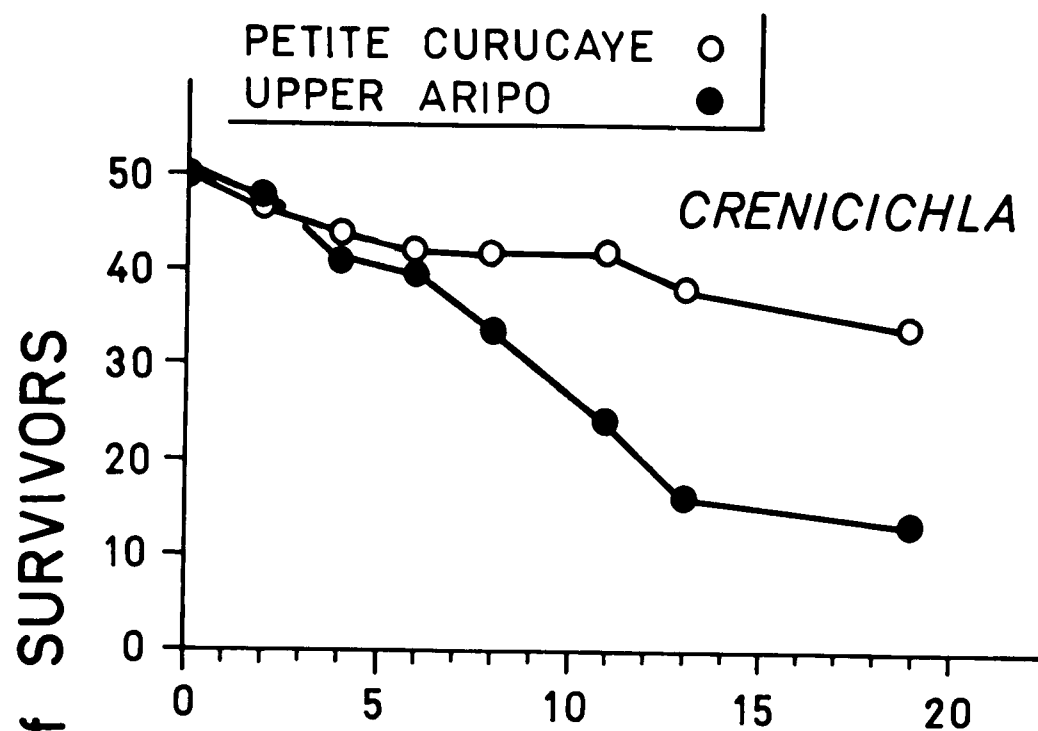
vulnerable (Table 21 E).

On the average it would appear that about twice as many guppies taken from Rivulus habitats were eaten than those from characid-cichlid habitats. Unfortunately at the time these experiments were done I was unable to collect enough Guayamare adults to compare their survival with Lower Aripo fish. This would have been a nice control experiment for the "predation hypothesis" because these streams differ in several abiotic features (temperature, water velocity, depth, shade, turbidity, substrate composition) and are separated by over 30 km; however, both have characid and cichlid predators. Their survival rates probably would have been more similar than Lower Aripo vs. Paria etc. (later in this chapter I do test laboratory stocks of Guayamare against Lower Aripo).

Besides survival experiments between populations exposed to different predator species (Table 21), it is of interest to determine if there are differences between populations exposed to different densities of the same predator species. The 2 outdoor experiments were designed for this purpose, Petite Curucaye (high Rivulus density) versus Upper Aripo (medium Rivulus density). In one test a large Crenicichla was the predator, and in the other 2 large Hoplias were used.

The results (Figure 18) of the 2 tests were quite similar: a parallel decrease for the first 4 to 5 days

FIGURE 18. Relative survival of wild-caught female guppies of two populations exposed to either a single Crenicichla or two Hoplias. Tests were carried out in adjacent 1600 liter outdoor pools.



followed by a reduction in mortality of Petite Curucaye guppies. The test was more decisive when Crenicichla was used.

It should be recalled, however, that Upper Aripo guppies occur above a major barrier to gene flow whereas Petite Curucaye fish are isolated only by distance. The superior survival of the latter might thus reflect the influx of genes from the Santa Cruz R. where characids and cichlids are abundant. On the other hand, the philopatry described in Chapter 3 might favor local adaptation if selection is intense. But this is prejudging the issue--I have thus far presented no data to show that genetics has anything to do with these differences in survival.

Conclusion for survival experiments with wild-caught guppies

The overall results of these experiments, which used a variety of predators, different methods, and different sizes of experimental environments, all point to the conclusion that fish taken from populations exposed to apparently heavy predation pressure enjoy better survival when tested competitively against samples taken from populations exposed to only moderate or light predation (or a different type of predation--there is no reason to assume a priori that Rivulus predation differs from characid-cichlid predation only in a quantitative manner).

The question that immediately arises from these

results is why survival should differ among populations, especially in experiments that control for morphological differences among populations (size and colour). The obvious factor appears to be behavior. I have stressed earlier that under natural conditions guppies exposed to different predators show several conspicuous differences in behavior that appear to make functional sense if viewed as antipredator devices. It seems reasonable that some of these differences may have been operative in the aquarium and pool survival experiments albeit that the environment was more homogeneous than in nature (no shoreline depth gradient, little substrate variation, no water current, finite volume for escape, etc.). It may be concluded then that some of the behavioral differences occurring in natural populations do have survival value, even in a "simplified" predator-prey encounter situation.

The crucial question I asked at this point was: Are the differences in survival a function of phenotypic differences in behavior resulting from differential exposure to predators (avoidance conditioning), or are they a reflection of microevolutionary (i.e. genetic) differences in behavior. I have already remarked upon the lability of avoidance thresholds; for this reason it was necessary to control for experiential factors.

Experiment 6.2 Relative survival of laboratory stocks.

Introduction

The study of behavioral genetics has revealed that the majority of heritable behavior patterns (or differences between similar patterns) are controlled by complex polygenic systems (Caspari, 1967). Only rarely have adaptive or potentially adaptive patterns been found to be under simple (one or two factor) genetic control (e.g. Bastock, 1956; Rothenbuhler, 1964). This does not mean that direct genetic evidence for a polygenic system cannot be obtained--it is just more difficult (Franck, 1969; see Lindzey et al, 1971, for a review of research methodology).

In the present study, I was not directly concerned with obtaining an accurate heritability estimate of some behavioral trait (DeFries, 1967; Van Oortmerssen, 1970); my chief aim was to determine if phenotypic differences observed in nature (and reflected in the survival tests with wild fish) would persist in animals raised under identical conditions in the laboratory.

If such differences still persist (especially after several generations of laboratory breeding) then it is safe to assume that genetic differences are present. This is termed an innate difference by the majority of European ethologists (e.g. Tinbergen, 1959; Hinde, 1959); defined as above, this term is also acceptable to American experimental psychologists (Lehrman, 1970).

When genetic differences are found in natural populations, they are normally assumed to have arisen in response to selective pressures (adaptive genetic difference), though at the molecular level there is much debate about the possibility of neutral genes and "non-Darwinian" evolution.

Methods

The procedures for the establishment of the laboratory stocks have already been outlined (Chapter 2). For the 5 stocks I studied in detail, especially large cultures were maintained. In order to replicate predation experiments where strict control is placed on the size, sex, experience, and geographic origin of the fish, one must raise literally thousands of adult animals.

The experiments were performed using the "standard survival test" procedure as outlined in Chapter 4.

Individual Crenicichla (C-1, C-2, C-3, C-4) were used in 3 adjacent aquaria fitted as described before (12 light-12 dark photoregime). The guppies were at least second generation laboratory stock and the majority were probably fourth and fifth generation fish. All had been maintained under identical laboratory conditions and were predator-naive (excluding attempted cannibalism which was rarely observed).

Two parallel sets of experiments were run. One used only females, the other males. All tests were of the competitive type with Lower Aripo as the reference stock

against which the other 4 were tested, one at a time. In order to rule out possible bias resulting from size-selective predation, each pair of stocks tested was matched as closely as practically feasible with respect to the mean, range, and variance of body sizes (for relevant statistics, refer to Appendix, Table 4). Stocks were recognized by Trypan Blue marks on the caudal peduncle.

This 4 x 2 design (stocks x sex) was also repeated under a different light regime. Preliminary observations on Crenicichla predation in ad libitum tests (predator not starved experimentally) revealed that the majority of predation occurred in dim or dark light. Therefore, to insure that the prey were able to see the predator, I mounted a 7-watt incandescent bulb 65 cm over the water surface. This provided dim, even illumination during the period when the main lights were off.

In each experiment the initial density was always 100, 50 of each stock. Survival was followed initially at 24-hour intervals and then at longer intervals until the density approached or reached zero. The total number of fish used in the entire experiment was 1600, i.e. 800 per light regime.

To facilitate statistical comparisons, I compared the relative numbers of each stock that had been eaten at the census period closest to 50% survival (50 out of 100 fish remain). The observed mortality and that expected

under the null hypothesis (random predation) were tested for significance using the χ^2 statistic (corrected for continuity).

Results

In the experiments with a 12 light-12 dark photoregime, mortality did not differ significantly from chance levels except for Petite Curucaye females (Table 22 A). Surprisingly, Petite Curucaye guppies showed better survival than Lower Aripo (the trend for Petite Curucaye males was in the same direction as the females but not significant).

In the 12 light-12 dim photoregime, there were several notable deviations from random predation (Table 22 B). Lower Aripo fish showed less mortality than Paria or Upper Aripo, confirming the results with wild-caught fish (Table 21 A and 21 B respectively). The differences between Lower Aripo vs. Guayamare, and Lower Aripo vs. Petite Curucaye were not significant, though in the latter case, the trend was still in favour of the Petite Curucaye stock.

When the results of the Lower Aripo vs. Petite Curucaye tests with wild-caught fish (Table 21 C) are compared in a 2 x 2 contingency table with the results obtained for the same test with laboratory stock (sexes pooled), the difference is highly significant (wild vs. laboratory 12 light-12 dark: $\chi^2_{(1 \text{ df})} = 12.61, p < .001$; wild vs. laboratory 12 light-12 dim: $\chi^2_{(1 \text{ df})} = 7.88, p < .01$); when

TABLE 22. Relative mortality of naive laboratory stocks of guppies exposed to Crenicichla. A = 12 light--12 dark photoregime; B = 12 light--12 dim light photoregime; C = males and females pooled (NS = not significant; S = significant at 5% level or less).

A. Test	Predator	Number eaten at approx. 50% mortality		Total	X ²	p		
(females)								
1	C-3	Lower Aripo	26	Guayamare	17	43	1.49	> .20
2	C-1	Lower Aripo	31	P. Curucaye	14	45	5.69	< .02
3	C-1	Lower Aripo	26	Paria	17	43	1.49	> .20
4	C-1	Lower Aripo	28	Upper Aripo	29	57	0.00	-
Total			111		77	188		
(males)								
1	C-1	Lower Aripo	18	Guayamare	25	43	0.84	> .30
2	C-2	Lower Aripo	27	P. Curucaye	17	44	1.84	> .10
3	C-1	Lower Aripo	23	Paria	21	44	0.023	> .80
4	C-2	Lower Aripo	26	Upper Aripo	22	48	0.188	> .50
Total			94		85	179		
B. (females)								
1	C-1	Lower Aripo	30	Guayamare	20	50	1.62	> .20
2	C-4	Lower Aripo	31	P. Curucaye	26	57	0.281	> .50
3	C-4	Lower Aripo	16	Paria	32	48	4.69	< .05
4	C-1	Lower Aripo	16	Upper Aripo	34	50	5.78	< .02
Total			93		112	205		
(males)								
1	C-4	Lower Aripo	26	Guayamare	37	63	1.59	> .20
2	C-1	Lower Aripo	36	P. Curucaye	24	60	2.02	> .10
3	C-2	Lower Aripo	15	Paria	33	48	6.02	< .02
4	C-4	Lower Aripo	18	Upper Aripo	31	49	2.94	< .10
Total			95		125	220		
C. <u>12 light-12 dark</u>								
				X ²		p		
	Lower Aripo	X	Guayamare	0.012		> 0.90 (NS)		
			Petite Curucaye	7.59		< 0.01 (S)		
			Upper Aripo	0.038		> 0.80 (NS)		
			Paria	1.15		> 0.20 (NS)		
<u>12 light-12 dim light</u>								
	Lower Aripo	X	Guayamare	0.00		> 0.90 (NS)		
			Petite Curucaye	2.19		> 0.10 (NS)		
			Upper Aripo	9.09		< 0.005 (S)		
			Paria	11.34		< 0.001 (S)		

only the light treatments for laboratory stock are compared there is no significant difference ($\chi^2_{(1 \text{ df})} = 1.01$, $p > .30$).

Discussion of results of survival experiments

Some casual night observations of Crenicichla feeding behavior during Exp. 6.2 should be brought to bear on the difference in the results obtained with and without supplementary light (i.e. Table 22 A vs. 22 B). From numerous observations it was clear that the Crenicichla were not feeding during the day. Daytime activity consisted almost entirely of "resting" in a pit under the flower pot that had been provided. Though visually isolated in adjacent aquaria, all 4 Crenicichla dug pits in exactly the same location between the front glass and the underside of the flower pot. This seemed puzzling at first but I soon realized that this location provided the best concealment from the prey.

When the lights automatically turned off at night, the predators would become active, first moving into the flower pot and then striking at guppies in all regions of the tank. It was obvious that the guppies were more "helpless" under conditions of nocturnal predation; evasive action was often taken when the predator had almost engulfed the prey. This suggests that Crenicichla has a visual advantage under low light intensity. Since under my

experimental conditions the guppies had no refuge, the inter-stock differences in escape behavior (except Petite Curucaye) may have been inconsequential when the light levels at night were very low.

When light was added though, differences did appear in the predicted direction and closely resembled the results obtained with wild-caught fish. Although no dim light was provided at night in tests with wild-caught guppies, this probably would not have affected the results because the Crenicichla used in these tests were day-active; at night they were quiescent. Since in nature I have seen Crenicichla attack guppies at midday, it is possible that the nocturnal habits observed in Exp. 6.2 were unnatural. The most reasonable explanation is that the Crenicichla learned that guppies are much easier to capture in darkness.

Night observations in nature have revealed that in streams with characids and cichlids, guppies are distributed in a narrow ribbon along the stream edge (shallow water). This might function to reduce the accessibility of guppies to nocturnal predators (e.g. Hoplias). This habitat selection may be effective enough to force Crenicichla to stalk or ambush unwary fish during the day when guppies typically swim further from shore. Apparently this extra "work" was not required in Exp. 6.2 (dark nights) so the predators shifted to nocturnal predation.

The addition of a dim light at night, however,

allowed the guppies to use their vision more effectively (increase in reaction distance) and the experimental results obtained are probably more representative of natural predation. The Crenicichla used in Exp. 6.1 (day-active) apparently had not been in captivity long enough to learn that in the absence of a shallow refuge, guppies are more accessible at night.

This raises the question of why laboratory stocks of Petite Curucaye guppies were the least vulnerable under conditions of nocturnal predation. First it is important to recall that in nature, the founders of this stock suffer from severe Rivulus predation. Also, in the Petite Curucaye R., Rivulus normally feeds under dim light or dark conditions; unless conditioned otherwise, it does the same in the laboratory. Since it is a small predator, the shore offers no refuge for guppies. Thus habitat selection is ineffectual for avoiding Rivulus at night and other survival strategies are required. Though I am not certain what these are, the evidence from Exp. 6.2 suggests Petite Curucaye guppies may have superior scotopic vision. This could be a function of several factors including an increase in the lens diameter or number of rod cells. This might be at the expense of visual acuity (cone vision) but in a very small stream such as Petite Curucaye, this may not be too critical. As in other nocturnal fish, the acoustico-lateralis system may also be better developed.

The hypothesis that intense Rivulus predation in a confined environment has been responsible for the micro-evolution of specialized behavioral and sensory adaptations demands additional experimental testing. I plan to pursue this in the future, looking in detail at the photobehavioral and anatomical features of this population.

Returning to the overall results of the survival experiments with laboratory stocks, it may be concluded that when the vision of guppies is not impaired by very low light levels, Lower Aripo and Guayamare stocks are less vulnerable to Crenicichla than Upper Aripo and Paria stocks. Since these differences persisted in predator-naive stocks that had been bred for several generations under identical laboratory conditions, they are almost certainly genetically determined.

Geographic variation in the antipredator behavior of laboratory stocks

The overall question in this section was prompted by 4 previous findings:

1. Behavioral differences in antipredator behavior occur in natural populations (observational evidence).
2. Differences in vulnerability to predators occur among samples taken from natural populations (Exp. 6.1).
3. Differences in vulnerability to predators occur among samples taken from laboratory stocks (Exp. 6.2).
4. The results of (2) and (3) are positively

correlated (except for Petite Curucaye).

These results raise the question: What behavioral differences occur among laboratory stocks? In effect, this is asking what differences have been inherited from the wild-caught founders of the laboratory stocks.

To answer this question I made detailed measurements of the behavioral parameters that showed geographic variation in nature: (a) microhabitat selection, (b) reaction distance to predators, (c) escape motor patterns of individual fish, (d) schooling behavior.

(a) Microhabitat selection

The field observations have shown that populations of guppies that are exposed to heavy predation by characid and cichlid fish are more restricted to the shoreline. In sections of stream with deep water and low to moderate velocity, the fish who do leave the shoreline occur near the surface.

I wished to test if there are also differences among the 5 laboratory stocks in their tendency to remain close to the shore, or near the surface, or both. Since movement up a shore gradient is also movement towards the water surface, it was necessary to separate these 2 factors in separate experiments, one with and one without a shore depth gradient.

Based on the field observations (Table 19), I expected that Paria, Upper Aripo, and Petite Curucaye would

spend less time near the shore edge and more time in deeper water than either Lower Aripo or Guayamare. Since in nature, Upper Aripo and Petite Curucaye showed less distinct spreading across the stream and less adhesion to the substratum, I predicted these stocks would fall intermediate between Paria and Lower Aripo/Guayamare.

Experiment 6.3 Depth preference of 5 laboratory stocks of guppies.

Methods

In this experiment I measured the depth of spontaneous swimming and exploratory behavior. A 200 liter aquarium (bare except for a sand substrate) was filled to a depth of 35 cm. The water column was divided into 5 depth strata of 7 cm each by a grid on the front pane of the tank. Observations were made from 1 m in front of the tank with the viewing level across from the middle stratum (error due to parallax was small).

The procedure consisted of recording the position of each fish with reference to the 5 strata. Groups of 10 fish per stock (5 males and 5 females of uniform size) were used at a time. They were selected at random from stock aquaria, placed in a plastic basket in the test tank for 10 min and then released (prior to selection all subjects were fed). After an additional 5 min of free swimming in the tank, the position of each animal was recorded at 1-minute intervals

for a total of 30 min. The order of testing for each stock was randomized. Five replicates were completed. Thus 250 fish (5 stocks x 5 replicates per stock x 10 fish per replicate) were used in this experiment with 30 observations per fish.

Results

The results are shown as a depth profile from the surface stratum to the bottom (Figure 19). Even with casual analysis it is clear that the 5 stocks did not spend equal time at the same depth. The data were treated statistically by comparing the frequency of occurrence of fish in either the top or bottom depth stratum. Each stock was compared with the other 4 using the Mann-Whitney U Test; the results are shown as a matrix of probabilities (Table 23). Where pairs of stocks were not predicted to differ (Lower Aripo vs. Guayamare; Petite Curucaye vs. Upper Aripo), two-tailed probabilities are given; all other tests are one-tailed.

The results reveal that Paria had a definite preference for the bottom (deep water); this was significantly different from the other 4 stocks. Lower Aripo spent significantly more time near the surface than Upper Aripo but no difference was observed for the bottom stratum. Petite Curucaye spent significantly more time near the bottom than either Lower or Upper Aripo. All other comparisons showed differences that could have occurred by chance.

FIGURE 19. Depth profile of spontaneous swimming behavior of 5 laboratory stocks of guppies. Refer to Table 23 for statistical analysis.

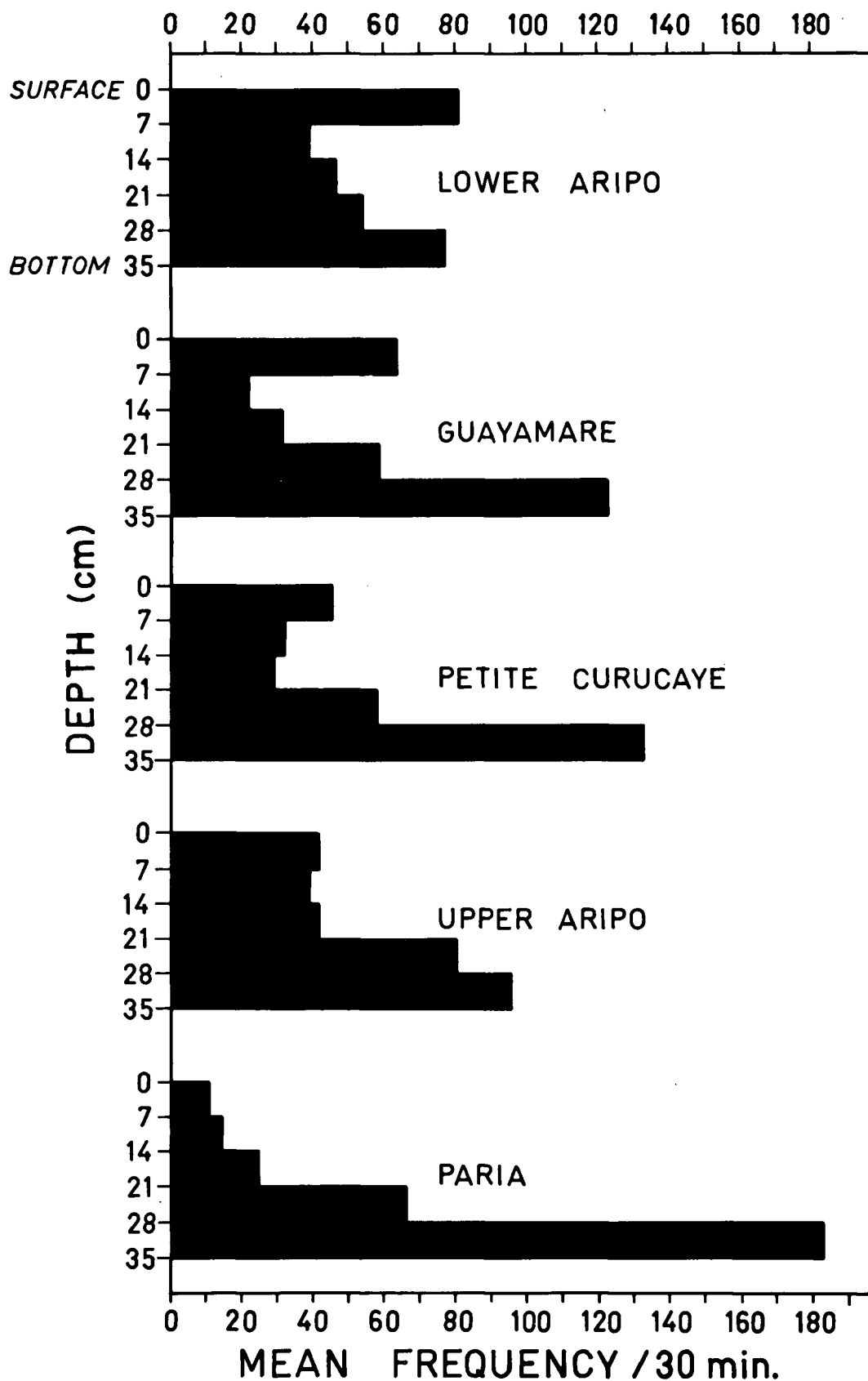


TABLE 23. A comparison of the frequency of occurrence of 5 stocks of guppies in relation to water depth. Values in the matrix are probabilities for each pair of comparisons (Mann-Whitney U Test).

A. Top depth stratum		$H_1: LA/Guay > PCur/UA > Par$				
	Mean frequency	LA	Guay	PCur	UA	Par
Lower Aripo	81.4	-				
Guayamare	63.6	.42	-			
P. Curucaye	46.2	.075	.345	-		
Upper Aripo	41.8	.048	.421	1.00	-	
Paria	10.8	.004	.004	.004	.048	-

B. Bottom depth stratum		$H_1: LA/Guay < PCur/UA < Par$				
	Mean frequency	LA	Guay	PCur	UA	Par
Lower Aripo	77.8	-				
Guayamare	123.2	.15	-			
P. Curucaye	133.2	.048	.345	-		
Upper Aripo	96.2	.500	.210	.008	-	
Paria	183.2	.004	.016	.016	.004	-

When both the upper and lower stratum comparisons are viewed together, it is clear the Paria is distinct from the other 4 stocks. Lower Aripo and Guayamare do move freely in all depth strata but spend a considerable time near the surface. The other stocks fall into an intermediate category. These results support the predictions rather well and suggest that even in the absence of a shore gradient, water current, or predators, some of the differences observed in nature persist in laboratory stocks.

A similar depth measurement made 24 hr after the introduction of the fish to the test tank showed no stock differences. Most fish were near the bottom feeding on the substrate. It appears therefore, that the depth preferences are important when fish are in a "wary state", as was the case when they were taken from their familiar holding tanks and introduced into the test tank. Since all 5 stocks were handled identically, the differences observed in the first 30 min in the new environment could not have been due to differential stress applied by the experimenter, but must have reflected a real difference in the portion of the habitat that was explored. Presumably once no real "danger" was found, the responses waned and the main determinant of habitat choice became the region of food abundance. No feeding behavior was observed during the first 30-minute test for any of the 250 fish. Also there was no difference between the sexes in depth preference.

Experiment 6.4 Shore preference of 5 laboratory stocks of guppies.

In this experiment I wanted to determine if there were any stock differences in the choice of habitat, given a gradient from deep water to dry land. Because fish are placed in a wary condition when they are transferred to a new (and possibly dangerous) environment, it might be expected that stocks derived from populations that are found restricted to the stream edge (see Table 19) may show a habitat preference even in the absence of experience with predators. (Preliminary tests had shown that guppies quickly learn to remain in safe regions of an aquarium when exposed to predators.)

Methods

I tested the 5 stocks in a simulated cross-section of a stream (tank dimensions: L = 240, W = 58, D = 30 cm). One side of the tank had a stream "bank" made of rocks, gravel, and sand. This bank was formed to provide a smooth depth gradient from 20 to 0 cm. This gradient section was divided from the rest of the aquarium by a tight-fitting clear glass partition. Tests with guppies were done on the "shore" side of the partition while the deep side had a predator complement of 2 Crenicichla, 4 Aequidens, and 3 Astyanax. An opaque cover was placed over the clear glass partition to visually isolate the predators from the prey.

On the prey side of the partition, a reference grid

was marked on the front pane; four 20 cm sections were delineated with section 1 as the deep zone next to the partition (and also the predators), sections 2 and 3 as the transitional zones from deep to shallow water, and section 4 as the shallowest section terminating at the shoreline.

For each test, 5 males and 5 females were selected at random from stock tanks. I attempted to keep the sizes of the fish in each group as similar as possible. The fish were first placed in a floating basket for 10 min and then released into section 1 (deep portion). The fish were allowed another 5 min to move about the tank; following this their position was recorded every minute for 30 min (Test A). Another 30-minute recording was made 6 hr later (Test B). After this the opaque partition was removed so that the predators and guppies were in visual communication. The fish were allowed a 5-minute "settling down" period and then another 30-minute recording was made (Test C). A final 30-minute observation was made 24 hr after the removal of the cover (Test D).

Therefore, 1 hour of data was collected on the positions of the guppies in isolation and 1 hour with only a clear glass partition separating predator and prey. All species of predators made numerous approaches to the partition once the cover was removed. Thus sections 1 and 2 were closest to the predators and guppies in these sections encountered a visual threat from the predators.

All 5 stocks were tested in the same manner with 2 replicates per stock (total of 100 fish used).

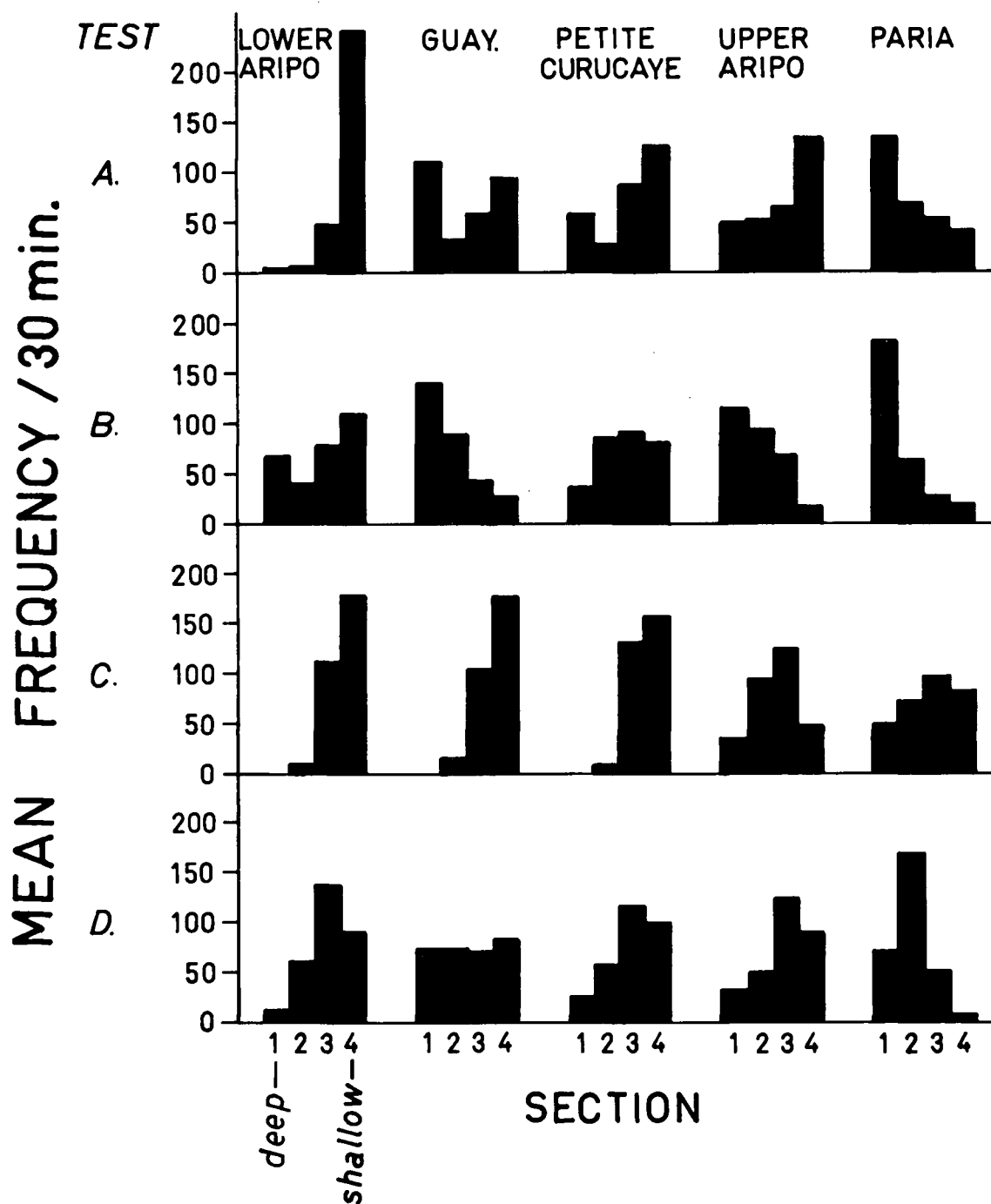
Results

The results are presented as the mean frequency distribution along the depth gradient for each stock during each of the 4 tests (Figure 20). Although the replication was insufficient to permit a statistical comparison among the 5 stocks, some noteworthy trends were evident. In Test A (initial habitat choice), Paria and Lower Aripo showed opposite responses: Paria fish remained mainly in the deep section (1) whereas Lower Aripo fish spent virtually all their time in the extreme shallow zone (4). The other 3 stocks showed intermediate responses with less clear-cut preference.

By Test B the fish had already spent 6 hr exploring the new environment and were now less wary. There was an overall shift away from the shore by all stocks. This was most evident for Lower Aripo.

Tests C and D incorporated a second component into the habitat selection of guppies: the visual presence of predators. Though all guppies that were used had never been exposed to predators before, they all showed a major shift to shallower water (i.e. away from the predators). Of the 10 groups of fish tested (2 per stock), 8 moved to shallower water when the opaque cover was removed. Two groups (1 Lower Aripo and 1 Petite Curucaye) moved to

FIGURE 20. Habitat selection of 5 laboratory stocks of guppies placed in a depth gradient. Refer to text for details.



slightly deeper water but these fish were already restricted to sections 3 and 4 before the predators were made visible, i.e. they were already in a "safe" zone.

It is noteworthy that although Upper Aripo and Paria fish did exhibit a shift towards the shore once the predators were exposed, they nevertheless spent considerable time in sections 1 and 2 (mean observations per fish in sections 1 and 2: Upper Aripo = 12.9; Paria = 12.4). This is in sharp contrast to the other 3 stocks (Lower Aripo = 1.2; Guayamare = 1.8; Petite Curucaye = 0.9).

By Test D (24 hr of exposure to predators), 8 out of the 10 groups shifted to deeper water again. Both Upper Aripo replicates, however, moved to shallower water. This cannot be explained by differences in predator activity since I also had a complete record of the position of each predator relative to the partition (these were made every 30 sec after recording the guppies' positions). The predator records indicate a normal number of visits to the area within 30 cm of the partition. Since the Upper Aripo difference between Test C and D was small, it is safe to conclude that after 24 hr exposure to predators without negative reinforcement (except occasional attacks by Crenicichla which terminated when the predator struck the glass partition), the guppies became habituated to the predators and were less restricted to the shoreline. However, they did remain closer to the shore than before the

predators were visible (Test B). This suggests that the presence of large moving objects elicits an avoidance response that is not contingent upon actual attacks.

Some casual observations made up to 1 week after exposure to predators revealed that the habitat distribution observed after 24 hr of exposure was stable. Even the Paria fish moved only slightly closer to the partition. For all stocks, the distribution observed at Test B did not reappear.

The main conclusion from this experiment is that Lower Aripo and Paria guppies have different habitat preferences. The shore adhesion of Lower Aripo is very reminiscent of the behavior seen under natural conditions, suggesting again that genetic factors may be important in determining these differences. The absence of a well-defined habitat response in Guayamare guppies is curious since in nature these fish are also found very close to shore. The Guayamare habitat does differ from Lower Aripo in 2 main ways: the water is turbid and in most places there is no gentle gradient to deep water--the drop-off is sudden.

With the appearance of predators (Tests C and D), other behavioral components come into play to modify the initial habitat choice (reaction distance and habituation to predators). The responses of the different stocks to the appearance of the predators gives the first hint that the presence of predators does not have the same impact on the behavior of all stocks of guppies. In this experimental

design, however, there were too many variables and not enough replicates to be certain of this difference. The following experiment was designed to more accurately measure the behavioral responses to a predator.

(b) Reaction distance to predators

Experiment 6.5 Reaction distance of 5 laboratory stocks of guppies to a predator.

It has been pointed out that geographic variation in the reaction distance to threatening stimuli is correlated with predation. Fish from streams which have few predators respond at a smaller distance than those exposed to intense predation. The aim of this experiment was to determine if these differences are retained in predator-naive laboratory stocks. If differences observed in the field are the product of differential experience with predators, no differences should appear in these tests (null hypothesis). If genetic factors are involved, the laboratory stocks should resemble to a certain extent the responsiveness of the wild founders, i.e. reaction distance of Lower Aripo/ Guayamare>Petite Curucaye/Upper Aripo>Paria. (Recall the latter could be caught by hand in the field.)

Methods

The experiment was conducted in a long shallow tank filled to a depth of 8.0 cm (tank dimensions same as in Exp. 6.4). The bottom was covered with light brown sand. Illumination was even over the entire length of the tank

(40-watt cool white fluorescent tubes mounted 50 cm above the water). In one end of the tank I placed a small 21 liter glass aquarium containing a preserved 190 mm Crenicichla (same individual as in photograph, Figure 4).

This "predator" was suspended by monofilament so that it hung in midwater. This was arranged on a pulley system which made it possible to either suspend the fish motionless in midwater, or to give it an up-and-down movement manually (amplitude = 2 cm, 1 cycle per sec).

At the other end of the long outer tank, a bottomless white plastic pail was connected to a separate second pulley system. This pail was pushed into the sand and held the test fish before it was released into the tank for an "encounter" with the predator. The distance of release was 200 cm directly in front of the predator. With these 2 pulley systems, therefore, it was possible to release the prey, measure the reaction distance, and "animate" the predator, all from a position directly in front of the predator.

It should be noted that the predator was preserved in formalin but retained most of the markings and colour of a live fish. Tests with a live predator proved unfeasible because of the high degree of variability in the behavior of the predator and its distance from the prey. For this reason I resorted to the use of a preserved predator because its "behavior" and distance to the prey could be

rigidly controlled.

To insure that all fish had identical pre-test experience, the following procedure was strictly enforced. Twenty-four hours before a test, lots were drawn to determine which of the 5 stocks would be tested. Ten fish (5 males and 5 females) of this stock were subsequently selected at random, measured, and placed in individual, numbered glass beakers with 300 ml of water. A small floating Ceratopteris plant was added to each beaker.

For the test, each individual in turn was placed in the opaque pre-release pail for 5 min. After this I moved to the observation area in front of the predator and gently pulled the pail up and out of the water with the pulley, thereby releasing the test fish. In the majority of cases, the fish immediately began exploring the large tank (exploration area = 56 x 216 cm). During this exploratory period, the fish would eventually swim to the other end of the tank where the predator was located. In most cases, the fish perceived the predator and gave an avoidance response (one of the 6 types described previously). The distance from the predator (to nearest 1 cm), behavior, and time since release were recorded for this response. As soon as the response had been elicited, the fish was removed and returned to its beaker and the next fish was tested in the same manner.

Certain fish did not respond to the predator but

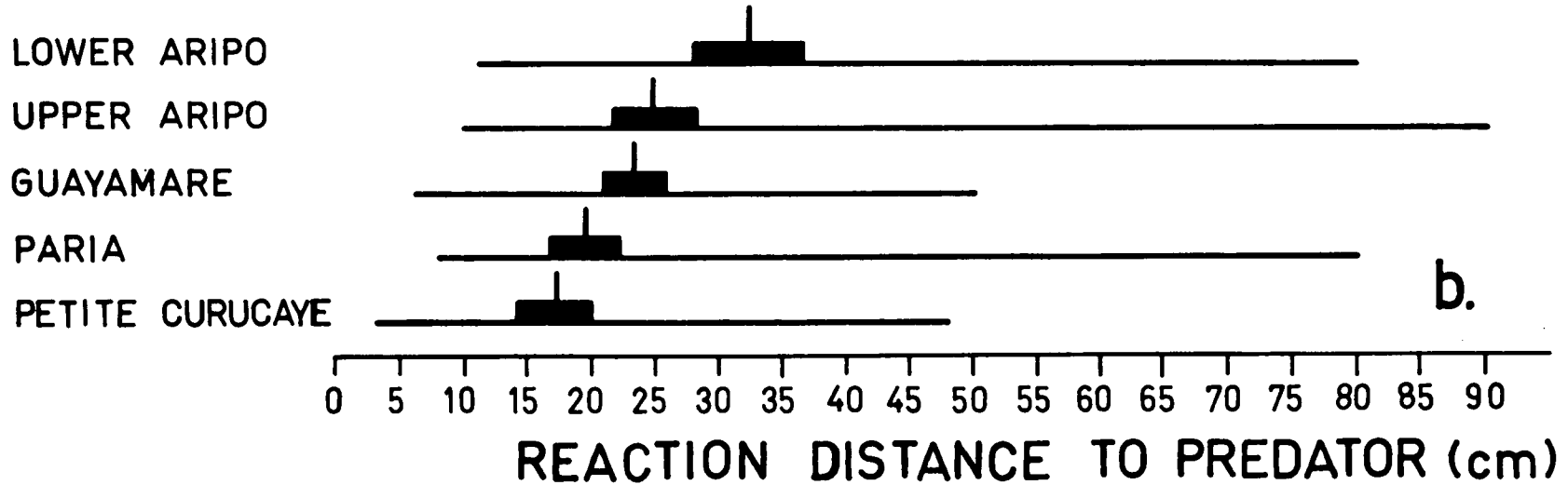
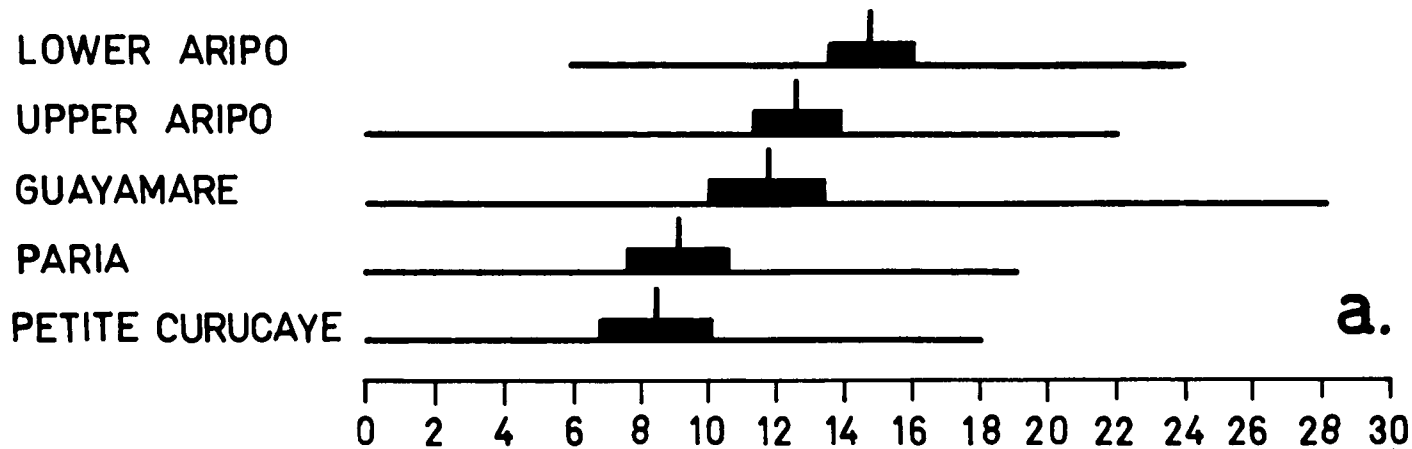
swam right up to the glass and moved along or up and down in front of the predator; this was recorded as "no response". A few fish failed to swim close enough to the predator in the maximum time allotment of 15 min. These fish were retested 24 hr later.

Each group of 10 fish constituted a "set" for that test day. A total of 5 sets were completed for each of the 5 stocks (total of 250 fish used). Two tests were conducted on each fish: the first with the predator motionless and the second with the predator moving in the fashion described above. The first test (A) was completed for 1 set of each of the 5 stocks before they were retested with the moving predator. The same order of testing was followed in test B as in A, thus assuring that each fish had equal residency in the beakers between tests. Each individual therefore, was tested twice, once at test A and once at test B. After this they were discarded.

Results

The results of the experiment in terms of the reaction distance to the predator are given in Figure 21. Non-overlap of the 95% confidence limits indicates a significant difference between any 2 pairs compared (see Eberhardt, 1968 for a useful discussion of this procedure). In test A (predator still) there was a significant difference between Lower Aripo and each of: Guayamare, Paria, and Petite Curucaye. Upper Aripo showed a significantly larger

FIGURE 21. Reaction distance of 5 laboratory stocks of guppies to a dead Crenicichla. (a) predator not moving; (b) predator moving. Vertical line shows the mean, horizontal line the total range, black rectangle the 95% confidence limits.



reaction distance than Petite Curucaye and Paria.

When the predator was moving (test B) the same rank order persisted (perfect rank correlation: $r_s = +1.00$, $p = .01$, one-tailed test). The reaction distances were approximately doubled with the addition of movement. (A non-parametric test of reliability for each of the 250 fish in test A and B revealed that in spite of the individual variability in reaction distance, the method of measurement was a reliable indicator of an individual's reactivity ($\chi^2_{(1 \text{ df})} = 4.07$, $p < .05$), i.e. an individual that responded at a distance greater than the median (for its respective stock) at test A, was likely to respond similarly when retested several days later at test B.)

In test B, Lower Aripo fish were significantly different from all the other stocks. Upper Aripo, Guayamare, and Paria did not differ significantly from each other but Upper Aripo and Guayamare were significantly different from Petite Curucaye.

With the exception of Lower Aripo and Paria, these results do not fit the predictions too closely (Guayamare and Petite Curucaye had shorter reaction distances than expected and Upper Aripo had a greater reaction distance). It is unlikely that this was due to experimental error because the sample size was large and the distance measurements were accurate. It is more probable that experience with predators enhances the reaction distance of Guayamare

fish relatively more than Upper Aripo. It would be interesting to measure the reaction distance of Guayamare vs. Upper Aripo after prolonged exposure to a live Crenicichla.

Relative to Lower Aripo, the shorter reaction distances of Petite Curucaye and Guayamare might reflect habitat differences; the small size of the Petite Curucaye R. and the turbidity of the Guayamare R. would presumably make long reaction distances superfluous.

Finally, reaction distance is an unweighted measure of an animal's avoidance behavior. One animal may respond vigorously at a short distance whereas another may show mild avoidance at a large distance. The results of Exp. 6.2 suggest that although Upper Aripo has a reaction distance equal to Guayamare, Guayamare is less vulnerable when exposed to a real Crenicichla. It is probably safest to use several measures of antipredator behavior in arriving at a prediction of relative vulnerability.

(c) Escape motor patterns of individual fish

Under natural conditions, guppies use a variety of different escape motor patterns to avoid predators. I have stated before that the stimulus strength required to elicit certain of these patterns tends to vary with the exposure to predators. Thus, under the same stimulus conditions, Lower Aripo and Guayamare quickly revert to rapid darts and even surface jumps whereas Paria guppies may only show a mild response. This difference was not measured accurately

in the field; for this reason it is of great interest to know if naive fish show different responses to the same stimulus. If differences do occur in the predicted direction, it gives credibility to the hypothesis that microevolutionary differences in behavior occur among natural populations.

Experiment 6.6 Escape motor patterns of 5 laboratory stocks of guppies exposed to a preserved Crenicichla.

The behavioral responses of the 5 stocks of guppies to a preserved motionless and moving Crenicichla were quantified at the same time as the reaction distances were measured (see Exp. 6.5 for methods).

Results

The responses of the 250 fish under each treatment are tabulated in Table 24. Surface skim was not observed in any of the 500 tests suggesting that it is elicited under conditions of real pursuit.

For the rest of the patterns, it is clear that there is considerable variation within each stock. For test A (predator not moving) the modal response for Lower Aripo, Guayamare, and Upper Aripo was avoidance drift (72%, 48%, and 76% respectively); Paria and Petite Curucaye most often showed weak avoidance drift (46% and 42% respectively). Pooling the results for all 5 stocks in test A, exactly 50% (125/250) of the animals showed avoidance drift.

TABLE 24. Escape motor patterns of 5 laboratory stocks of guppies exposed to a dead Crenicichla. (n = 50 fish per stock)

	STOCK					
Behavior pattern	L. Aripo	Guayamare	U. Aripo	Paria	P. Curucaye	Total
<u>A. Predator not moving</u>						
no response	0	2	2	6	10	20
weak avoidance drift	6	14	5	23	21	69
avoidance drift	36	24	38	13	14	125
turn around	1	4	2	2	1	10
rapid dart	7	6	2	6	2	23
surface jump	0	0	1	0	2	3
Total	50	50	50	50	50	250
<u>B. Predator moving</u>						
no response	0	0	0	0	0	0
weak avoidance drift	8	12	11	21	24	76
avoidance drift	17	13	18	18	15	81
turn around	20	17	16	11	7	71
rapid dart	5	5	5	0	3	18
surface jump	0	3	0	0	1	4
Total	50	50	50	50	50	250

It is noteworthy that 8% of the fish did not respond to the predator at all when it was motionless. A lack of response was most common in Petite Curucaye and Paria where this group made up 20% and 12% of the sample respectively. In contrast, all Lower Aripo guppies responded to the motionless predator.

In test A no inter-stock trends were evident for the patterns turn around, rapid dart, and surface jump because they were elicited so infrequently. The main inter-stock difference was in the relative proportions of animals showing either a weak avoidance drift or avoidance drift.

In test B (predator moving), all animals showed a response and there was a general shift to a more reactive type of behavior pattern. The main effect of moving the predator was to decrease the number of fish showing avoidance drift and increase the number showing the next "higher" category of reaction: turn around (turn around test A = 10; test B = 71). There was no discernible change in the rapid dart and surface jump categories, suggesting again that actual pursuit, and not movement alone, trigger these responses. A turn around in this experiment was sufficient to "escape" from the predator.

For Lower Aripo and Guayamare fish, turn around was the most common response elicited by the moving predator (40% and 34% respectively) while weak avoidance drift remained the modal pattern for Paria and Petite Curucaye

(42% and 48% respectively). Therefore, the only effect of moving the predator on the last 2 stocks was to double the reaction distance; the motor patterns remained virtually unchanged.

Experiment 6.7 Escape motor patterns of 5 laboratory stocks of guppies exposed to Rivulus.

Introduction

Exp. 6.5 and 6.6 have suggested that reaction distance and the proportion of fish showing certain escape motor patterns may be significant measures for comparing stocks, even when the fish are not actively attacked. However, under these conditions most of the antipredator behavior is of an explorative or mildly avoiding nature. These patterns clearly are important as a first line of defense but would not be very effective under real attack.

Therefore, I made simultaneous quantitative observations on the attack behavior of Rivulus and the antipredator behavior of the 5 stocks. Rivulus is an ideal predator for this type of test because it is relatively small and after a period of handling and starvation (training period) will attack prey repeatedly until they are successfully captured.

It should be emphasized that this experiment was an attempt to make comparative measurements of the prey's behavior. Therefore, I had to control the behavior of the predator as much as possible; hence only a single female

Rivulus (75 mm) was used in all tests.

Methods

All tests were conducted between 1500 and 1945 hr in a 60 liter glass aquarium filled to a depth of 20 cm (tank dimensions: L = 61 cm, W = 32, D = 31). Except for a light-brown sand substrate and a "hiding" rock for the predator, the tank was bare. Illumination (12 light-12 dark) was provided by a 40-watt cool white fluorescent tube mounted 35 cm over the water surface.

Sixty-five male guppies, 13 of each stock were used. They were all predator-naive and ranged from 19.0 to 23.5 mm. Since it was known from Exp. 5.4 that prey body size is an important variable in the capture success of Rivulus, great care was taken to match the stocks for body size ($\bar{x} \pm SD$: Lower Aripo = 21.7 ± 1.07 ; Guayamare = 21.0 ± 1.36 ; Petite Curucaye = 20.8 ± 1.52 ; Upper Aripo = 21.9 ± 1.13 ; Paria = 20.8 ± 1.44).

The procedure consisted of placing a single guppy with the predator and recording all the interactions between prey release and prey kill. An Esterline-Angus event recorder (see Chapter 2) was used to quantify the following behavioral measures.

Predator behavior

1. total time, prey release to kill
2. predator orientation time
3. approach frequency
4. attack frequency
5. capture

Prey behavior

1. avoidance drift frequency
2. rapid dart frequency
3. surface skim frequency
4. surface jump frequency
5. hiding duration

Except "predator orientation time", all these patterns have been described before or are self-explanatory. Predator orientation time is the total time spent by the predator orienting and chasing the prey, i.e. total time, prey release to kill, minus periods of predator inactivity (rest pause, etc.). It is a good index of the escape ability of the prey.

Before the prey was released, it was placed behind an opaque partition in one corner of the aquarium for 5 min. Since the predator had been conditioned to receive guppies near the partition, it was usually swimming in this vicinity prior to prey release. To "give the prey a chance" I confined the predator to its hiding place located 50 cm from the partition. This insured that all the guppies had equal time to react to the first few approaches by the predator.

The guppies were presented to the predator at the rate of 1 per 24 hr; no other food was provided so that the hunger level was always the same prior to each test. The sequence of presentation was arranged so that a fish of the same stock would be tested every fifth day (e.g. Day 1: UA; Day 2: Par; Day 3: Guay; Day 4: LA; Day 5: PCur; Day 6: UA; etc.). This design called for 65 consecutive days of prey presentation to meet the 24-hour interval criterion. Unfortunately I was unable to conduct a test on 6 days interspersed throughout the 65-day

schedule. Thus on 6 tests (3 Paria, 2 Lower Aripo, 1 Upper Aripo) the predator had not received a guppy for 48 hr. This did not appear to bias the results.

In 4 tests (3 Lower Aripo, 1 Guayamare) the predator was unable to capture the prey even though it made numerous attempts. On these occasions the recordings were terminated at 15 min and the prey discarded. To keep the hunger level the same for the test the following day, a small amount of Tubifex was fed to the predator.

Results

The results of the experiments showed that several of the behavior patterns occurred so infrequently that they were not amenable to statistical analysis (avoidance drift, surface skim, prey hiding). The low occurrence of avoidance drift supports the idea that it serves mainly an explorative function. Since the predator did not remain motionless or permit the close approach of the prey, this behavior was observed in only 3 out of the 65 fish.

Five of the measures were analyzed for differences between all 10 possible pairs of stocks (Table 25). The Mann-Whitney U Test was used to test the hypothesis that for each measure: Lower Aripo/Guayamare > Petite Curucaye/Upper Aripo > Paria. Comparisons among these 3 groups used the one-tailed probabilities while comparisons within groups (LA vs. Guay; PCur vs. UA) used the two-tailed test because no prediction was made.

TABLE 25. Comparison of predatory success and prey escape behavior for 5 laboratory stocks of guppies. Values in the matrix are the significance levels for each pair of comparisons (Mann-Whitney U Test); NS = $p > .05$.

	Mean frequency	LA	Guay	PCur	UA	Par
<u>A. Frequency of predator approaches to prey</u>						
Lower Aripo	17.8	-				
Guayamare	10.9	.05	-			
P. Curucaye	9.7	.001	NS	-		
Upper Aripo	9.7	.001	NS	NS	-	
Paria	8.7	.001	NS	NS	NS	-
<u>B. Frequency of predator attacks at prey</u>						
Lower Aripo	5.7	-				
Guayamare	3.5	.02	-			
P. Curucaye	3.1	.001	.05	-		
Upper Aripo	2.9	.001	NS	NS	-	
Paria	2.8	.001	.05	NS	NS	-
<u>C. Mean predator orientation time (sec)</u>						
Lower Aripo	64.6	-				
Guayamare	37.7	NS	-			
P. Curucaye	27.9	.05	NS	-		
Upper Aripo	27.1	.025	NS	NS	-	
Paria	20.2	.001	.01	.025	.05	-
<u>D. Frequency of rapid darts away from predator</u>						
Lower Aripo	13.4	-				
Guayamare	8.7	NS	-			
P. Curucaye	7.9	.01	NS	-		
Upper Aripo	7.6	.01	NS	NS	-	
Paria	5.5	.001	.01	.01	.01	-
<u>E. Frequency of surface jumps away from predator</u>						
Lower Aripo	5.1	-				
Guayamare	2.9	NS	-			
P. Curucaye	2.2	.025	.05	-		
Upper Aripo	2.6	.025	NS	NS	-	
Paria	1.5	.001	.001	.05	.01	-

The 4 fish that were not caught by the predator were dropped from the analysis of the attack frequency, thus for this measure the sample size for Lower Aripo was 10, for Guayamare, 12. For all other comparisons the U statistic was calculated for $n_1 = 13$, $n_2 = 13$.

The results of this analysis reveal that the predictions fit the observed data rather closely. The main deviation from the expected relationship among the stocks was the difference between Guayamare and Lower Aripo. By all measures, Lower Aripo fish showed a superior ability to escape predation and in "predator effort per kill" (approach and attack frequency) were statistically distinct from Guayamare. Though the mean predator orientation times between Lower Aripo and Guayamare were not significant, a trend is apparent (64.6 vs. 37.7 sec respectively).

No significant difference was found between Lower Aripo and Guayamare for the 2 motor patterns used in escape. The difference between the means suggests that a statistically significant difference might have been detected if a larger sample size had been used.

For the other comparisons, my hypothesis predicted the outcome more closely. No difference could be detected between Petite Curucaye and Upper Aripo for any of the 5 measures--the mean scores were virtually identical (in sharp contrast to the reaction distance to the preserved Crenicichla, see Fig. 21).

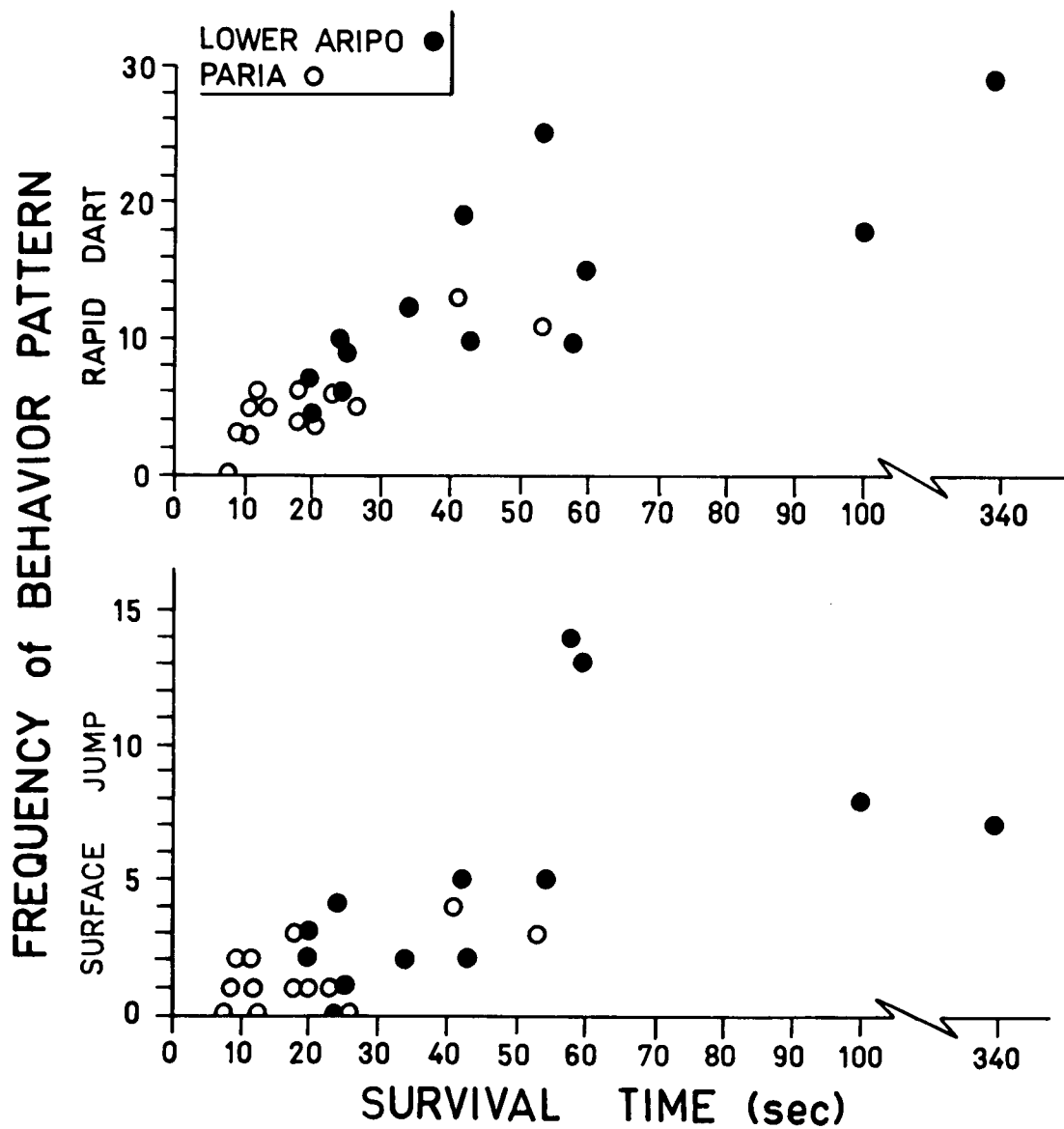
As expected, Paria guppies showed the poorest escape ability of the 5 stocks; the predator orientation time differed significantly from the other 4 stocks. This difference was reflected in the low frequency of rapid darts and surface jumps. The Paria fish were usually approached by the predator to within easy striking range. On the average, it took twice as many attacks (strikes) to capture a Lower Aripo guppy than a Paria. The Paria fish did not appear to dart away as rapidly once struck. The Lower Aripo guppies would thus appear to be more fit in several ways: reaction distance, escape motor patterns, and overall endurance. Three Lower Aripo guppies in fact completely exhausted the predator and were not caught while a fourth endured 14 attacks before it was captured.

As an aid to visualizing the differences between the 2 extreme stocks (Lower Aripo and Paria), I have plotted the frequency of rapid darts and surface jumps against the predator orientation time (= survival time, Figure 22). If I can assume that the behavior of the Rivulus was the same for these tests (I have no evidence to the contrary), then the survival value of the behavior is obvious.

These results suggest that populations of guppies that have been subject to intense selection by characid and cichlid predators are also superior in avoiding Rivulus predation. Whether this applies under natural conditions can only be decided by a transplant experiment. Conceivably

199a

FIGURE 22. Relationship of escape motor patterns to survival time in two laboratory stocks of guppies exposed to Rivulus.



the hyperreactivity of Lower Aripo fish may actually be a disadvantage in a Rivulus-infested stream (e.g. Petite Curucaye). In contrast to the Lower Aripo situation, good escape behavior in Rivulus-infested streams does not guarantee immunity simply by allowing an animal to reach a refuge--Rivulus is quite capable of pouncing on a guppy in only a few mm of water (and even on dry land!). Crenicichla, Hoplias, or Astyanax on the other hand, do not continue to pursue guppies that have eluded them and reached shallow water.

Experiment 6.8 Responses of Guayamare and Paria guppies to
a simulated aerial predator.

Introduction

The behavior pattern "down-dart" has already been described for wild guppies. This response is elicited by a disturbance above the water surface and may be evoked by either the passage of an object or its shadow over a fish. The adaptive value of such behavior is probably linked to the risk of predation from non-aquatic predators such as birds, bats, and perhaps terrestrial animals "fishing" from shore.

It is often stated (but rarely demonstrated) that the antipredator behavior of an animal is adapted to the hunting strategy of its most common predator (e.g. Keenleyside, 1955). For any given species or population of prey, the main threat may stem from one or a few

predators (Larson, 1960; Curio, 1970a, 1970b, 1970c) but more commonly a host of predators with a variety of strategies (Crane, 1952; Thorpe, 1963:69; Kruuk, 1964, 1972; Walther, 1969; Ghiselin and Ricklefs, 1970; Snyder and Snyder, 1971).

As an hypothesis it would be predicted that if a wide variety of predators (with dissimilar hunting methods) have had a selective impact on a prey population, the anti-predator response might be quite non-specific and applicable to all strange moving objects of appropriate size ("generalist").

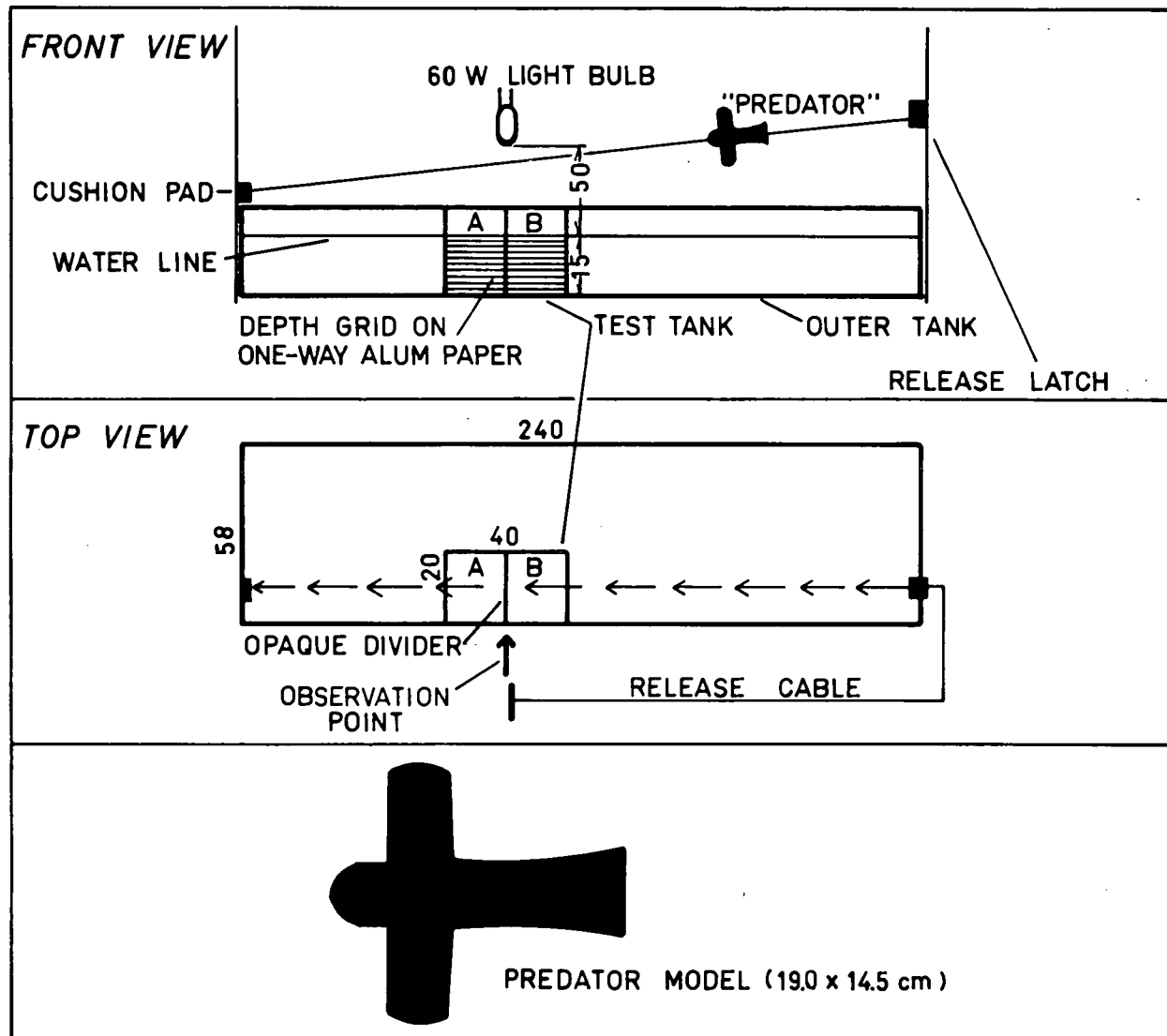
Contrariwise, a response might be very specific if only one species of predator, or group of species (with similar hunting methods) must be avoided ("specialist").

The virtual absence of aquatic predators from the Paria R. presents a unique opportunity to test for the responsiveness to aerial predators since these are presumably the most important in this environment. I have already described differences between wild Paria and Guayamare guppies in their response to an overhead object; I now wish to compare the responses of laboratory stocks of Paria and Guayamare fish.

Methods

The experiment was conducted with the apparatus illustrated in Figure 23. The aim of the experiment was to quantify the response of single, isolated, naive fish to a

FIGURE 23. Test apparatus used for measuring the responsiveness of laboratory stocks of Guayamare and Paria guppies to a simulated aerial predator. All measurements in cm.



standard aerial, escape-evoking stimulus. The aerial "predator" was a black cardboard, bird-shaped model which was attached to a clear monofilament line that ran over the entire length of the outer tank at an angle of 12° (no special significance is attached to the shape of the model). This angle was sufficient to allow the predator to slide down the line with a uniform velocity over the inner tank (test tank--refer to figure).

As the predator passed over the test tank, it produced a shadow about twice its size. The duration of the stimulus was 0.5 sec for each half of the test tank (velocity at this point was 40 cm/sec).

The test tank was enclosed on the sides and rear with black plexiglas and was divided in the middle by a black plexiglas partition. The front of the tank was covered with one-way Alum paper; thus the fish were able to see outside the test tank only directly upwards. A depth grid was marked off on the Alum paper in fifteen 1 cm intervals corresponding to the depth strata from zero to 15 cm.

All tests were conducted in the evening in a darkened room. Only a 60 watt incandescent bulb directly over the mid-point of the test tank provided illumination. Water temperature was held constant at 25.0 C.

Twenty male Guayamare and Paria guppies were selected randomly from stock aquaria and assigned to

individual beakers containing 300 ml of water and a small Sagittaria plant. All fish were measured to assure size homogeneity (Guayamare: $\bar{x} = 17.7 \text{ mm} \pm 0.95 \text{ SD}$; Paria: $\bar{x} = 18.0 \text{ mm} \pm 0.72 \text{ SD}$).

Fish were tested in a paired design which was randomized with respect to the individual of Paria that was paired with the individual of Guayamare. The compartment (A or B) was decided by flipping a coin. Paria was in A nine times, Guayamare 11 times. Five pairs were tested on each night for 4 consecutive nights (total of 20 pairs).

Each fish was placed in its respective compartment (A or B) for 15 min; by this time the guppies were always actively swimming about the compartment. A trial was started as soon as both fish (visually isolated from each other) were above the 10 cm grid mark (i.e. in the top two-thirds of the water column). The predator was released at this time by a cable which I held while viewing the fish directly in front of, and in midline of the grid on the test tank.

The moment the "bird flew" over the right-hand corner of compartment B, the position of both fish was noted (d_1). The maximum downward position of movement of the fish was noted as soon as the predator had passed over both compartments (d_2). The difference ($d_2 - d_1$) is the distance moved in the down-dart.

At the same time as the d_1 depths were recorded,

2 stopwatches were started to measure the duration of "freezing" (i.e. the swimming inhibition of each fish; I defined this period operationally as the length of time each fish remained motionless in its d_2 position).

To reduce variation arising from differences in the position of the fish at the moment of stimulus presentation, I repeated the test 5 times for each pair; a 5-minute interval was allotted between successive tests. Thus for each fish, the distance of the down-dart and the duration of swimming inhibition are based on the means of these 5 subsamples.

Results

To test for the possibility of bias resulting from a stock depth difference at d_1 , I calculated the means and 95% confidence limits for these pre-stimulus depths. The means differed by only 0.7 cm (Paria: 9.0 cm, Guayamare: 8.3 cm) and there was complete overlap in the 95% confidence intervals suggesting that the pre-stimulus conditions were identical for both stocks.

Also, a Sign Test was used to test if there was an effect of the selection of the compartment on the swimming inhibition time (the water surface of compartment A is closer to the predator) but this also proved to be untrue ($p = .504$).

The only significant differences that could be detected were the differences between stocks (Table 26).

TABLE 26. Responses of naive Guayamare and Paria guppies to a simulated aerial predator. The data for each pair of fish are the means for 5 separate trials.

Pair	Down-dart distance (cm)		Duration of swimming inhibition (sec)	
	Paria	Guayamare	Paria	Guayamare
1	3.4	1.0	10.6	7.4
2	3.8	4.2	8.2	3.8
3	2.6	1.0	9.8	5.0
4	3.2	3.2	6.8	6.4
5	3.4	2.8	9.2	9.4
6	5.2	6.2	7.6	13.4
7	2.2	2.2	8.0	4.0
8	3.4	3.8	8.4	3.4
9	4.6	2.0	8.6	3.6
10	5.8	5.4	10.4	8.0
11	3.0	0.8	5.8	2.8
12	3.8	4.2	14.8	3.0
13	4.2	3.0	8.0	6.2
14	4.2	2.0	6.0	4.2
15	4.0	3.0	3.2	3.6
16	3.8	5.2	7.4	2.6
17	8.6	5.8	5.8	8.6
18	4.8	7.8	9.8	12.0
19	4.6	4.2	5.4	3.0
20	4.2	2.2	2.6	0.2
Mean	4.14	3.50	7.82	5.53
Wilcoxon T	41.5		38.5	
N	18		20	
p	< .05		< .01	

A Wilcoxon matched-pairs signed-ranks test was used to test the hypothesis that Paria would have a greater down-dart distance and greater swimming inhibition duration than Guayamare (in conformity with the field observations). This hypothesis could not be falsified for either behavioral measure (down-dart distance: $T = 41.5$, $p < .05$; swimming inhibition time: $T = 38.5$, $.01 > p > .005$).

These results suggest that in the absence of significant aquatic predation, guppies who, in response to an aerial threat, move deeper in the water column and remain motionless longer are favored by natural selection. To validate this hypothesis further it would be necessary to test the 2 stocks with real aerial predators but I did not have the time or facilities to do this.

The hypothesis assumes that moving into deeper water and remaining motionless decreases the likelihood of detection or capture, or both, by aerial predators. From what little is known about the feeding habits of fish-eating bats (see discussion in Chapter 3) and kingfishers, these assumptions are not unreasonable. Though the fishing habits of Trinidad kingfishers have not been studied, observations on the European (Eastman, 1969) and Belted (White, 1936; Salyer and Lagler, 1949; Hyatt, pers. comm.) kingfisher indicate that these predators are most efficient in clear shallow water. However, Eastman (1969) has shown that when the European kingfisher launches an attack from a perch,

the aim at the perch is decisive because the eyes are closed underwater. The "Paria response" would probably be less effective than a rapid dart in countering this strategy. Nevertheless, the response would appear to be admirably suited for the kingfisher's alternate fishing strategy--the hovering attack.

(d) Schooling behavior

Differences among the wild populations of guppies in the tendency to form cohesive groups have been described previously in a general qualitative manner (Table 20). I suggested that a correlation exists between the intensity of aquatic predation and the development of schooling behavior. Since schooling behavior is often thought to have evolved as an antipredator strategy (for reviews see Breder, 1967; Shaw, 1970), I felt it was vital to determine if there were differences in the cohesive properties of groups of naive guppies originating from natural populations that suffer from different predation pressures. Experiments of this type have not been carried out previously and are necessary to understanding the adaptive significance of schooling behavior.

Experiment 6.9 Schooling behavior of 5 laboratory stocks of guppies.

Methods

The tests were conducted in a bare, blue-green

wooden tank (48 x 110 cm) filled to a depth of 3 cm with aged tap water maintained at 25 ± 2 C. Illumination was held constant with a 40-watt cool white fluorescent tube mounted 60 cm above the water surface. The shallow water flattened the school to a depth of one or two fish. A reference grid of ten 528 cm^2 squares was marked on the bottom of the tank to record the position of each fish.

Observations were made by looking into a mirror (mounted on the rear wall of the tank 35 cm above the water and held at a 45° angle to the water) through a small hole in a blind covering the top of the tank. In this way I could observe fish without disturbing them (disturbance increases cohesion).

All recordings were made on groups of 10 mature fish, 5 of each sex chosen randomly from stock aquaria. The size distributions of all test fish were the same. They were fed prior to, and discarded after testing. The testing schedule was randomized with respect to the day and time of day each stock was tested (845-1745 hr). Ten replicates for each of the 5 stocks were completed for a total of 50 groups (total of 500 fish used).

Each group was placed in the test tank for 10 min. After this the position of each of the 10 fish was recorded at 1-minute intervals for 30 min (test A). This was repeated 5 hr later after the fish had explored their new environment (test B). For each 30-minute test I calculated

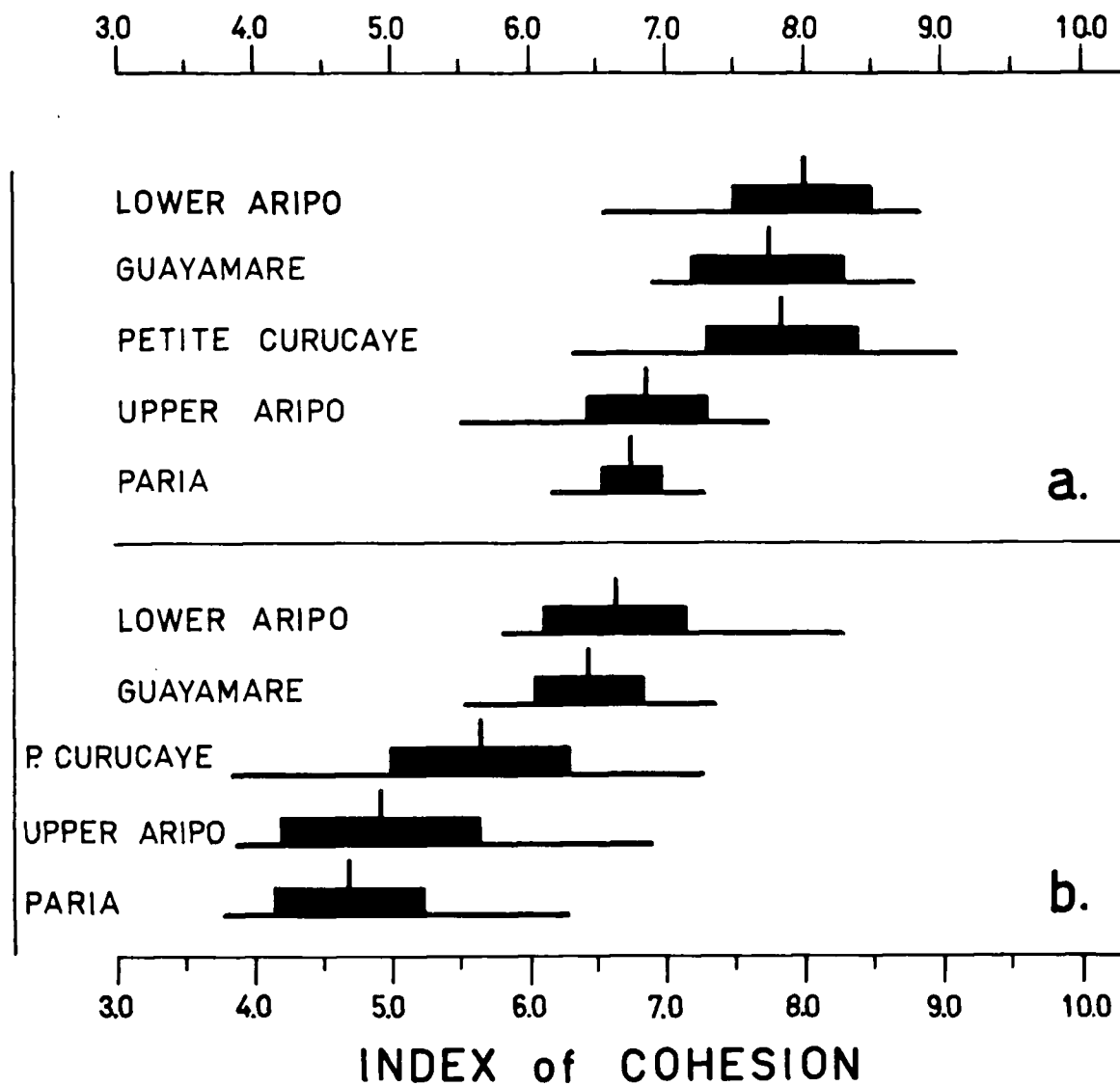
an "index of cohesion" in the following way: for each 1-minute observation the maximum density for any of the 10 grid squares was recorded. The index is the mean maximum density for 30 of these observations. The index has a theoretical minimum of 1 (1 fish in each of the 10 squares) and a maximum of 10 (entire group in the area of 1 square or less). Although this method of quantifying schooling behavior is less precise than those used by other workers (e.g. Williams, 1964; Cullen et al, 1965; Hunter, 1966; Symons, 1971), it appeared to be adequate for the comparative purposes desired here.

Results

Within each stock I found a significant decrease in the index from test A to test B (Figure 24); all stocks were less cohesive once they had explored the tank. However, the degree of dissociation (school spread, or break-up) was greater for Petite Curucaye, Upper Aripo, and Paria than either Lower Aripo or Guayamare.

More noteworthy were the differences among the 5 stocks at both test periods. The trend in the mean index score from a maximum at Lower Aripo to a minimum at Paria closely parallels the variation I observed in nature (Table 20). The stocks derived from rivers where predation is intense have a significantly higher score than those obtained from rivers with low predation. Statistically, Petite Curucaye is inseparable from Guayamare or Lower

FIGURE 24. Mean index of cohesion for five laboratory stocks of guppies. Vertical line shows the mean, horizontal line the total range, black rectangles the 95% confidence limits. (a) = Test A; (b) = Test B



Aripo, though in test B, there is greater overlap with Upper Aripo and Paria. Petite Curucaye thus appears to be a transitory schooler, very cohesive in strange surroundings but quick to spread out once the new environment has been thoroughly explored (< 5 hr). Guayamare and Lower Aripo on the other hand are still quite cohesive after 5 hr.

I have no evidence that the method of measuring schooling resulted in any stock-specific bias. Though all stocks did spend more time in the 4 corner grids than in the others, inspection of individual recordings revealed no tendency for any stock to be more attracted to one region of the test tank than the other. In addition, light intensity and feeding motivation (2 variables known to affect cohesion in the guppy) were controlled and the entire testing schedule was randomized (controlling for potential circadian differences). I conclude that the methods did not introduce a systematic bias favoring greater or lesser cohesiveness in any one stock.

Overall, the results of this experiment confirm the hypothesis that differences in the tendency to school are controlled by underlying genetic factors. But this is only a partial answer to the question of the adaptive significance of differences in schooling in natural populations of guppies. Although the correlation with predation may be a causal one, at the interspecific level it has often been warned that, "we are making a serious error in forcing

ourselves to find a single adaptive feature for schooling (Shaw, 1970: 471)."

However, I have some field and laboratory evidence that suggests schooling may reduce predation. On several occasions I observed an elliptical school of 20 to 50 guppies moving upstream towards a potential predator (large Aequidens) concealed near the shore. The guppies at the front of the school perceived the predator and their evasive movements (rapid darts and surface skim away from the shore) were transmitted to the rear of the school. Thus the advance warning of potential danger was advantageous to the fish who did not see the predator.

Also, in the field I have observed predators (Rivulus) switching their attack course behind a fleeing school of guppies; the same phenomenon has been seen in outdoor pools containing guppies and Hoplias or Crenicichla. These predators do appear to have less difficulty attacking an isolated fish than one in a school ("confusion effect"). It is particularly revealing to watch the eye movements of a Crenicichla lurking near a school of guppies. Guppies that dart away from the school periphery are quickly fixated and if they wander too far are often attacked. The predator appears to be inhibited from darting into the centre of a school. In the laboratory a hungry Crenicichla may wait for hours before an opportune moment arises to strike an unwary guppy. This anecdotal evidence suggests

that schooling behavior functions to increase the perceptual awareness of a group of guppies and also serves to thwart the decision-making apparatus of the predator.

Experiment 6.91 Predatory behavior of Rivulus exposed to Lower Aripo and Paria guppies.

Introduction

If a well-developed schooling response does reduce the risk of predation, then Paria fish should suffer greater relative mortality than Lower Aripo fish when groups of both stocks are exposed to a predator. This experiment has already been done for wild-caught guppies (Table 21 A) and laboratory stocks (Table 22 C) but in both experiments no behavior was recorded so it is not known why Paria suffered greater mortality. As subsequent experiments have revealed, Lower Aripo and Paria stocks differ in many ways other than schooling behavior. Even when schooling is impossible (Exp. 6.7), Lower Aripo guppies are less vulnerable than Paria. Thus schooling behavior per se may not have been responsible for the survival trends observed in the original series of experiments.

To check this I made direct observations of Rivulus attacking mixed groups of Paria and Lower Aripo guppies.

Methods

Two Rivulus of approximately 70 mm were placed in a 40 liter aquarium and starved for 24 hr. Five Paria and

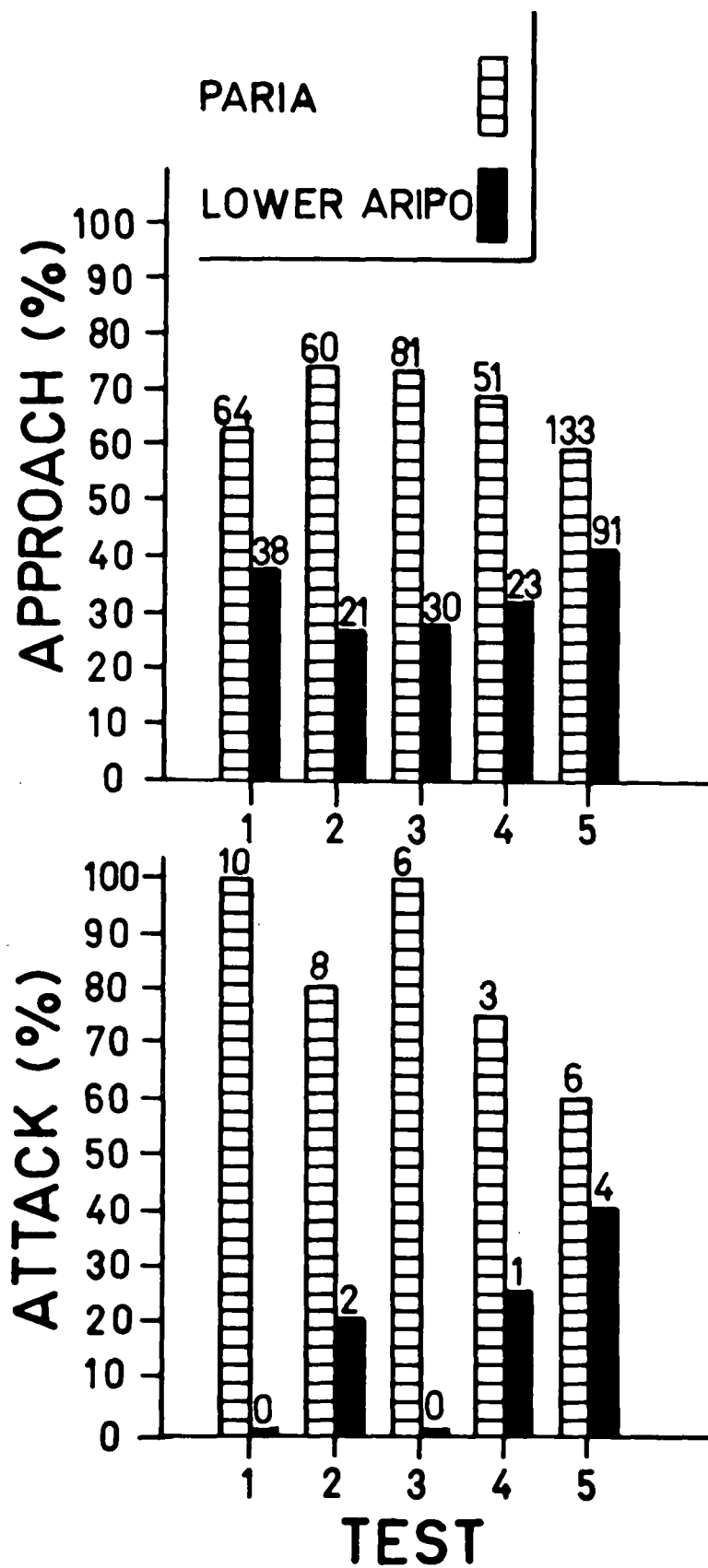
5 Lower Aripo male guppies of equal size were added to this tank and the frequency of approaches, attacks, and captures were recorded in the standard manner (test duration 1 hour). This was replicated 5 times, each test preceded by a 24 hr starvation period. All guppies were predator-naive and second or third generation laboratory stock.

Results

When the 2 stocks of guppies were first added to the predators' tank, they formed a loose collective school in the centre. At first the hungry Rivulus were observed to dash wildly into the school usually without success. The "confusion effect" was clearly seen as the predators shifted their approaches from one prey to the next. Sometimes a Rivulus did appear to get a "fix" on one guppy and would chase it incessantly, quite oblivious of the other guppies that would appear in its path.

Eventually the Rivulus ceased darting into the school but remained some distance away near the bottom. The fish that were approached and attacked were usually the ones that dropped out of the school; in most cases these were Paria guppies. This is reflected in the greater number of approaches and attacks directed towards Paria fish (Figure 25). Because the stock difference in the percentage of attacks (contacts) is greater than in approaches, it suggests Lower Aripo guppies were also better at eluding the predators (cf. Exp. 6.7).

FIGURE 25. Predatory behavior of Rivulus exposed to equal numbers of Paria and Lower Aripo guppies (laboratory stock). Values above each histogram are the actual scores.



It is concluded that schooling behavior reduces the vulnerability of guppies to fish predators. A precise measure of its efficacy is difficult to obtain because the tendency to school is linked to other behavioral traits that also serve to reduce predation.

Conclusion and general discussion of geographic variation in behavior

In this chapter I have attempted to determine the significance of natural variation in the escape responses of certain populations of the guppy. I have also evaluated the adaptive role of population differences in habitat selection and schooling behavior.

The field observations revealed a close correlation between the development of presumed antipredator behavior and the distribution and abundance of predaceous fish. Predation experiments with wild-caught fish demonstrated that samples of guppies taken from a stream infested with characid and cichlid predators (Lower Aripo) had a significant survival advantage when tested against one of several populations exposed to only Rivulus hartii (Petite Curucaye, Upper Aripo, Paria). Since most of these experiments were conducted with female guppies of comparable body size, relative vulnerability was certainly based upon behavioral differences.

In addition, tests with Petite Curucaye and Upper Aripo females suggested that differences in relative

vulnerability exist even between populations exposed to the same predator. This seems to be related to the abundance of predators (and hence presumed predation intensity) within a given stream. Because the rate of gene flow between Petite Curucaye and Santa Cruz Rivers (isolation by distance only) is probably greater than between Upper and Lower Aripo (major waterfall barrier), these results may not have entirely reflected local adaptations to Rivulus predation. Tests on a population exposed to intense Rivulus predation but completely isolated from streams with characids and cichlids will be necessary to establish if Rivulus predation alone can account for the behavioral characteristics of the Petite Curucaye population.

The most significant finding in this chapter was that samples of laboratory populations of guppies that had been bred and reared under identical conditions were taken non-randomly by predators. Because there was good agreement between the tests with wild-caught and naive laboratory stocks, the hypothesis that selective predation has been responsible for the microevolution of behavior is supported.

These experiments also revealed the importance of light in predator avoidance. With the notable exception of the Petite Curucaye stock, selective predation could not be demonstrated under conditions of near total darkness. When dim light was provided, however, (Table 22 B) relative survival closely paralleled the results of experiments

conducted under bright illumination (Exp. 6.1 and 6.7). The reason(s) for the superior survival of Petite Curucaye guppies under dark or dim light conditions is unknown but it is tempting to speculate that this population has a more light-sensitive visual apparatus. I hypothesize that this is a consequence of selection by Rivulus in a confined environment devoid of an adequate refuge.

A comparative study of the lower limits of scotopic vision using the technique of Lang (1967) might be informative (Lang's domestic guppies had a lower scotopic limit of 7×10^{-6} lux, compared to man's lower limit of 7×10^{-7} lux), together with an anatomical investigation (cf. Werner, 1969; King, 1970).

The discovery of survival differences among laboratory stocks initiated a series of detailed behavioral experiments aimed at qualitatively and quantitatively establishing the differences among stocks. Overall there is good concordance among measures taken on independent samples from each of the 5 stocks (Table 27).

Behaviorally, Lower Aripo and Paria are the 2 most distinct stocks (least overlap). I have argued that much of the behavior of Lower Aripo guppies (and also Guayamare and probably many others) has evolved in response to selective predation by characid and cichlid fish. The low relative vulnerability of Lower Aripo fish to predators (Exp. 6.1, 6.2, 6.7) clearly illustrates the survival value

TABLE 27. Concordance of behavioral measures taken on 5 stocks of guppies. For each experiment the mean scores are ranked from greatest to least. The experiment number is given in brackets.

Behavioral measurement	Stock				
	LA	Guay	PCur	UA	Par
Depth preference (6.3) ^a	1	2	3	4	5
Shore preference (6.4) ^b	1	4	3	2	5
Reaction distance (6.5)	1	3	5	2	4
Survival time (6.7)	1	2	3	4	5
Response to aerial predator (6.8) ^c	-	2	-	-	1
Schooling behavior (6.9) ^d	1	2	3	4	5
Sum of ranks	5	13	17	16	24
Mean = 15					
Kendall W = 0.76; s = 190; k = 5; p < .01					

^abased on Table 23A.

^bbased on mean number of observations in section 4 at Test A (refer to Fig. 20).

^comitted from calculation of concordance.

^dbased on mean of Test A and B (refer to Fig. 24).

of the behavioral differences.

For the Paria population, the threat from fish predators is virtually absent. In all tests with real fish predators, this stock was the most vulnerable. These results are attributable in part to a high escape threshold, inappropriate microhabitat selection (in or over deep water, away from shore), and the absence of schooling behavior.

Though Paria guppies are clearly unadapted to fish predation, I suggest that some of their behavior does serve an antipredator function. The tendency for these fish to live dispersed in deep water may offer some safety from aerial predators such as kingfishers and bats which presumably operate most effectively in shallow water. The superior adaptation of Paria guppies to non-aquatic predators is speculative; it is essential that this hypothesis be tested with real aerial predators, preferably in situ.

In the stocks showing intermediate predator avoidance, there are several exceptions to the "predator hypothesis": the long reaction distance of Upper Aripo guppies and the schooling response of Petite Curucaye (at least in test A of Exp. 6.9). I have already discussed the question of reaction distance elsewhere and emphasized that antipredator behavior must always be viewed within the framework of the physical environment. In very small or turbid streams there is no avenue for selection to operate for avoidance at great distances--aquatic predators appear suddenly at a

close range. The best strategy for escape must be the selection of a predator-free microhabitat (if available) or the effective use of escape motor patterns (e.g. rapid dart, surface skim and jump). Clearly, it is important to rely on several measures as an estimate of population differences in antipredator behavior. For example, if reaction distance alone is used, the actual vulnerability of Petite Curucaye and Upper Aripo would be over and underestimated, respectively.

I suggest that schooling is also an antipredator mechanism in the guppy. Field and laboratory observations indicated that this behavior operates as a counteradaptation to a predator's strategy of fixating a single prey fish moments before an attack. Furthermore, guppies in a school share a greater sphere of visual awareness of potential danger in the surrounding environment. Because fish predators occur along every few metres of stream, it is unlikely that schooling protects guppies simply by reducing the probability of a predator encountering a prey (Brock and Riffenburgh, 1960; Olson, 1964). At Lower Aripo, the presence of Crenicichla near concentrations of guppies suggests that both predator and prey may be aware of each other's whereabouts; searching is probably not as important as for pelagic fish predators, or terrestrial predators hunting for concealed prey (Croze, 1970). My observations on the adaptive significance of schooling more closely parallel the descriptions of Manteifel' and Radakov (1961),

Eibl-Eibesfeldt (1962), and Hobson (1968).

Though numerous workers have documented differences in the antipredator behavior of closely related species (Crane, 1952; Cullen, 1957; Hoogland et al, 1957; Hoar, 1958; Hailman, 1965; Benzie, 1965; Heatwole, 1968; Robinson, 1969; Curio, 1969; Baker, 1970), a detailed study of geographic (i.e. intraspecific) variation in antipredator behavior has not been carried out previously (at least in fish). Furthermore, in some reports of geographic differences (e.g. Williamson, 1952) a genetic basis for the differences has not been established. Since the tendency to escape may vary within the lifetime of an individual as a consequence of maturation, habituation, and direct experience of attacks by predators, it is usually necessary to conduct some breeding experiments and control the prior history of the offspring.

Population differences in antipredator adaptations have been described for several species. Much of the recent work on fish has centered on the stickleback family (Gasterosteidae). McPhail (1969) discovered an innate difference between the escape responses (including reaction distance) of the larvae of 2 forms (taxonomic status unclear) of threespine stickleback. He related this difference to the presence or absence of differential selection on fry by a small carnivorous fish, Novumbra hubbsi.

Moodie (1972a, 1972b) argued that on the Queen Charlotte Islands, selective predation (mainly by cutthroat trout) has been responsible for the evolution of a "black" pelagic race of large, long-spined Gasterosteus aculeatus (cf. McPhail, 1969; Semler, 1971).

In a laboratory study of selection by squawfish and cutthroat trout, Moodie, McPhail, and Hagen (unpubl.) found that samples of sticklebacks from one population remained in a shelter longer and were consequently less vulnerable. However, this difference in behavior may actually be correlated with the number of lateral plates of the stickleback (see also Hagen and Gilbertson, 1972). If this is confirmed, it would be an example of behavioral polymorphism rather than a population difference of the type I have described for the guppy.

Little work has been done on geographic variation in the antipredator behavior of other animals. Curio (1961) described racial differences between the mobbing behavior of Spanish and German populations of the pied flycatcher. The German population mobbed both the redbacked shrike and the owl (Strix aluco) whereas the Spanish population mobbed only the latter. This difference coincides with the distribution of these predators since the shrike is not sympatric with the Spanish population.

Johnson (1970) found that 2 geographically isolated subspecies of the lizard Sceloporus occidentalis showed

differences in the distance at which they fled at the approach of a human "predator". The subspecies with the shortest reaction distance was also the most cryptically coloured. A similar difference was found between 2 species of anoline lizards by Heatwole (1968).

The conclusion that can be drawn from these studies (including the present one) is that predation can be a diversifying force in the evolution of behavior. At the intraspecific level this diversity is usually expressed as a quantitative or (less commonly) a qualitative difference in behavior. Behavioral differences are often linked to other antipredator mechanisms such as cryptic or aposematic coloration, spines, protective armour, etc. In the guppy, however, microevolutionary changes in behavior are less easily related to visible morphological differences, i.e. it is very difficult to predict the responsiveness of a guppy to a predator simply by its coloration (of course with females it is impossible). I suspect that there are many other examples of geographic variation in antipredator behavior that have remained undiscovered because there are no morphological clues to population differentiation. For example, had the guppy been a monomorphic species over its entire geographic range, the present study might never have been initiated in the first place.

Predation may also have effects on behavior not directly related to predator avoidance (Cullen, 1957; Wilz,

1971; Ballin, 1973). This is not surprising because predator avoidance often necessitates a major change in the ecological niche of a population. It is tempting to speculate that in this manner predation may result in incipient speciation.

Ballin's (1973) results are very important in this regard because he has discovered a tendency for virgin females of the Paria, Upper Aripo, and Guayamare stocks to respond selectively to their own males. Because Ballin found no support for the hypothesis (Haskins et al, 1961) that females prefer the humanly more conspicuously-colored males, he inferred a behavioral basis for this interstock discrimination.

In another series of tests Ballin uncovered quantitative differences in the frequency and intensity of courtship and intermale aggressive behavior. Some of these differences are probably an indirect result of intense or relaxed predation pressure. Presumably only in a relatively predator-free stream (e.g. Paria, Upper Aripo) can male guppies "afford" to spend a large proportion of their time engaged in conspicuous courtship and aggressive displays.

Summary of Chapter 6

1. Natural populations of guppies exposed to predation by characid and cichlid fish:

- (a) are more restricted to shallow water near the stream shore.

(b) are more cohesive (show a greater tendency to school).

(c) avoid a potential predator at a greater reaction distance.

(d) show escape motor patterns at a lower alarm threshold.

2. To elude a potential fish predator, an individual guppy may use one or several of 6 main escape motor patterns. These patterns appear to be arranged on a scale of increasing reactivity ranging from "precautionary" to "emergency" behavior. No qualitative differences were observed in these motor patterns in fish taken from different populations.

3. The primary sensory modality used in predator avoidance is vision.

4. Experiments with natural predators demonstrated that guppies taken from a population exposed to characid and cichlid predators were less vulnerable than fish taken from populations exposed to Rivulus.

5. Experiments with Crenicichla and predator-naive guppies (second to approximately fifth generation laboratory stock) derived from 5 natural populations revealed differences in relative vulnerability parallel to those found among wild-caught specimens. With one exception (Petite Curucaye), stock differences were only evident when the scotopic vision of guppies was not greatly impaired by

very low light levels. It is hypothesized that Petite Curucaye guppies have evolved a more light-sensitive visual system to contend with nocturnal predation by Rivulus.

6. Paria stock preferred to remain near the bottom of a deep aquarium significantly more than Lower Aripo, Upper Aripo, Guayamare, and Petite Curucaye. There was a tendency for Guayamare and Lower Aripo stocks to spend relatively more time near the surface.

7. Lower Aripo and Paria stocks selected different microenvironments when placed in a simulated shore gradient. The former tended to remain closer to shore and the latter away from shore.

8. The mean reaction distance of the 5 stocks of guppies to a motionless, dead Crenicichla was (in rank order), Lower Aripo>Upper Aripo>Guayamare>Paria>Petite Curucaye. When the "predator" was animated, the reaction distance doubled for all 5 stocks, however, the same rank order persisted.

9. When Lower Aripo, Guayamare, and Upper Aripo stocks were exposed to a motionless, dead Crenicichla, the modal escape motor pattern was avoidance drift; for Petite Curucaye and Paria stocks the modal pattern was weak avoidance drift. The motionless predator elicited no response from 20% of Petite Curucaye and 12% of Paria fish. The addition of motion to the predator evoked a more reactive escape motor pattern from the majority of Lower

Aripo, Guayamare, and Upper Aripo fish but had a lesser effect on Paria and Petite Curucaye stock.

10. The survival time of individual Lower Aripo males placed with a hungry Rivulus was significantly greater than that of Petite Curucaye, Upper Aripo, and Paria males. Of the 5 stocks tested, Paria was caught in the shortest time. Survival time was positively correlated with the frequency of rapid darts and surface jumps.

11. The Paria stock was significantly more responsive to a simulated aerial predator than Guayamare.

12. Groups of Lower Aripo, Guayamare, and Petite Curucaye guppies swam about a strange environment in a more cohesive school than Upper Aripo or Paria fish. All stocks were less cohesive after a 5-hour exploration period but this dispersion was less pronounced in Lower Aripo and Guayamare fish.

13. Field and laboratory observations suggested that schooling behavior functions to increase the perceptual awareness of a group of guppies and also serves to thwart the decision-making mechanisms of the predator ("confusion effect"). It is difficult to demonstrate the survival value of population differences in schooling behavior because the tendency to school is linked to other behavioral traits which also serve as antipredator mechanisms.

14. I conclude that much of the geographic variation in antipredator behavior is caused by genetic differences

that have evolved in response to differential predation pressures. It appears that in some cases (e.g. Lower Aripo and Guayamare vs. Petite Curucaye) these microevolutionary behavioral differences are maintained without a major barrier to gene flow.

CHAPTER 7

GENERAL DISCUSSION AND CONCLUSIONS

Organic diversity is impressive, wonderful, fascinating, or exasperating, according to one's tastes and temperament. Does it have some biological function and meaning?

(Dobzhansky, 1970: 24)

The purpose of this chapter is to review the major findings of this thesis. I shall also indicate the areas where additional research might be most productive. Because this study is the first of a series (Liley and Seghers, unpubl.; Ballin, 1973; Liley, unpubl.) concerned with geographic variation in the behavior and morphology of Trinidad guppies, it was necessary to provide a sufficiently broad background for future work. Consequently only a few aspects have been investigated intensively and many questions left unanswered.

The Environment

In Chapter 3 I described the biotic and abiotic environment of the guppy with special reference to the natural history, diversity, and discontinuous distribution of potential predators, and the major and minor barriers to gene migration. I concluded that Hoplias malabaricus, Crenicichla alta, Astyanax bimaculatus, and Rivulus hartii

are the most important present-day predators of the guppy. Because the distribution of Rivulus normally does not overlap with the distribution of the other predators, it provided a unique opportunity to search for potential geographic variation in antipredator adaptations.

Field observations and collections quickly revealed a wealth of interpopulation diversity in the sex ratio, coloration, body size, and behavior of guppies. The object of this thesis was to assess to what extent (if any) this variation was attributable to differential selection by predators. The most conclusive evidence was obtained for behavioral variation. Although this may be related in part to the greater effort expended on this aspect, I feel that behavior is likely the most sensitive indicator of a micro-evolutionary response to predation.

Coloration and the sex ratio

The primary concern of Chapter 4 was to re-evaluate the "Haskins hypothesis", viz. that variation in coloration is governed by the conflicting pressures imposed by predation and sexual selection. I discovered that this hypothesis was difficult to test satisfactorily.

I attempted to understand the significance of differences in coloration by quantifying variation in the colour patterns of populations living in a diverse range of habitats (Liley and Seghers, unpubl.) and by studying

the mechanisms responsible for sex ratio differences. Extreme departures from a theoretical 1:1 Mendelian sex ratio were correlated with the presence of dense populations of Rivulus. Laboratory experiments revealed that unbalanced ratios (in favour of females) were not caused by genetic differences in the sex determination system. In addition, sex ratios were not correlated with sexual dimorphism in colour.

Predation experiments with Rivulus demonstrated that male guppies were not selectively attacked but were less adept than females at avoiding capture. Size-selective predation by Rivulus probably places males at a selective disadvantage as well. Though I was unable to test the Haskins hypothesis adequately, the circumstantial evidence suggests that the colour of male guppies is not a great liability. Because Ballin (1973) could find no evidence of a female preference for conspicuously-colored males, the question of the significance of geographic variation and polymorphism in coloration is still open.

Although I concluded that sex ratios have little bearing on the problem of the adaptive significance (if any) of colour pattern variation, the work in Chapter 4 does raise some interesting questions regarding sex ratios per se. There is a voluminous literature on this topic, however, most investigators have just tabulated sex ratios without comment, some have offered plausible hypotheses for

sex ratio differences, but very few have ever attempted to test these ideas. Similarly, my study has singled out predation as an important environmental variable determining the sex ratios of guppies, but this was considered only within a narrow frame of reference (colour variation). An interesting question is what effect an extremely unbalanced sex ratio (e.g. Tompire Tributary, Petite Curucaye) has on the population dynamics and mating system of the guppy (and perhaps even the rate of evolutionary change, cf. Giesel, 1972). In addition, the questions raised in Chapter 4 concerning geographic variation in the sex differences in antipredator behavior, and the possibility of mimicry in the guppy-Rivulus system, are worthy of further exploration. For example, how does Rivulus discriminate between its own fry and guppy fry?

Body size

The striking geographic variation in the body size of adult guppies was considered in Chapter 5. This topic was treated only cursorily because controlled experiments (Liley, unpubl.) were still underway to determine if there was a genetic basis for population differences in mean adult body size. Nevertheless, preliminary field observations suggested that genetic differences were very probable (Liley's experiments subsequently confirmed this for several populations) even though a substantial portion of the size

differences was undoubtedly a direct phenotypic response to environmental differences, primarily temperature (I found a significant negative correlation between body size and stream temperature).

Of a multitude of potential selective factors that might be responsible for genetic differences in body size, I investigated only one, size-selective predation. Field and laboratory evidence supported the hypothesis that large guppies enjoy an advantage with respect to Rivulus predation but are more vulnerable to large predators such as Hoplias and Crenicichla. In the laboratory, one aspect of size-selective predation, the handling efficiency of the predator, was shown to be an important factor determining the relative vulnerability of guppies of different sizes. Under natural conditions, however, several additional size-specific mechanisms presumably interact to determine relative vulnerability (e.g. the conspicuousness, accessibility, and escape behavior of the prey; the prior experience and hunger of the predator).

Although size-selective predation is a plausible explanation for the evolution of differences in body size, it does not adequately explain all the size trends. Detailed work will be required on other selective factors such as sexual selection, water velocity, and temperature.

Antipredator behavior

In Chapter 6 I attempted to assess the functional

and evolutionary significance of geographic variation in antipredator behavior. Field observations revealed that where characid and cichlid predators were present, guppies were more restricted to the stream shore, showed a greater tendency to school, avoided a potential predator at a greater distance, and had a lower "alarm threshold". I formulated the hypothesis that these behavioral traits served as antipredator mechanisms and that the geographic variation reflected microevolutionary (i.e., genetic) differences. To test this I conducted predation experiments with samples of wild-caught and predator-naïve (laboratory-bred) guppies that originated from 5 natural populations. These populations encompassed the range of predation pressure (by fish), and the degree of geographic isolation found in the Northern Range.

Overall, the results of these experiments supported my hypothesis, i.e. fish either taken, or descended from, populations exposed to characids and cichlids were relatively less vulnerable than those exposed to Rivulus. To determine why some of the predator-naïve stocks were taken non-randomly, I compared their habitat preferences, schooling behavior, reaction distance, and escape motor patterns. Significant interstock differences were found for several of the measures; generally, these were consistent with the field observations and also the performance of each stock in "competitive" predation experiments. For example, Lower

Aripo, one of the least vulnerable stocks, had a long reaction distance to a standard "predator", was a cohesive schooler, and responded (on first exposure) to a real predator with vigorous escape motor patterns. Paria guppies, on the other hand, tended to be less cohesive, had a shorter reaction distance, and responded weakly to a predator. Consequently they were more vulnerable.

It is important to stress that these generalizations apply only to a "typical" individual of any given stock (behavior scores falling near the mean); in all behavior tests there was some overlap between even the most divergent stocks (Paria vs. Lower Aripo). Presumably some of this intrastock variability also has a genetic basis and forms the raw material upon which stabilizing and directional selection can operate.

Although the main purpose of Chapter 6 was to test the hypothesis that behavioral differences among natural populations of guppies are heritable and serve as anti-predator mechanisms, little attention was given to the role of learning in predator avoidance. It is well-known, however, that the tendency to show antipredator behavior is dependent to some degree on experiential factors (Nice and Ter Pelkwyk, 1941; Hinde, 1954; Schleidt, 1961; Melzack, 1961; George, 1960; Veselov, 1964; Popov, 1953, cited in Manteifel' and Radakov, 1961; Benzie, 1965; Curio, 1969; Dill, 1972).

Though learning undoubtedly occurred during the

course of my long-term experiments, I implicitly assumed that the rates of change in behavior were the same between stocks. This may not be correct. In Exp. 6.9 schools of Petite Curucaye fish tended to dissociate more rapidly than either Lower Aripo or Guayamare. Also in Exp. 6.2, it appeared that after a few hours of exposure to a Crenicichla, Lower Aripo and Guayamare guppies became relatively more reactive (increase in reaction distance and decrease in the threshold for surface jumps, etc.) than Paria guppies.

These observations might reflect different stock (i.e. genotype) x environment interactions. Whether such differences occur between populations of fish is unknown at present, but in view of studies on, (a) strain differences in avoidance conditioning in mice and rats (Collins, 1964; Bovet et al, 1969; Wahlsten, 1972), (b) responses of domestic rat populations to artificial selection for high and low rates of avoidance conditioning (Bignami, 1965) and (c) differences between species of fish in the effect of experience with a predator (Benzie, 1965) or other stimuli (Wodinsky et al, 1962) it seems at least possible that guppy populations may differ in their rates of avoidance conditioning and habituation to predators. Preliminary investigations of habituation (Russell, 1967a) and avoidance conditioning (Werboff and Lloyd, 1963) have been carried out using domestic guppies but not within the framework of a realistic predator-prey system. Perhaps these techniques

could be modified for a comparative study of learning by using more natural "predator" stimuli and quantifying a change in behavior known to be adaptive.

One of the major weaknesses of this work is that I have studied antipredator behavior primarily in a "simplified" laboratory environment devoid of much of the spatial and temporal heterogeneity known to occur in nature (Appendix, Tables 1 and 2). Some of these factors, notably stream morphometry, water velocity, turbidity, substrate composition, light intensity, and the presence of other fish species (non-predators), may play an important role in the way antipredator behavior contributes to survival. More specifically, they may explain some of the anomalous results I obtained in certain tests. For example, the mean reaction distance of Guayamare and Petite Curucaye stock was relatively shorter than I had expected on the basis of predation pressure alone. However, these results are less surprising when it is known that these streams are, respectively, turbid and very small.

Suggestions for future research

With the completion of this thesis, the analysis of the causation of intraspecific variation in guppies has only begun. I hope the following suggestions may stimulate further work on this topic.

1. For each of the principal populations of guppies,

"predation pressure" (i.e. predator biomass x frequency of predation on guppies) should be estimated quantitatively. This will require an intensive sampling program (including suspected aerial predators) to be carried out in all seasons.

2. The survival value of behavior should be demonstrated by transplant experiments in the field. This might answer some of the questions concerning the role of light in predator avoidance and the existence of predator-specific antipredator adaptations.

3. As an expansion of suggestion 2, several large-scale (and long-term!) transplant experiments could be conducted. The streams flowing off the northern face of the Northern Range (e.g. Paria, Marianne, Yarra) are devoid of characids and cichlids (see Figure 3). Nevertheless the ecological conditions for such species as Crenicichla and Astyanax appear to be ideal. A revealing experiment would be to stock the Paria R. with Crenicichla and monitor the immediate and long-term changes (if any) in (a) the coloration of males, (b) the sex ratio, (c) body size, (d) anti-predator behavior--especially habitat selection, reaction distance, and schooling behavior, (e) courtship and aggressive behavior (cf. Ballin, 1973), and (f) population parameters such as age structure and fecundity.

A parallel-flowing stream (Yarra) also devoid of Crenicichla could serve as a control. (The colour patterns

[Liley and Seghers, unpubl.] and antipredator behavior [personal observations] of guppies in this stream are very similar to Paria.) If, after the introduction of Crenicichla, Paria guppies tended to converge towards a Lower Aripo phenotype, this would be convincing proof of the efficacy of a piscivore in natural selection.

Another revealing experiment would be to stock a small stream devoid of fish (there are several along the north coast of Trinidad) with a founder population of Lower Aripo guppies and monitor the same morphological and behavioral parameters as above.

With a natural generation time of 3-4 months, I suspect that over several years there would be ample opportunity (through segregation and recombination) for concealed variation to be released and exposed to new selection pressures, viz. relaxed or intensified predation. Mather (1970) has stated that it is not uncommon in artificial selection experiments for the mean of a quantitative trait to be pushed beyond the range of the original population in about 12 generations. Whether selection can proceed this rapidly in a well-buffered natural population remains to be demonstrated.

4. The work in Chapter 6 should be extended into a more systematic study of the genetics of antipredator behavior. There is a considerable psychological literature on the behavior genetics of "fear" and "boldness" in mammals

(Dawson, 1932; Foster, 1959; Whitney, 1970; Plutchik, 1971) but information for fish (and lower vertebrates in general) is virtually non-existent. As Thiessen (1972: 116) rightly points out, "Behavioral evolution has been left primarily to the ethologists, who, in spite of their impressive accomplishments, lack genetic sophistication. . . ." Nevertheless, both Bruell (1967) and Thiessen (1972) stress that the kinds of information which are now available for the guppy (adaptive significance of behavior differences) are a necessary prerequisite for future progress in the science of behavior genetics.

I propose that a detailed study be carried out on the behavior of the hybrids (F_1 , F_2 , and backcrosses) of some of the more divergent stocks (e.g. Lower Aripo x Paria). Another approach would be to deliberately select (or let a predator select) for animals showing superior escape behavior. It would be of considerable theoretical interest, for example, to determine if artificial selection for increased reaction distance has any effect on other, possibly tightly-linked, or "coadapted" behavioral traits such as schooling behavior, habitat preference, courtship display, etc.

5. The final suggestion is that work should be done on the proximate factors responsible for the behavioral differences. Why, on its first meeting with a hungry Rivulus, does a Lower Aripo guppy respond with a rapid dart

followed by a series of surface jumps, when under the same circumstances a Paria guppy hardly responds at all? Why did 20% of Petite Curucaye and 12% of Paria guppies not "recognize" the motionless Crenicichla in Exp. 6.6?

These questions have a direct bearing on the search for the elusive "innate releasing mechanism" (IRM) underlying antipredator behavior. Though interesting hypotheses can be generated from ethological analyses (e.g. Curio, 1969), it will eventually be necessary to translate overt behavioral differences into genetic effects on the structure or function of the nervous system (Manning, 1967). This could pose quite a challenge because the neural correlates of avoidance behavior have been studied in only a few animals (e.g. Ewert, 1970); furthermore very little is known about the genetic aspects of intraspecific variation in sensory mechanisms (Marler and Hamilton, 1966).

In the guppy, some of the geographic differences in behavior may be caused simply by differences in the structure or sensitivity of the eye. Nevertheless, it is very probable that other peripheral centres and even the CNS are differentially responsive to stimuli associated with predators.

LITERATURE CITED

- Allen, K. R. 1969. Distinctive aspects of the ecology of stream fishes: a review. J. Fish. Res. Bd. Canada 26: 1429-1438.
- Alm, G. 1959. Connection between maturity, size, and age in fishes. Rept. Inst. Freshw. Res. Drottningholm 40: 5-145.
- Anderson, F. S. 1961. Effect of density on animal sex ratio. Oikos 12: 1-16.
- Baerends, G. P., R. Brower, and H. T. Waterbolk. 1955. Ethological studies on Lebistes reticulatus Peters. I. Analysis of the male courtship pattern. Behaviour 8: 249-334.
- Bainbridge, R. 1960. Speed and stamina in three fish. J. Exp. Biol. 37: 129-153.
- Baker, R. R. 1970. Bird predation as a selective pressure on the immature stages of the cabbage butterflies, Pieris rapae and P. brassicae. J. Zool., Lond. 162: 43-59.
- Ballin, P. J. 1973. M. Sc. Thesis. Univ. British Columbia.
- Barlow, G. W. 1961. Causes and significance of morphological variation in fishes. Syst. Zool. 10: 105-117.
- Bastock, M. 1956. A gene mutation which changes a behavior pattern. Evolution 10: 421-439.
- Beebe, W. 1925. Studies of a tropical jungle; one quarter of a square mile of jungle at Kartabo, British Guiana. Zoologica 6: 5-193.
- Beebe, W. 1952. Introduction to the ecology of the Arima Valley, Trinidad, B. W. I. Zoologica 37: 157-183.
- Belcher, C., and G. D. Smooker. 1936. On the birds of the Colony of Trinidad and Tobago. Ibis 6: 792-813.
- Benzie, V. L. 1965. Some aspects of the anti-predator responses of two species of stickleback. D. Phil. Thesis. Univ. Oxford. 149 p.

- Berry, R. J., and J. H. Crothers. 1968. Stabilizing selection in the dog-whelk (Nucella lapillus). J. Zool., Lond. 155: 5-17.
- Bertalanffy, L. von. 1938. A quantitative theory of organic growth. Human Biol. 10: 181-213.
- Beukema, J. J. 1968. Predation by the three-spined stickleback (Gasterosteus aculeatus L.). The influence of hunger and experience. Behaviour 31: 1-126.
- Beyerle, G. B., and J. E. Williams. 1968. Some observations of food selectivity by northern pike in aquaria. Trans. Amer. Fish. Soc. 97: 28-31.
- Bignami, G. 1965. Selection for high rates and low rates of avoidance conditioning in the rat. Anim. Behav. 13: 221-227.
- Bloedel, P. 1955. Hunting methods of fish-eating bats, particularly Noctilio leporinus. J. Mammal. 36: 390-399.
- Boeseman, M. 1960. The fresh-water fishes of the island of Trinidad. Stud. Fauna Curaçao Caribb. Isl. 10: 72-153.
- Boeseman, M. 1964. The fresh-water fishes of the island of Trinidad: addenda, errata, et corrigenda. Stud. Fauna Curaçao Caribb. Isl. 20: 52-57.
- Bovet, D., F. Bovet-Nitti, and A. Oliverio. 1969. Genetic aspects of learning and memory in mice. Science 163: 139-149.
- Brawn, V. M. 1969. Feeding behaviour of cod (Gadus morhua). J. Fish. Res. Bd. Canada 26: 583-596.
- Breder, C. M., Jr. 1967. On the survival value of fish schools. Zoologica 52: 25-40.
- Breder, C. M., Jr., and C. W. Coates. 1932. A preliminary study of population stability and sex ratio of Lebistes. Copeia 1932: 147-155.
- Breder, C. M., Jr., and C. W. Coates. 1935. Sex recognition in the guppy, Lebistes reticulatus Peters. Zoologica 19: 187-207.

- Brett, J. R., J. E. Shelbourn, and C. T. Shoop. 1969. Growth rate and body composition of fingerling sockeye salmon, Oncorhynchus nerka, in relation to temperature and ration size. J. Fish. Res. Bd. Canada 26: 2363-2394.
- Brock, V. E., and R. H. Riffenburgh. 1960. Fish schooling: a possible factor in reducing predation. J. Conseil Int. Exploration Mer 25: 307-317.
- Brooks, J. L. 1968. The effects of prey size selection by lake planktivores. Syst. Zool. 17: 273-291.
- Brooks, J. L. 1971. The evolutionary effects of predation. Trans. Amer. Micros. Soc. 90: 105-106.
- Brooks, J. L. and S. I. Dodson. 1965. Predation, body size, and composition of plankton. Science 150: 28-35.
- Brower, J. V. Z. 1960. Experimental studies of mimicry. IV. The reactions of starlings to different proportions of models and mimics. Amer. Natur. 94: 271-282.
- Brown, M. E. 1957. Experimental studies of growth, p. 361-400. In M. E. Brown (ed.) Physiology of fishes. Vol. 1. Academic, New York.
- Bruell, J. H. 1967. Behavioral heterosis, p. 270-286. In J. Hirsch (ed.) Behavior-genetic analysis. McGraw-Hill, New York.
- Cain, A. J., and P. M. Sheppard. 1954. Natural selection in Cepaea. Genetics 39: 89-116.
- Calaprice, J. R. 1969. Production and genetic factors in managed salmonid populations, p. 377-388. In T. G. Northcote (ed.) Symp. on Salmon and Trout in Streams. Institute of Fisheries, Univ. British Columbia.
- Caspari, E. 1967. Gene action as applied to behavior, p. 112-134. In J. Hirsch (ed.) Behavior-genetic analysis. McGraw-Hill, New York.
- Clark, E., and L. R. Aronson. 1951. Sexual behavior in the guppy, Lebistes reticulatus (Peters). Zoologica 36: 49-66.
- Collins, R. L. 1964. Inheritance of avoidance conditioning in mice: a diallel study. Science 143: 1188-1190.

- Comfort, A. 1961. The longevity and mortality of a fish (Lebistes reticulatus Peters) in captivity. Gerontologia 5: 209-222.
- Crane, J. 1952. A comparative study of innate defensive behavior in Trinidad mantids (Orthoptera, Mantoidea). Zoologica 37: 259-293.
- Croze, H. 1970. Searching image in Carrion crows. Z. Tierpsychol., Suppl. 5: 1-86.
- Cullen, E. 1957. Adaptations in the Kittiwake to cliff-nesting. Ibis 99: 275-302.
- Cullen, J. M., E. Shaw, and H. A. Baldwin. 1965. Methods for measuring the three dimensional structure of fish schools. Anim. Behav. 13: 534-543.
- Curio, E. 1961. Rassenspezifisches Verhalten gegen einen Raubfeind. Experientia 17: 188-189.
- Curio, E. 1965. Zur Geographischen Variation des Feinderkennens einiger Darwinfinken (Geospizidae). Zool. Anz., Suppl. 28: 466-492.
- Curio, E. 1969. Funktionweise und Stammesgeschichte des Flugfeinderkennens einiger Darwinfinken (Geospizinae), Z. Tierpsychol. 26: 394-487.
- Curio, E. 1970a. Die Selektion dreier Raupenformen eines Schwärmers (Lepidopt., Sphingidae) durch einen Anolis (Rept., Iguanidae). Z. Tierpsych. 27: 899-914.
- Curio, E. 1970b. Validity of the selective coefficient of a behaviour trait in hawkmoth larvae. Nature 228: 382.
- Curio, E. 1970c. Die Messung des Selektionswertes einer Verhaltensweise. Verh. Dt. Zool. Ges. 64: 348-352.
- Davis, C. C. 1968. Quantitative feeding and weight changes in Poecilia reticulata. Trans. Amer. Fish. Soc. 97: 22-27.
- Dawson, W. M. 1932. Inheritance of wildness and tameness in mice. Genetics 17: 296-326.
- De Fries, J. C. 1967. Quantitative genetics and behavior: overview and perspective, p. 322-339. In J. Hirsch (ed.) Behavior-genetic analysis. McGraw-Hill, New York.

- Dill, L. 1972. Ph. D. Thesis. Univ. British Columbia.
- Dingle, H. 1966. The effect of population density on mortality and sex ratio in the milkweed bug, Oncopeltus, and the cotton stainer, Dysdercus (Heteroptera). Amer. Natur. 100: 465-470.
- Dobzhansky, T. 1970. Genetics of the evolutionary process. Columbia Univ. Press, New York, 505 p.
- Eastman, R. 1969. The kingfisher. Collins, London.
- Eberhardt, L. L. 1968. An approximation to a multiple-comparison test. Copeia 1968: 314-319.
- Ehrlich, P. R., and P. H. Raven. 1969. Differentiation of populations. Science 165: 1228-1232.
- Eibl-Eibesfeldt, I. 1962. Freiwasserbeobachtungen zur Deutung des Schwarmverhaltens verschiedener Fische. Z. Tierpsychol. 19: 165-182.
- Estes, R. D., and J. Goddard. 1967. Prey selection and hunting behavior of the African wild dog. J. Wildl. Mgmt. 31: 52-70.
- Ewert, J.-P. 1970. Neural mechanisms of prey-catching and avoidance behavior in the toad (Bufo bufo L.). Brain Behav. Evol. 3: 36-56.
- Feinberg, E. H., and D. Pimentel. 1966. Evolution of increased "female sex ratio" in the blowfly (Phaenicia sericata) under laboratory competition with the housefly (Musca domestica). Amer. Natur. 100: 235-244.
- Fisher, R. A. 1958. The genetical theory of natural selection. 2nd revised ed. Dover, New York, 291 p.
- Foster, D. D. 1959. Differences in behavior and temperament between two races of the deer mouse. J. Mammal. 40: 496-513.
- Franck, D. 1969. Genetische Grundlagen der Evolution tierischer Verhaltensweisen. Zool. Anz. 183: 31-46.
- Fryer, G. 1965. Predation and its effects on migration and speciation in African fishes: a comment. Proc. Zool. Soc. Lond. 144: 301-322.

- Galbraith, M. G., Jr. 1967. Size-selective predation on Daphnia by rainbow trout and yellow perch. Trans. Amer. Fish. Soc. 96: 1-10.
- Geodakyan, V. A., and V. I. Kosobutskii. 1972. Control of the sex ratio in fish by the feedback mechanism, p. 122-130. In B. I. Cherfas (ed.) Genetics, selection, and hybridization of fish. Israel Program Sci. Transl., Jerusalem.
- George, C. J. W. 1960. Behavioral interaction of the pickerel (Esox niger Le Sueur and Esox americanus Le Sueur) and the mosquitofish (Gambusia patruellis [Baird and Girard]). Ph. D. Thesis. Harvard Univ.
- Gery, J. 1969. The fresh-water fishes of South America, p. 828-848. In E. J. Fittkau, J. Illies, H. Klinge, G. H. Schwabe, and H. Sioli (ed.) Biogeography and ecology in South America. Vol. 2. Junk, The Hague.
- Ghiselin, J., and R. E. Ricklefs. 1970. Prey population: a parsimonious model for evolution of response to predator species diversity. Science 170: 649-651.
- Gibson, M. B., and B. Hirst. 1955. The effect of salinity and temperature on the pre-adult growth of guppies. Copeia 1955: 241-243.
- Giesel, J. T. 1972. Sex ratio, rate of evolution, and environmental heterogeneity. Amer. Natur. 106: 380-387.
- Göz, H. 1941. Über den Art- und Individual-geruch bei Fischen. Z. Vergl. Physiol. 29: 1-45.
- Gunter, G. 1950. Correlation between temperature of water and size of marine fishes on the Atlantic coast. Copeia 1950: 298-304.
- Hagen, D. W., and L. G. Gilbertson. 1972. Geographic variation and environmental selection in Gasterosteus aculeatus L. in the Pacific Northwest, America. Evolution 26: 32-51.
- Hailman, J. P. 1965. Cliff-nesting adaptations of the Galapagos swallow-tailed gull. Wilson Bull. 77: 346-362.
- Hamilton, T. H. 1961. The adaptive significance of intra-specific trends of variation in wing length and body size among bird species. Evolution 15: 180-195.

- Hanson, A. J., and H. D. Smith. 1967. Mate selection in a population of sockeye salmon (Oncorhynchus nerka) of mixed age-groups. J. Fish. Res. Bd. Canada 24: 1955-1977.
- Hartman, G. F. 1969. Reproductive biology of the Gerrard stock rainbow trout, p. 53-67. In T. G. Northcote (ed.) Symp. on Salmon and Trout in Streams. Institute of Fisheries, Univ. British Columbia.
- Haskins, C. P., and E. F. Haskins. 1949. The role of sexual selection as an isolating mechanism in three species of poeciliid fishes. Evolution 3: 160-169.
- Haskins, C. P., and E. F. Haskins. 1950. Factors governing sexual selection as an isolating mechanism in the poeciliid fish Lebistes reticulatus. Proc. Nat. Acad. Sci. 36: 464-476.
- Haskins, C. P., and E. F. Haskins. 1951. The inheritance of certain color patterns in wild populations of Lebistes reticulatus in Trinidad. Evolution 5: 216-225.
- Haskins, C. P., and E. F. Haskins. 1954. Note on a "permanent" experimental alteration of genetic constitution in a natural population. Proc. Nat. Acad. Sci. 40: 627-635.
- Haskins, C. P., E. F. Haskins, J. J. A. McLaughlin, and R. E. Hewitt. 1961. Polymorphism and population structure in Lebistes reticulatus, p. 320-395. In W. F. Blair (ed.) Vertebrate Speciation. Univ. Texas Press, Austin.
- Haskins, C. P., P. Young, R. E. Hewitt, and E. F. Haskins. 1970. Stabilised heterozygosis of supergenes mediating certain Y-linked colour patterns in populations of Lebistes reticulatus. Heredity 25: 575-589.
- Hastings, R. W., and R. W. Yerger. 1971. Ecology and life history of the diamond killifish, Adinia xenica (Jordan and Gilbert). Amer. Midl. Nat. 86: 276-291.
- Heatwole, H. 1968. Relationship of escape behavior and camouflage in anoline lizards. Copeia 1968: 109-113.
- Herklots, G. A. C. 1961. The Birds of Trinidad and Tobago. Collins, London. 287 p.

- Hester, F. J. 1964. Effects of food supply on fecundity in the female guppy, Lebistes reticulatus (Peters). J. Fish. Res. Bd. Canada 21: 757-764.
- Hinde, R. A. 1954. Factors governing the changes in strength of a partially inborn response, as shown by the mobbing behaviour of the chaffinch (Fringilla coelebs), II. The waning of the response. Proc. Roy. Soc. B, 142: 331-358.
- Hinde, R. A. 1959. Behaviour and speciation in birds and lower vertebrates. Biol. Rev. 34: 85-128.
- Hoar, W. S. 1958. The evolution of migratory behaviour among juvenile salmon of the genus Oncorhynchus. J. Fish. Res. Bd. Canada 15: 391-428.
- Hobson, E. S. 1968. Predatory behavior of some shore fishes in the Gulf of California. U. S. Bureau Sport Fish. Wildl. Res. Rep. 73: 1-92.
- Holcomb, L. C., and G. Twiest. 1970. Growth rates and sex ratios of red-winged blackbird nestlings. Wilson Bull. 82: 294-303.
- Holling, C. S. 1966. The functional response of invertebrate predators to prey density. Mem. Ent. Soc. Can. 48: 1-86.
- Hoogland, R., D. Morris, and N. Tinbergen. 1957. The spines of sticklebacks (Gasterosteus and Pygosteus) as means of defence against predators (Perca and Esox). Behaviour 10: 205-236.
- Hubbs, C. L. 1940. Speciation of fishes. Amer. Natur. 74: 198-211.
- Huffaker, C. F., K. P. Shea, and S. G. Herman. 1963. Experimental studies on predation: Complex dispersion and levels of food in an acarine predator-prey interaction. Hilgardia 34: 305-330.
- Humphries, D. A., and P. M. Driver. 1970. Protean defence by prey animals. Oecologia (Berl.) 5: 285-302.
- Hunter, J. R. 1966. Procedure for analysis of schooling behavior. J. Fish. Res. Bd. Canada 23: 547-562.
- Hynes, H. B. N. 1970. The ecology of running waters. Univ. Toronto Press, Toronto. 555 p.

- Ivlev, V. S. 1961. Experimental ecology of the feeding of fishes. Yale Univ. Press, New Haven. 302 p.
- Jackson, P. B. N. 1961. The impact of predation, especially by the tigerfish (Hydrocyon vittatus) on African freshwater fishes. Proc. Zool. Soc. Lond. 136: 603-622.
- Jackson, P. B. N. 1965. Reply to G. Fryer and P. H. Greenwood. Proc. Zool. Soc. Lond. 144: 313-321.
- Johnson, C. R. 1970. Escape behavior and camouflage in two subspecies of Sceloporus occidentalis. Amer. Midl. Natur. 84: 280-282.
- Junge, G. C. A., and G. F. Mees. 1961. The avifauna of Trinidad and Tobago. Rijksmuseum van Natuurlijke Historie, Leiden. 172 p.
- Kallman, K. D. 1965. Genetics and geography of sex determination in the poeciliid fish, Xiphophorus maculatus. Zoologica 50: 151-190.
- Keenleyside, M. H. A. 1955. Some aspects of the schooling behaviour of fish. Behaviour 8: 183-248.
- Kettlewell, H. B. D. 1961. The phenomenon of industrial melanism in Lepidoptera. Annu. Rev. Entomol. 6: 245-262.
- Kimura, M., and G. Weiss. 1964. The stepping stone model of population structure and the decrease of genetic correlation with distance. Genetics 49: 561-576.
- King, J. A. 1970. Ecological psychology: an approach to motivation, p. 1-33. In W. J. Arnold and M. M. Page (ed.) Nebraska symposium on motivation. Univ. Nebraska Press, Lincoln.
- Krumholz, L. A. 1963. Relationships between fertility, sex ratio, and exposure to predation in populations of the mosquitofish Gambusia manni Hubbs at Bimini, Bahamas. Int. Rev. Ges. Hydrobiol. 48: 201-256.
- Kruuk, H. 1964. Predators and anti-predator behaviour of the black-headed gull (Larus ridibundus L.). Behaviour, suppl. 11: 1-130.
- Kruuk, H. 1972. The spotted hyena. Univ. Chicago Press, Chicago. 335p.

- Lack, D. 1954. The natural regulation of animal numbers. Clarendon, Oxford.
- Lang, H.-J. 1965. Eine lichtmikroskopische Untersuchung der Guppy-Netzhaut. Zeiss-Mitt. 3: 415-438.
- Lang, H.-J. 1967. Über das Lichttrückenverhalten des guppy (Lebistes reticulatus) in farbigen und farblosen Lichtern. Z. Vergl. Physiol. 56: 296-340.
- Larson, S. 1960. On the influence of the Arctic fox Alopex lagopus on the distribution of Arctic birds. Oikos 11: 276-305.
- LeBrasseur, R. J. 1969. Growth of juvenile Chum salmon (Oncorhynchus keta) under different feeding regimes. J. Fish. Res. Bd. Canada 26: 1631-1645.
- Lehrman, D. S. 1970. Semantic and conceptual issues in the nature-nurture problem, p. 17-52. In L. R. Aronson, E. Tobach, D. S. Lehrman, and J. S. Rosenblatt (ed.) Development and evolution of behavior. Freeman, San Francisco.
- Liley, N. R. 1966. Ethological isolating mechanisms in four sympatric species of poeciliid fishes. Behaviour, Suppl. 13: 1-197.
- Lindzey, G., J. Loehlin, M. Manosevitz, and D. Thiessen. 1971. Behavioral genetics. Annu. Rev. Psych. 22: 39-94.
- Liu, R. K., and R. L. Walford. 1966. Increased growth and lifespan with lowered ambient temperature in the annual fish Cynolebias adolffi. Nature 212: 1277-1278.
- Lowe-McConnell, R. H. 1964. The fishes of the Rupununi savanna district of British Guiana. Part I. Ecological groupings of fish species and effects of the seasonal cycle on the fish. Zool. J. Linn. Soc. 45: 103-144.
- Lowe-McConnell, R. H. 1969a. The cichlid fishes of Guyana, South America, with notes on their ecology and breeding behaviour. Zool. J. Linn. Soc. 48: 255-302.
- Lowe-McConnell, R. H. 1969b. Speciation in tropical freshwater fishes. Biol. J. Linn. Soc. 1: 51-75.

- Maher, W. J. 1970. The pomarine jaeger as a brown lemming predator in northern Alaska. *Wilson Bull.* 82: 130-157.
- Manning, A. 1967. Genes and the evolution of insect behavior, p. 44-60. In J. Hirsch (ed.) *Behavior-genetic analysis*. McGraw-Hill, New York.
- Manteifel', B. P., and D. V. Radakov. 1961. The adaptive significance of schooling behavior in fishes. *Russ. Rev. Biol.* 50: 338-345.
- Marler, P., and W. J. Hamilton III. 1966. *Mechanisms of animal behavior*. Wiley, New York. 771 p.
- Mason, L. G. 1965. Prey selection by a non-specific predator. *Evolution* 19: 259-260.
- Mather, K. 1970. The nature and significance of variation in wild populations. *Symp. Zool. Soc. Lond.* 26: 27-39.
- McPhail, J. D. 1969. Predation and the evolution of a stickleback (*Gasterosteus*). *J. Fish. Res. Bd. Canada* 26: 3183-3208.
- Melzack, R. 1961. On the survival of mallard ducks after "habituation" to the hawk-shaped figure. *Behaviour* 17: 9-16.
- Miller, R. B. 1957. Have the genetic patterns of fishes been altered by introduction or by selective fishing? *J. Fish. Res. Bd. Canada* 14: 797-806.
- Moodie, G. E. E. 1972a. Predation, natural selection and adaptation in an unusual threespine stickleback. *Heredity* 28: 155-168.
- Moodie, G. E. E. 1972b. Morphology, life-history, and ecology of an unusual stickleback (*Gasterosteus aculeatus*) in the Queen Charlotte Islands, Canada. *Can. J. Zool.* 50: 721-732.
- Needham, P. R., and J. G. Needham. 1962. *A guide to the study of fresh-water biology*. Holden-Day, San Francisco. 108 p.
- Nice, M. M., and J. J. Ter Pelkwyk. 1941. Enemy recognition by the song sparrow. *Auk* 58: 195-214.

- Nikolskii, G. V. 1969. Theory of fish population dynamics as the biological background for rational exploitation and management of fishery resources. Oliver and Boyd, Edinburgh. 323 p.
- Noble, G. K. 1938. Sexual selection among fishes. Biol. Rev. 13: 133-155.
- Nyberg, D. W. 1971. Prey capture in the largemouth bass. Amer. Midl. Natur. 86: 128-144.
- Olson, D. P. 1965. Differential vulnerability of male and female canvasbacks to hunting. Trans. N. Amer. Wildl. Conf. 30: 121-135.
- Olson, F. C. W. 1964. The survival value of fish schooling. J. Conseil Int. Exploration Mer 29: 115-116.
- Oortmerssen, G. A. van. 1970. Biological significance, genetics and evolutionary origin of variability in behaviour within and between inbred strains of mice (Mus musculus). Behaviour 38: 1-92.
- Paloheimo, J. E., and L. M. Dickie. 1966. Food and growth of fishes. II. Effects of food and temperature on the relation between metabolism and body weight. J. Fish. Res. Bd. Canada 23: 869-908.
- Parker, R. R. 1971. Size selective predation among juvenile salmonid fishes in a British Columbia inlet. J. Fish. Res. Bd. Canada 28: 1503-1510.
- Pfeiffer, W. 1962. The fright reaction of fish. Biol. Rev. 37: 495-511.
- Plutchik, R. 1971. Individual and breed differences in approach and withdrawal in dogs. Behaviour 40: 302-311.
- Protasov, V. R. 1970. Vision and near orientation of fish. Israel Program Sci. Transl., Jerusalem. 175 p.
- Raleigh, R. F., and D. W. Chapman. 1971. Genetic control in lakeward migrations of cutthroat trout fry. Trans. Amer. Fish. Soc. 100: 33-40.
- Ray, C. 1960. The application of Bergmann's and Allen's rules to the poikilotherms. J. Morphol. 106: 85-108.

- Reed, J. R. 1969. Alarm substances and fright reaction in some fishes from the southeastern United States. Trans. Amer. Fish. Soc. 98: 664-668.
- Ricker, W. E. 1954. Stock and recruitment. J. Fish. Res. Bd. Canada 11: 559-623.
- Ricker, W. E. 1969. Effects of size-selective mortality and sampling bias on estimates of growth, mortality, production and yield. J. Fish. Res. Bd. Canada 26: 479-541.
- Roberts, T. R. 1972. Ecology of fishes in the Amazon and Congo basins. Bull. Mus. Comp. Zool. 143: 117-147.
- Robinson, M. H. 1969. Defenses against visually hunting predators, p. 225-259. In T. Dobzhansky, M. K. Hecht, and W. C. Steere (ed.) Evolutionary Biology. Vol. 3. Appleton-Century-Crofts, New York.
- Rohlf, F. J., and G. D. Schnell. 1971. An investigation of the isolation-by-distance model. Amer. Natur. 105: 295-324.
- Rose, S. M. 1959. Population control in guppies. Amer. Midl. Natur. 62: 474-481.
- Rosen, D. E., and R. M. Bailey. 1963. The poeciliid fishes (Cyprinodontiformes), their structure, zoogeography and systematics. Bull. Amer. Mus. Natur. Hist. 126: 1-176.
- Rosen, D. E., and A. Tucker. 1961. Evolution of secondary sexual characters and sexual behavior patterns in a family of viviparous fishes (Cyprinodontiformes: Poeciliidae). Copeia 1961: 201-212.
- Rothenbuhler, W. C. 1964. Behavior genetics of nest cleaning in honey bees. IV. Response of F_1 and backcross generations to disease-killed brood. Amer. Zool. 4: 111-123.
- Rüppell, G., and E. Gösswein. 1972. Die Schwärme von Leucaspius delineatus (Cyprinidae, Teleostei) bei Gefahr im Hellen und im Dunkeln. Z. Vergl. Physiol. 76: 333-340.
- Russell, E. M. 1967a. Changes in the behaviour of Lebistes reticulatus upon a repeated shadow stimulus. Anim. Behav. 15: 574-585.

- Russell, E. M. 1967b. The effect of experience of surroundings on the response of Lebistes reticulatus to a strange object. Anim. Behav. 15: 586-594.
- Salyer, J. C. II, and K. F. Lagler. 1949. The eastern belted kingfisher, Megaceryle alcyon alcyon (Linnaeus), in relation to fish management. Trans. Amer. Fish. Soc. 76: 97-117.
- Schleidt, W. M. 1961. Reaktionen von Truthühnern auf fliegende Raubvögel und Versuche zur Analyse ihrer AAM's. Z. Tierpsychol. 18: 534-560.
- Schutz, F. 1956. Vergleichende Untersuchungen über die Schreckreaktion bei Fischen und deren Verbreitung. Z. Vergl. Physiol. 38: 84-135.
- Selander, R. K. 1965. On mating systems and sexual selection. Amer. Natur. 99: 129-141.
- Semler, D. E. 1971. Some aspects of adaptation in a polymorphism for breeding colours in the threespine stickleback (Gasterosteus aculeatus). J. Zool., Lond. 165: 291-302.
- Shaw, E. 1970. Schooling in fishes: critique and review, p. 452-480. In L. R. Aronson, E. Tobach, D. S. Lehrman, and J. S. Rosenblatt (ed.) Development and evolution of behavior. Freeman, San Francisco.
- Shoemaker, H. H. 1944. A laboratory study of fish populations. Trans. Amer. Fish. Soc. 74: 350-359.
- Shontz, C. J. 1962. The effects of altitude and latitude on the morphometry, meristics, growth and fecundity of the eastern blacknose dace, Rhinichthys atratulus atratulus (Hermann). Ph. D. Thesis. Univ. Pittsburgh. 161 p.
- Snyder, N. F. R., and H. A. Snyder. 1971. Defenses of the Florida apple snail Pomacea paludosa. Behaviour 40: 175-215.
- Soulé, M. 1966. Trends in the insular radiation of a lizard. Amer. Natur. 100: 47-64.
- Sterba, G. 1962. Freshwater fishes of the world. Vista Books, London. 878 p.
- Struhsaker, J. W. 1968. Selection mechanisms associated with intraspecific shell variation in Littorina picta (Prosobranchia: Mesogastropoda). Evolution 22: 459-480.

- Suthers, R. A. 1965. Acoustic orientation by fish-catching bats. *J. Exp. Zool.* 158: 319-348.
- Symons, P. E. K. 1971. Estimating distances between fish schooling in an aquarium. *J. Fish. Res. Bd. Canada* 28: 1805-1806.
- Thiessen, D. D. 1972. A move toward species-specific analyses in behavior genetics. *Behav. Genet.* 2: 115-126.
- Thornhill, E., L. Ramcharitar, D. Herrera, R. Aaron, R. Rudder, C. Thomas, K. James, and R. Moore. 1966. Ecological study of a part of the Maracas River. Univ. West Indies, St. Augustine, Trinidad. 25 p.
- Thorpe, W. H. 1963. Learning and instinct in animals. Methuen, London. 558 p.
- Tinbergen, N. 1959. Behaviour, systematics and natural selection. *Ibis* 101: 318-330.
- Veselov, E. A. 1964. Simplest methods for studies on some reflexes of fish larvae and fry, p. 297-311. In E. N. Pavlovskii (ed.) *Techniques for the investigation of fish physiology*. Israel Program Sci. Transl., Jerusalem.
- Wahlsten, D. 1972. Genetic experiments with animal learning: a critical review. *Behav. Biol.* 7: 143-182.
- Waldman, B. 1969. Do guppies see colors? *The Aquarium* 3: 12-16, 58-61.
- Walther, F. R. 1969. Flight behaviour and avoidance of predators in Thomson's gazelle (*Gazella thomsoni* Guenther 1884). *Behaviour* 34: 184-221.
- Ware, D. M. 1971. The predatory behaviour of rainbow trout (*Salmo gairdneri*). Ph. D. Thesis. Univ. British Columbia. 158 p.
- Weatherley, A. H. 1966. Ecology of fish growth. *Nature* 212: 1321-1324.
- Werboff, J., and T. Lloyd. 1963. Avoidance conditioning in the guppy (*Lebistes reticulatus*). *Psychol. Rep.* 12: 615-618.

- Werner, Y. L. 1969. Eye size in geckos of various ecological types (Reptilia: Gekkonidae and Sphaerodactylidae). Israel J. Zool. 18: 291-316.
- White, H. C. 1936. The food of kingfishers and mergansers on the Margaree river, Nova Scotia. J. Biol. Bd. Canada 2: 299-309.
- Whitney, G. 1970. Timidity and fearfulness of laboratory mice: an illustration of problems in animal temperament. Behav. Genet. 1: 77-85.
- Williams, G. C. 1964. Measurement of consociation among fishes and comments on the evolution of schooling. Pub. Mus. Mich. State Univ., Biol. Ser. 2: 351-383.
- Williams, G. C. 1966. Adaptation and natural selection. Princeton Univ. Press. Princeton, N. J. 307 p.
- Williamson, K. 1952. Regional variation in the distraction displays of the oystercatcher. Ibis 94: 85-97.
- Wilz, K. J. 1971. Comparative aspects of courtship behavior in the ten-spined stickleback, Pygosteus pungitius (L.). Z. Tierpsychol. 29: 1-10.
- Winge, Ø. 1922a. A peculiar mode of inheritance and its cytological explanation. J. Genet. 12: 137-144.
- Winge, Ø. 1922b. One-sided masculine and sex-linked inheritance in Lebistes reticulatus. J. Genet. 12: 145-162.
- Winge, Ø. 1927. The location of eighteen genes in Lebistes reticulatus. J. Genet. 18: 1-42.
- Winge, Ø. 1934. The experimental alteration of sex chromosomes into autosomes and vice versa, as illustrated by Lebistes. C. R. Lab. Carlsberg 21: 1-50.
- Winge, Ø., and E. Ditlevsen. 1947. Colour inheritance and sex determination in Lebistes. Heredity 1: 65-83.
- Wodinsky, J., E. R. Behrend, and M. E. Bitterman. 1962. Avoidance-conditioning in two species of fish. Anim. Behav. 10: 76-78.
- Worth, B. 1967. A naturalist in Trinidad. Lippincott, Philadelphia. 291 p.

Zaret, T. M., and A. S. Rand. 1971. Competition in tropical stream fishes: support for the competitive exclusion principle. Ecology 52: 336-342.

APPENDIX

Contents

TABLE		Page
1	Physical and chemical characteristics of streams in the Northern Range region	262
2	The families and species of fish collected at 23 sites in the Northern Range region . . .	268
3	Summary of the diets of the main potential predators of the guppy	269
4	Size statistics of guppies used in Exp. 6.2. . .	270

FIGURE

1	Map of the experimental section of the Petite Curucaye R.	271
2	Size distribution of the entire guppy population of the experimental section of Petite Curucaye River	272
3	Size distribution of the entire <u>Rivulus</u> population of the experimental section of Petite Curucaye River	273

TABLE 1. Physical and chemical characteristics of streams in the Northern Range region. The measurements were made in 1969. For methods refer to Chapter 2. For stream locations and abbreviations refer to Chapter 2, Figure 1.

Key to symbols

Weather: S - full sun; no large clouds
 SP - sunny periods; no rain but intermittent cloud cover
 SPR - sunny periods with rain
 (L) - light rain
 (H) - heavy rain
 C - cloudy with no sunny periods
 CR - cloudy with rain (L) or (H)
 HR - heavy continuous rain

Characteristics of stream:

 W - mean width of stream in metres
 D - mean depth of stream in metres
 a - a constant used in calculating volume of flow
 0.9 - smooth sand, silt, mud
 0.8 - rough rocks, large pebbles
 R - volume of flow (discharge) in m³/sec
 V - water velocity of main portion of stream in m/sec
 T - water temperature in °C
 DH - total degrees of water hardness in ppm
 C - degree of overhead cover (shade)
 0 - no shade
 1 - small amount of shade restricted mainly to streambank
 2 - medium shade (50% cover)
 3 - medium to dense (75% cover with few exposed parts)
 4 - very dense cover with virtually complete shading
 Turb - water turbidity
 0 - always clear
 1 - turbid only after heavy rains
 2 - turbid throughout year

TABLE 1 (Continued)

Stream	Date	Time	Weather	W	D	a	R	V	T	pH	DH	C	Turb
PCur	20 Mar	1100	S	0.60	0.05	0.9	0.0072	0.266	24.7	7.5	300	4	0
	28	1100	SP	0.40	0.04		0.0052	0.363	24.5	7.3	300		
	14 Apr	1400	SPR(H)	0.60	0.05		0.0054	0.200	25.0	7.2	280		
	27	1530	--	0.60	0.05		0.0064	0.235	25.0	7.2	--		
	16 May	1645	SP	--	--		--	--	25.1	7.2	260		
	17	1400	S	0.70	0.04		0.0048	0.190	25.1	7.2	250		
	19	1630	S	--	--		--	--	25.0	7.2	250		
	20	1030	SP	0.75	0.06		0.0202	0.500	25.1	7.2	240		
	20	1630	HR	--	--		--	--	24.8	7.2	220		
	22	0915	S	--	--		--	--	24.8	7.2	220		
	22	1445	S	--	--		--	--	25.5	7.2	220		
	27	1545	SPR(H)	--	--		--	--	25.1	7.2	200		
	7 Jun	1400	C	0.90	0.07		0.0142	0.250	25.0	7.2	210		
	13	1100	SPR	--	--		--	--	25.3	7.2	230		
	18	1315	S	0.50	0.05		0.0064	0.286	25.2	7.2	180		
	1 Jul	1500	SP	0.60	0.04		0.0066	0.308	25.1	7.2	150		
	5	1730	S	--	--		--	--	25.0	7.2	170		
	Means			0.63	0.05		0.0085	0.289	25.0	7.2	230		
GCur	22 Apr	1100	SP	1.00	0.15	0.8	0.0128	0.107	25.1	7.0	--	3	0
	20 May	0920	C	--	--		--	--	25.1	7.2	200		
	Means			1.00	0.15		0.0128	0.107	25.1	7.1	200		
UCur	13 Mar	1330	S	--	--	0.8	--	--	24.8	7.4	305	3	0
TT	25 Mar	1400	SP	0.90	0.11	0.9	0	0	26.2	7.2	--	4	0

TABLE 1 (Continued)

Stream	Date	Time	Weather	W	D	a	R	V	T	pH	DH	C	Turb
UA(N)	29 Mar	1030	SP	--	--	0.9	--	--	24.5	7.5	210	3	0
	13 Apr	1120	S	--	--		--	--	24.8	7.6	190		
	3 Jul	1100	SP	--	--		--	--	24.5	7.4	150		
	Means								24.6	7.5	183.3		
UA(X)	29 Mar	0930	SP	--	--	0.8	--	--	24.0	7.6	220	3	0
	13 May	1330	SP	3.00	0.08		0.0480	0.250	28.1	7.6	190		
	3 Jun	1500	SPR(L)	--	--		--	--	28.8	7.6	130		
	15	1300	CR(H)	4.00	0.22		0.7040	1.000	24.1	7.0	110		
	21	1200	CR(H)	--	--		--	--	26.3	7.2	90		
	3 Jul	1200	SP	4.00	0.08		0.1024	0.400	26.4	7.6	150		
	Means			3.67	0.13		0.2848	0.550	26.3	7.4	148.3		
UTac	15 Apr	1630	SP	1.30	0.15	0.8	0.0780	0.500	26.9	7.0	100	2	0
	24 May	1520	S	1.10	0.09		0.0297	0.375	27.8	7.0	90		
	Means			1.20	0.12		0.0539	0.438	27.4	7.0	95		
UArouc	2 May	1500	S	2.00	0.06	0.9	0.0348	0.323	26.8	7.0	--	2	0
UGuan	2 May	1200	S	5.00	0.10	0.8	0.2667	0.667	26.5	7.0	--	2	0

TABLE 1 (Continued)

Stream	Date	Time	Weather	W	D	a	R	V	T	pH	DH	C	Turb
BB	6 May	1220	SP	0.80	0.08	0.8	0.0279	0.545	24.9	7.6	--	3	0
	29	1300	SPR(L)	--	--		--	0.500	24.9	7.4	140		
	24 Jun	1400	SPR(L)	--	--		--	--	25.2	7.6	150		
	Means			0.80	0.08		0.0279	0.523	25.0	7.5	145		
Mar	19 Mar	1530	SP	4.50	0.20	0.8	0.360	0.500	23.5	7.1	140	2	0
	29 Jun	1200	SP	5.00	0.11		0.259	0.588	25.1	7.2	100		
	Means			4.75	0.16		0.309	0.544	24.3	7.2	120		
Oro	8 Mar	1500	S	--	--	0.8	--	--	26.0	7.8	250	2	0-1
	25 Apr	1030	SP	6.00	0.20		1.129	1.176	26.8	7.2	--		
	10 May	1200	S	--	--		--	--	27.2	7.4	190		
	Means			6.00	0.20		1.129	1.176	26.7	7.5	220		
Guay	16 Mar	1000	SP	--	--	0.9	--	--	28.0	7.1	110	1	2
	7 Apr	1115	SP	--	--		--	--	29.2	7.0	120		
	9	1600	S	--	--		--	--	30.9	7.0	130		
	12	1100	C	--	--		--	--	28.6	7.0	120		
	17	1500	S	--	--		--	--	30.9	7.0	--		
	30	0630	S	--	--		--	--	28.5	7.0	--		
	12 May	1400	S	--	--		--	--	31.0	7.0	130		
	10 Jun	1500	S	--	--		--	--	27.9	6.8	70		
	30	1330	SPR(H)	2.00	1.50		1.08*	0.400	27.0	7.0	80		
	Means			2.00	1.50		1.08	0.400	29.1	7.0	108.6		

*rough approximation--the volume of flow of lowland rivers is very variable depending on season and weather.

TABLE 1 (Continued)

Stream	Date	Time	Weather	W	D	a	R	V	T	pH	DH	C	Turb
Cap	24 Apr	1100	S	--	--	0.9	--	--	28.0	7.0	--	0	2
	17 Jun	1600	CR(H)	--	--		--	--	26.2	6.8	110		
	30	1200	CR(H)	5.00	1.50		2.25*	0.333	26.5	7.0	110		
	Means			5.00	1.50		2.25	0.333	26.9	6.9	110		
Car	7 Apr	1100	SP	--	--	0.9	--	--	30.2	7.0	170	0	2
	30 Jun	1400	SPR(H)	25.0	3.00		22.50*	0.333	27.2	7.0	90		
	Means			25.0	3.00		22.50	0.333	28.7	7.0	130		
Yar	19 Mar	1700	SP	0.75	0.07	0.8	0.0280	0.667	24.0	7.1	140	2	0
	29 Jun	1400	CR(L)	0.80	0.13		0.0289	0.348	26.0	7.0	80		
	Means			0.78	0.10		0.0285	0.508	25.0	7.1	110		
Par	18 Apr	1215	SPR	3.00	0.20	0.8	0.0828	0.172	25.0	7.1	--	2	0
	11 Jun	1400	SP	--	--		--	--	26.5	7.0	100		
	22	1300	SPR(L)	--	--		--	--	25.0	7.0	100		
	29	1600	CR(L)	4.00	0.14	0.8	0.0729	0.163	25.1	7.0	90		
	Means			3.50	0.17		0.0779	0.168	25.4	7.0	96.7		

TABLE 1 (Continued)

Stream	Date	Time	Weather	W	D	a	R	V	T	pH	DH	C	Turb
LA	2 Apr	1115	SP	5.00	0.12	0.8	0.3200	0.667	26.5	7.5	230	2	0-1
	3 Jun	1600	CR	--	--		--	--	25.5	7.2	90		
	21	1600	SPR(L)	--	--		--	--	25.6	7.0	90		
	3 Jul	1300	SP	7.00	0.13		0.4040	0.555	26.4	7.2	100		
	Means			6.00	0.13		0.3620	0.611	26.0	7.2	127.5		
LTac	5 Apr	1000	CR	3.00	0.15	0.8	0	0	28.4	8.0+	190	2	0-1
	10	1330	--	--	--		--	--	33.5	8.0+	190		
	3 Jul	1530	SP	3.00	0.15		0.150	0.417	30.0	7.6	100		
	Means			3.00	0.15		0.075	0.209	30.6	--	160		
SC	13 Mar	1500	S	--	--	0.8	--	--	29.9	--	--	2	0-1
	28	1330	SP	--	--		--	--	28.4	7.2	300		
	14 Apr	1530	SPR(H)	8.00	0.20		0.768	0.600	27.4	7.2	280		
	16 May	1715	SP	--	--		--	--	27.5	7.0	250		
	5 Jul	1700	S	--	--		--	--	27.9	7.2	160		
	Means			8.00	0.20		0.768	0.600	28.2	7.2	247.5		

TABLE 2. The families and species of fish collected at 23 sites in the Northern Range region. For a list of stream abbreviations, refer to Fig. 1.

FAMILY AND SPECIES	UCur	PCur	SLR	MV	TT	GCur	UA	BB	Par	Yar	UArouc	UGuan	UTac	LPar	Mar	LCur	SC	Oro	LTac	LA	Cap	Car	Guay
F. Poeciliidae																							
<u>Poecilia reticulata</u>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
F. Cyprinodontidae																							
<u>Rivulus hartii</u>	+	+	+	+	+	+	+	+	+	+						+							
F. Cichlidae																							
<u>Aequidens pulcher</u>											+	+	+			+	+	+	+				
<u>Cichlasoma bimaculatum</u>																					+	+	
<u>Crenicichla alta</u>											+	+	+				+	+	+	b	b	+	+
<u>Tilapia mossambica</u>																						+	
F. Characidae																							
<u>Astyanax bimaculatus</u>																+	+	+	+	+			+
<u>Hemibrycon</u> sp.											+	+	+						+				
<u>Curimata argentea</u>																					+	+	
<u>Roboides dayi</u>																				+	+	+	
<u>Hoplias malabaricus</u>											a	a	+				+	+	+	a	+	+	+
<u>Corynopoma riisei</u>																		+	+	+	+	+	+
<u>Hemigrammus unilineatus</u>																							+
<u>Pristella riddlei</u> or																							
<u>Aphyocharax axelrodi</u>																	+			+	+	+	+
F. Loricariidae																							
<u>Hypostomus robinii</u>							+				a	+	+			+	+	+	+	+	+	+	+
<u>Ancistrus cirrhosus</u>																				+	+	+	+
F. Pimelodidae																							
<u>Rhamdia</u> sp.											+	+	+						+	+	+		+
F. Synbranchidae																							
<u>Synbranchus marmoratus</u>								+					+										
F. Callichthyidae																							
<u>Corydoras aeneus</u>																					+		+
F. Mugilidae																							
<u>Agonostomus monticola</u> (?)																							
F. Gobiidae																							
<u>Sicydium</u> sp. (?)									+	+				+	+								
Unidentified goby																							

a - species not seen or collected in this study but presence determined through personal communication with local residents.

b - species present in river according to Boeseman (1960).

TABLE 3. Summary of the diets of the main potential predators of the guppy. The data are expressed as the per cent frequency of occurrence of each food item.

	RIVULUS	CRENICICHLA	AEQUIDENS	HEMIBRYCON	ASTYANAX	HOPLIAS
Source	B. Basin Tomp. Trib. P. Curucaye	L. Aripo Maracas	L. Aripo L. Tacarigua Maracas	L. Aripo U. Arouca Maracas	L. Aripo L. Tacarigua	L. Aripo L. Tacarigua Maracas Caroni Guayamare
Number examined	259	14	42	64	28	7
Number empty	5	0	5	3	1	6
Size range (mm)	22-105	60-195	80-148	62-90	45-116	90-392
Contents						
<u>P. reticulata</u>	10.2	14.3		1.6	18.5	
Other fish		42.9 ^a			3.7 ^a	100 ^b
Oligochaeta					3.7	
Crustacea					3.7	
Mollusca		14.3	29.7			
Ephemeroptera		7.1		4.9	29.6	
Trichoptera	5.5	7.1	2.7	1.6	3.7	
Diptera	3.9			3.3	11.1	
Coleoptera	10.6			8.2	3.7	
Plecoptera	0.8					
Homoptera	0.8					
Hemiptera	7.1			3.3		
Arachnida	3.9				3.7	
Hymenoptera ^c	42.1		2.7	22.9	25.9	
Odonata	0.8			1.6	7.4	
Orthoptera	1.2					
Chilopoda	0.8					
Diplopoda	2.8					
Thysanoptera	0.4					
Others ^d	3.5	64.3	10.8	27.9	18.5	
Algae	3.1		2.7	27.9	40.7	
Vascular plants	5.9		2.7	3.3	25.9	

^aFish vertebrae, scales.

^bCrenicichla alta.

^cAll Formicidae.

^dUnidentified larval and adult insect parts.

TABLE 4. Size statistics of guppies used in Exp. 6.2. All sizes are in mm total length. n = 50 for each mean.

		A. 12 light--12 dark				B. 12 light--12 dim light			
		$\bar{x}$	SD	SE	Range	$\bar{x}$	SD	SE	Range
Females									
1.	Lower Aripo	20.34	1.61	0.23	17.0 - 24.0	24.40	3.32	0.47	18.5 - 30.0
	Guayamare	21.08	1.56	0.22	17.0 - 25.0	25.30	4.31	0.61	17.0 - 33.0
2.	Lower Aripo	21.54	2.32	0.33	18.0 - 25.5	20.76	1.84	0.26	16.0 - 25.0
	Petite Curucaye	22.07	2.16	0.31	18.0 - 25.5	20.59	2.19	0.31	17.0 - 26.5
3.	Lower Aripo	22.32	2.48	0.35	17.5 - 26.5	20.92	5.01	0.71	16.0 - 34.0
	Paria	21.43	3.67	0.52	17.0 - 28.5	21.95	4.13	0.58	16.0 - 31.5
4.	Lower Aripo	22.40	2.99	0.42	18.0 - 29.0	21.66	2.26	0.32	18.5 - 27.0
	Upper Aripo	22.21	2.43	0.34	18.0 - 28.0	20.68	1.97	0.28	17.0 - 25.5
Males									
1.	Lower Aripo	20.86	1.84	0.26	17.0 - 27.5	19.31	1.92	0.27	16.0 - 23.5
	Guayamare	20.64	1.70	0.24	18.0 - 25.0	18.54	2.02	0.29	15.0 - 23.0
2.	Lower Aripo	21.10	1.48	0.21	19.0 - 25.0	20.39	1.48	0.21	18.0 - 23.5
	Petite Curucaye	21.45	1.67	0.24	18.5 - 25.0	20.42	1.67	0.24	17.5 - 24.0
3.	Lower Aripo	21.00	1.43	0.20	18.0 - 24.0	20.74	1.42	0.20	18.0 - 23.5
	Paria	20.39	1.38	0.19	17.5 - 23.0	20.64	1.31	0.19	17.5 - 24.0
4.	Lower Aripo	21.19	1.18	0.17	18.5 - 24.0	18.79	2.62	0.37	16.0 - 25.0
	Upper Aripo	21.50	1.82	0.26	17.5 - 24.5	20.18	3.09	0.44	16.0 - 26.0

FIGURE 1. Map of the experimental section of the
Petite Curucaye R.

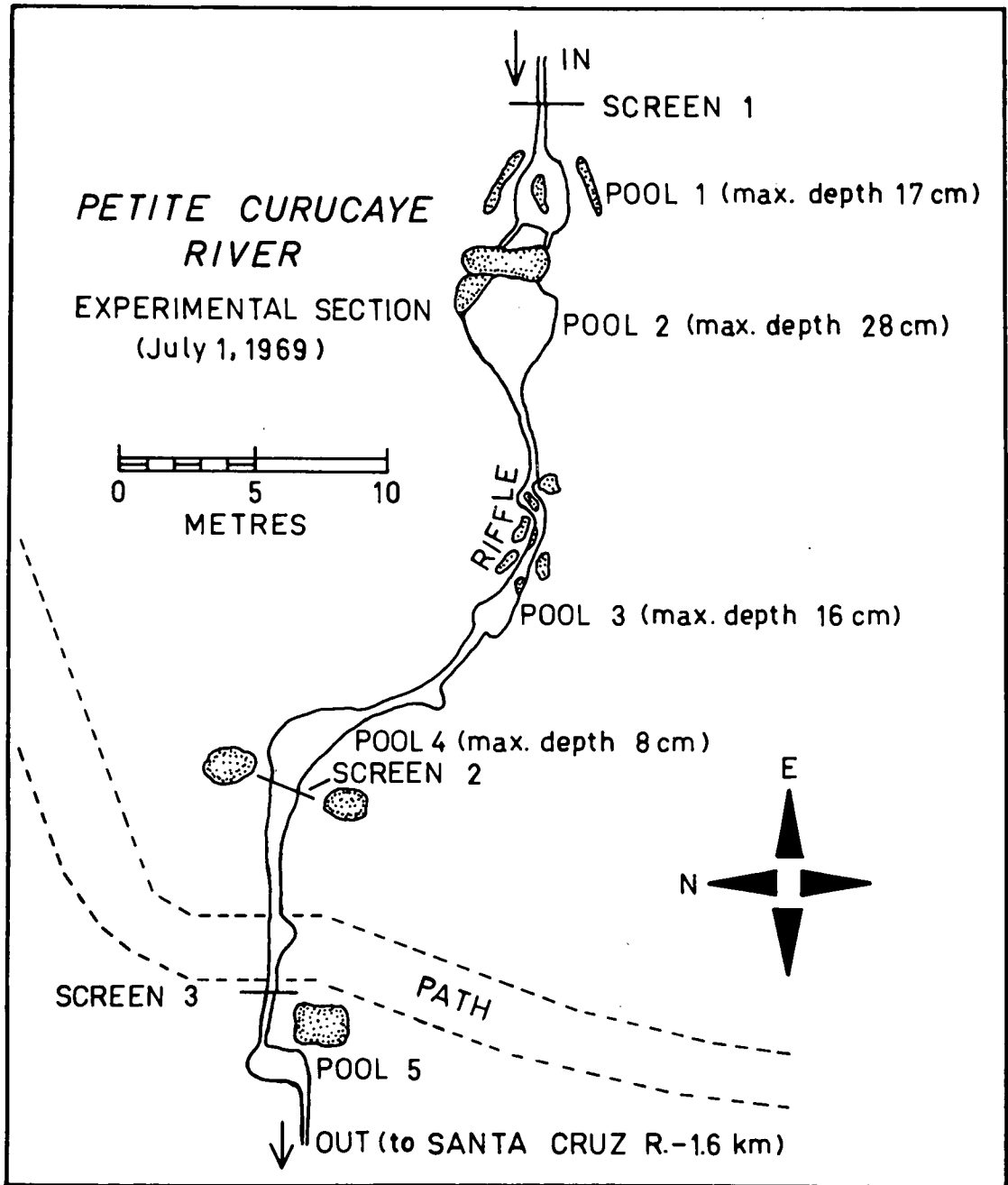


FIGURE 2. Size distribution of the entire guppy population of the experimental section of Petite Curucaye River.

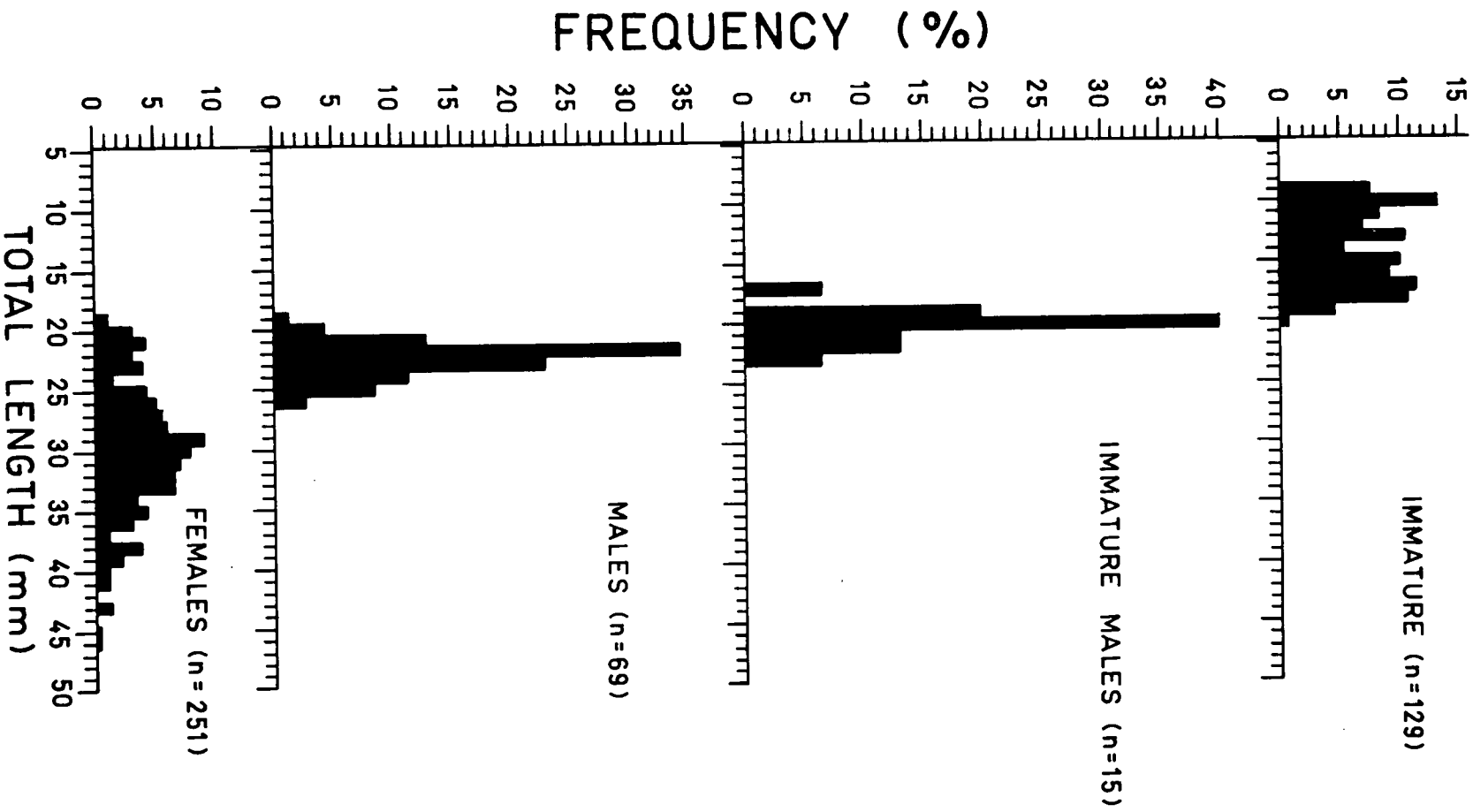


FIGURE 3. Size distribution of the entire Rivulus population of the experimental section of Petite Curucaye River.

